ANSI/IES RP-27-20
+ERRATA 1

Illuminating
ENGINEERING SOCIETY

RECOMMENDED PRACTICE: PHOTOBIOLOGICAL SAFETY FOR LIGHTING SYSTEMS

AN AMERICAN NATIONAL STANDARD

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ANSI/IES RP-27-20 ERRATA

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Additions will be posted to this list online as they become available. This errata list was last updated **March 18, 2021**.

Additions are shown in **bold** face. Deletions are shown as ~~strikeout~~.

In Section B.5 Bioeffect: Ultraviolet Cataract:
B.5.9 Experience with Lamps: Accidental injury is **not** known, even from exposure to xenon-arc lamps. Limited population exposed.

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Publication of this Recommended Practice
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by:
The IES Photobiology Committee

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Foreword

This Recommended Practice combines and updates the material from the previous ANSI/IES RP-27 series:

- *ANSI/IES RP-27.1-15, Recommended Practice for Photobiological Safety for Lamps and Lamp Systems – General Requirements*
- *ANSI/IES RP-27.2-00(R17), Recommended Practice for Photobiological Safety for Lamps and Lamp Systems – Measurement Techniques*
- *ANSI/IES RP-27.3-17, Recommended Practice for Photobiological Safety for Lamps and Lamp Systems – Risk Group Classification and Labeling*

1.0 Introduction and Scope

1.1 Introduction

Lamps were developed and produced in large quantities and became commonplace in an era when industry-wide safety standards were not common. The evaluation and control of lamp hazards is a more complicated subject than similar tasks for a single-wavelength laser system, where only one optical hazard might apply. The required radiometric measurements are quite involved, for they do not deal with the simple optics of a point source, but rather with an extended source that may or may not be altered by diffusers or projection optics. Also, the wavelength distribution of the lamp may be altered by ancillary optical elements such as diffusers and lenses, as well as variations in operating voltage.

To evaluate a broad-band optical source, such as an arc lamp, an incandescent lamp, a fluorescent lamp, LED (light emitting diode) singly or in an array, or a lamp system, three steps are necessary. First, the spectral distribution of optical radiation emitted from the source at an appropriate measurement location. The spectral distribution of the emission of interest for a lighting system may differ from that actually being emitted by the lamp alone, due to filtration by any optical elements (e.g., projection optics) in the light path. Second, the size, or projected size, of the source should be characterized in the retinal hazard spectral region. Third, it may be necessary to determine the variation of irradiance and projected radiance with distance.

The performance of the necessary measurements is not an easy task without sophisticated instruments. Photobiological action spectra, such as the ultraviolet (UV) Hazard Weighting Function, $S(\lambda)$, have rapidly changing values with slight changes in wavelength. Furthermore, sources such as lamps with glass envelopes have rapidly increasing output within the same ultraviolet wavelength band where $S(\lambda)$ is rapidly decreasing. (See **Annex E, Figure E-5**.) Hence, substantial inaccuracies in weighted results can arise from small measurement uncertainties. Users should normally rely upon the expertise of manufacturers for information on lamps and lamp systems. Safety requirements and reference measurement techniques for lamps and specific lamp systems are provided in this document.

The testing done for compliance with this Standard shall include a full analysis of the uncertainty in the results. This requirement leads to several corollary requirements:

- The testing shall be done by persons experienced in radiometry.
- The equipment used shall be fully characterized.
- The testing shall be scrutinized for all influences and sources of error.

This Standard recommends a double-monochromator system for measurements used in classifying sources, especially in the UV region, although such instrumentation may not be practical for some types of testing. The Standard therefore provides guidance on the use of other methods and when they may be appropriate. Alternative measurement methods described herein shall be used with full understanding of the limitations of each, and the method selected should be traceable back to spectral measurements. Further, the testing shall include a full analysis of the uncertainty of the results. The equipment used shall be fully characterized and major sources of error documented.

Finally, there are well known optical radiation hazards associated with some lamps and lamp systems. The purpose of this Standard is to inform the public and original equipment manufacturers (OEMs) about potential radiation hazards that may be associated with various lamps and lamp systems. It is also the purpose of this Standard to provide guidance, advice, and standard methods for evaluating and informing the user, both the public and the OEM, about the potential optical radiation hazards that may be associated with these products.

Note 1: Units of wavelength in this document are exclusively in nanometers (nm).

Note 2: Subtended angles are denoted by the full included angle, not the half angle.

1.2 Scope

This Recommended Practice covers the classification, labeling and informational requirements for lamps that emit optical radiation in the wavelength range from 200 nm to 3000 nm, with exception for LEDs used in optical fiber communication systems and for lasers. Lamps included are incandescent filament lamps including tungsten halogen types and incandescent heating sources, low pressure discharge lamps, high intensity discharge (HID) lamps, short arc lamps, carbon arcs, electroluminescent lamps, LEDs, organic LEDs (OLEDs), and laser-driven broadband sources. For the purposes of this document, induction lighting is classified under fluorescent lamps and plasma lighting is classified under HID lamps. Federal mandatory requirements for lamps subject to specific Federal Regulations take precedence over requirements included in this consensus standard.

Specific recommendations are included to provide consistency and to reduce test design time and effort. Further, this Recommended Practice is to be used by the radiometrist for guidance regarding special problems related to photobiological hazard measurements.

It is impractical for this Standard to teach all of the concepts or provide all experience needed to make accurate photobiological safety measurements.

2.0 References

2.1 Normative References

The following documents contain provisions which, through reference in this document, constitute provisions of the American National Standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this American National Standard are encouraged to investigate the possibility of applying the most recent editions of the standards listed here.

ANSI/IES LM-9-20, Approved Method: Electrical and Photometric Measurements of Fluorescent Lamps. New York: Illuminating Engineering Society; 2020.

ANSI/IES LM-20-20, Approved Method: Photometry of Reflector Type Lamps. New York: Illuminating Engineering Society; 2020.

ANSI/IES LM-45-20, Approved Method: Electrical and Photometric Measurements of General Service Incandescent Filament Lamps. New York: Illuminating Engineering Society; 2020.

ANSI/IES LM-51-20, Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps. New York: Illuminating Engineering Society; 2020.

ANSI/IES LM-54-20, Approved Method: IES Guide to Lamp Seasoning. New York: Illuminating Engineering Society; 2012.

ANSI/IES LM-58-20, Approved Method: Spectroradiometric Measurement Methods for Light Sources. New York: Illuminating Engineering Society; 2020.

ANSI/IES LM-66-20, Approved Method: Electrical and Photometric Measurements of Single-Based Fluorescent Lamps. New York: Illuminating Engineering Society; 2020.

ANSI/IES LM-79-19, Approved Method: Electrical and Photometric Measurements of Solid State Lighting

Products. New York: Illuminating Engineering Society; 2019.

ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: Illuminating Engineering Society; 2020. https://www.ies.org/standards/definitions/ (Accessed 2020 Feb 28).

ANSI/NCSL Z540-2-1997 (R2012), U.S. Guide to the Expression of Uncertainty in Measurement. Boulder, Colo.: National Conference of Standards Laboratories; 2012. (Equivalent is: ISO/IEC Guide 98-3, Evaluation of Measurement Data – Guide to the Expression of Uncertainty in Measurement, and its supplements.)

ASTM E387-84 (1995)e1, Standard Test Method for Estimating Stray Radiant Power of Spectrophotometers by Opaque Filter Method. Philadelphia: American Society for Testing Materials; 1995.

CIE 063-1984, The Spectroradiometric Measurement of Light Sources. Vienna: International Commission on Illumination; 1984.

CIE 105-1993, Spectroradiometry of Pulsed Optical Radiation Sources. Vienna: International Commission on Illumination; 1993.

CIE 134-1999, CIE Collection in Photobiology and Photochemistry. Vienna: International Commission on Illumination (CIE); 1999.

IEC 60825-1 ED.3.0:2014, Safety of Laser Products – Part 1: Equipment Classification and Requirements. International Electrotechnical Commission; 2014.

2.2 Informative References

Kostkowski HJ. Reliable Spectroradiometry. La Plata, MD: Spectroradiometry Consulting; 1997.

Ness JW, Zwick H, Stuck BE, Lund DJ, Lund BJ, Molchany JW, Sliney DH. Retinal image motion during deliberate fixation: Implications to laser safety for long duration viewing. Health Physics. 2000 Feb;78(2):131-142.

Sliney DH, Bergman R, O'Hagan J. Photobiological risk classification of lamps and lamp systems—History and rationale. LEUKOS. 2016;12(4):213-234.

Sliney DH, Wolbarsht ML. Safety with Lasers and Other Optical Sources. New York: Plenum Press; 1980.

3.0 Definitions

For standard nomenclature and definitions, radiometric and photometric quantities, and illuminating engineering terminology, the reader is referred to *ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering*. Online: www.ies.org/standards/definitions/. (Accessed 2019 Jul 31).

3.1 actinic
Capable of producing a photochemical effect.

3.2 assessment distance
Distance for Risk Group classification based upon reasonably foreseeable worst-case exposure. This is the *effective exposure distance* (see **Section 3.6**), which is time weighted and is not necessarily the measurement distance. *Note:* Radiometric measurements performed at one distance, e.g., 20 cm, can be transferred through calculations to the distance for risk assessment—e.g., the assessment distance of General Lighting Source (GLS) lamps where the illuminance is 500 lux—through use of the inverse-square law and source size evaluation.

3.3 attenuation coefficient, μ
The reduction in flux per unit distance in a given direction within a medium and defined by:

$$\Phi(r) = \Phi(0) \times e^{-\mu r},$$

where $\phi(r)$ is the flux at a distance r from a reference point having flux $\phi(0)$.

3.4 blue light hazard
Potential for a photochemically induced retinal injury resulting from radiation exposure at wavelengths primarily between 400 nm and 500 nm. This damage mechanism dominates over thermally induced retinal

injury for times exceeding 10 s (see **Annex B**, **Section B.3.3**, Ham reference).

3.5 continuous wave (CW) lamp

A lamp that is operated with a continuous output for a period of 0.25 s or longer, or a lamp with a modulation depth of 50% or less and a frequency greater than 100 Hz.

3.6 effective exposure distance

Nearest points of time-weighted human exposure consistent with the application of the lamp for which the time-weighted emission has been taken into account. The effective exposure distance for GLS lamps is the 500-lx distance.

3.7 emission limit

A limit defined for each Risk Group and potential optical radiation hazard (i.e., actinic UV, near UV, retinal thermal, blue light, and IR for cornea or lens), based on reasonably foreseeable conditions of exposure. It incorporates the concepts of both exposure duration and assessment distance and is derived from exposure limits. (See **Table 5-1** in **Section 5.3**.) *Note:* The *assessment distance* used for Risk Group classification is based on the worst-case *effective exposure distance* used for hazard evaluation.

3.8 erythema

The temporary reddening of the skin such as the delayed reddening produced by exposure to actinic UV radiation or the rapid reddening produced by heat, such as from exposure to intense infrared radiation (see **Section 3.14**).

Note: The degree of delayed erythema is used as a guide to dosages applied in ultraviolet therapy.

3.9 Exempt Group, RG-0

Risk Group classification for which there is no photobiological hazard. (See **Section 5.2.1**.)

3.10 exposure limit value

Maximum value of exposure to the eye or skin that is not expected to result in adverse biological effects.

3.11 general lighting source, GLS

A general term for lamps, nominally of white color, intended for lighting spaces that are typically occupied or viewed by people. Examples would be lamps for lighting offices, schools, homes, factories, or roadways. It does not include lamps for such uses as film projection, reprographic processes, "sun tanning," industrial processes, medical treatment, or searchlight applications. It also does not include lamps with outer envelopes constructed from quartz material that is not doped to block UV-B and UV-C (see **Section 3.31 ultraviolet radiation**). Although lamps with un-doped quartz outer envelopes may be used in locations typically occupied by people, their potential to emit ultraviolet radiation requires that they be more carefully evaluated. The process for classifying lamps is described in **Section 5.1**.

3.12 hazard distance

(See *ocular hazard distance* [**Section 3.20**].)

3.13 hybrid laser-lamp source

A lamp source that includes laser emission and may be combined with conventional lamp emissions; e.g., a laser-illuminated projector.

Note: This is not a laser-driven light source (see **Section 3.18**), since some laser emission is intended.

3.14 infrared (IR) radiation

Optical radiation for which the wavelengths are longer than those for visible radiation; from 780 nm to 10^6 nm (1 mm).

Note 1: In some applications, the infrared spectrum has also been divided into "near," "middle," and "far" infrared; however, the borders necessarily vary with the application (e.g., in meteorology, photochemistry, optical design, or thermal physics).

Note 2: For infrared radiation, the range between 780 nm and 10^6 nm is commonly subdivided into IR-A (780 nm to 1,400 nm), IR-B (1,400 nm to 3,000 nm), and IR-C (3,000 nm to 10^6 nm).

Note 3: Infrared radiation is generally evaluated in terms of its irradiance, i.e., the spectral total radiation incident per unit of the irradiated surface. Examples of applications of infrared radiation are industrial heating, drying, baking, and photoreproduction. Some applications, such as infrared viewing systems, involve detectors sensitive to a restricted range of wavelengths; in these cases, the spectral characteristics of the source and detector are of importance.

3.15 lamp

A generic term for a manufactured source created to produce optical radiation. As used in this standard, the term includes electrically powered devices that emit optical radiation from 200 to 3,000 nm.

Note 1: Devices that generate light and have integral components for optical control, such as lenses or reflectors, also are considered lamps. Examples include a lens-end lamp, and a PAR lamp with a reflector or lens cover.

Note 2: A device consisting of a lamp with shade, reflector, enclosing globe, housing, or other accessories is often called a "lamp." However, in this Standard, a lamp with such other components is termed a "lamp system" or "luminaire" to distinguish between the assembled unit and the light source within it.

3.16 lamp system

Any manufactured product or assemblage of components that incorporates, or is intended to incorporate, one or more lamps.

Note 1: A *luminaire* is the specific term for a lamp system used for general lighting.

Note 2: LEDs are almost always assembled into a lamp system (luminaire) for use in general lighting and are not separable into a lamp and a fixture.

3.17 lamp packaging

Any lamp carton, outer wrapping, or other means of containment that is intended for the storage, shipment, or display of a lamp(s) or that is intended to identify the contents or to recommend its use.

3.18 laser-driven light source

A lamp or lamp system that incorporates a laser but is designed to function as a conventional lamp, e.g., a phosphor-type-lamp system that employs a diode laser emitter rather than (or in addition to) an LED chip to excite the phosphor emission.

Note 1: Any leakage of coherent radiation must meet the requirements for a Class 1 laser product (U. S. Food and Drug Administration 21 Code of Federal Regulations 1040; ANSI Z136.1).

Note 2: Although laser-driven light sources are radiance limited and need not be tested for retinal thermal hazards, hybrid laser-lamp sources require such assessments. Furthermore, Risk Group 3 (RG-3) hybrid laser-lamp sources have an upper radiance limit required by International Electrotechnical Commission 60825-1:2014, which is:

$$L_T = (100\ W/\text{cm}^2\cdot\text{sr})/\alpha\ ,$$

where the angular subtense, α, of the optical source at 20 cm is limited to values between 0.005 rad and 0.1 rad. Thus, considering the minimum and maximum situations, for sources that subtend an angle of 0.005 rad or less (source diameter less than 1 mm seen at a 20-cm viewing distance), the applicable radiance criterion equals 20 kW/cm^2·sr, and for sources that are larger than 0.1 rad (source diameter greater than 20 mm seen at a 20-cm viewing distance), the applicable radiance criterion equals 1.0 kW/cm^2·sr.

3.19 ocular hazard distance

The distance from a source at which the projected radiance or irradiance for a given exposure duration equals the applicable ocular exposure limit.

Note: At distances greater than the ocular hazard distance, the exposure is less than the applicable limit.

3.20 photochemical effect

A biological effect produced by a chemical change in molecules, resulting from the absorption of photons. The changed molecules fail to function as before.

3.21 photokeratoconjunctivitis

A delayed inflammatory response of the cornea and conjunctiva following exposure to ultraviolet (UV) radiation.

Wavelengths shorter than 320 nm are most effective in causing photokeratoconjunctivitis. The peak of the action spectrum is approximately at 270 nm.

3.22 projected radiance

The radiance determined from measurements at the point of viewing (usually the human eye). This is not a universal term, but one defined for use in this Recommended Practice.

3.23 pulsed lamp

A lamp that delivers its energy in the form of a single pulse or a train of pulses, with each pulse having duration of less than 0.25 s, or a modulation depth greater than 50% and a frequency of less than 100 Hz.

Note 1: The duration of a lamp pulse is the time interval between the half-power points on the leading and trailing edges of the pulse.

Note 2: In this standard, GLS lamps are defined to be continuous wave lamps (see **Section 3.4**). Examples of pulsed lamps include photoflash lamps, flash lamps in photocopy machines, and strobe lights.

3.24 radiant exposure, H

$$H = \int E \cdot dt$$

The areal density of radiant energy incident upon a surface, where E is in watts per unit area. H is expressed in joules per unit area of the surface and is often termed dose.

3.25 retinal hazard region

The spectral region from 400 nm to 1,400 nm (visible plus IR-A) within which the normal ocular media transmit optical radiation to the retina.

3.26 retinal lesion

An unwanted change in the structure of the retina, caused by heating (thermal effect) or when photon absorption induces a chemical change (photochemical effect). Previously called *retinal burn*, *thermal retinal lesion*, or *photochemical lesion*.

3.27 Risk Group (RG)

A classification scheme for lamps and lamp systems that indicates the photobiological hazard. Four risk groups from Exempt to High Risk are categorized. (See **Section 5.1**.)

3.28 time-weighted-average (TWA)

The average value of the irradiance or radiance on the exposed subject over the applicable time. For risk evaluation it is usually assumed that the source emission is constant and that movement of the individual which occurs over the applicable time is taken into account.

3.29 ultraviolet (UV) radiation

Optical radiation for which the wavelengths are shorter than those for visible radiation.

Note 1: The range between 100 nm and 400 nm is commonly subdivided, as defined by the International Commission on Illumination (CIE), into: UV-A, from 315 nm to 400 nm; UV-B, from 280 nm to 315 nm; and UV-C, from 100 nm to 280 nm (CIE 134-1999). These designations for UV radiation should not be taken as precise limits, particularly for photobiological effects.

Note 2: A precise border between "ultraviolet" and "visible" cannot be defined, because visual sensation at wavelengths shorter than 400 nm is noted for very bright sources.

Note 3: Frequently in photobiology, the wavelength bands are taken as: UV-A from 320 nm to 400 nm, UV-B from 290 nm to 320 nm, and UV-C from 200 nm to 290 nm.

Note 4: Ultraviolet radiation at wavelengths shorter than approximately 180 nm is considered vacuum ultraviolet radiation.

3.30 visible radiation

Any optical radiation capable of causing a visual sensation directly

Note: There are no precise limits for the spectral range of visible radiation since they depend on the amount of radiant power reaching the retina and the responsivity of the observer. The lower limit is generally taken between 360 nm and 400 nm, and the upper limit between 760 nm and 830 nm.

3.31 visual angle

The angle subtended by an object or detail at the point of observation. It usually is measured in radians, milliradians, degrees, or minutes of arc. For a visible or near-infrared source, it is the angular subtense of the source.

3.32 wavelength

The distance between two successive points of a periodic wave in the direction of propagation in which the oscillation has the same phase. The commonly used units are:

NAME	SYMBOL	VALUE
micrometer	μm	$1\ \mu m = 10^{-6}$ m
nanometer	nm	$1\ nm = 10^{-9}$ m

4.0 Hazard Evaluation

4.1 General

Persons in the vicinity of lamps and lamp systems should not be exposed to harmful levels of ultraviolet, visible, and infrared radiation. Actual exposure depends on factors beyond those of the lamp or luminaire (lamp system); e.g., the spatial location of exposure and the duration of exposure. Consequently, lamp or lamp system evaluation and subsequent risk labeling of lamps must involve assumptions about the use of lamps. If it is necessary to determine the allowed exposure duration for an optical radiation hazard, the exposure limits to determine the potential hazard in a given setting are provided in **Annex A**. The lamp Risk Group emission limits were derived from these exposure limits.

4.2 Basis of Evaluation

Lamp or lamp system emission limits are used to evaluate the various photobiological hazards. Emission limits incorporate both the concept of exposure duration and exposure distance and are derived from exposure limits. Exposure limits are taken from the "Threshold Limit Values" of the American Conference of Governmental Industrial Hygienists (see **Annex A**) and are applied under either standardized typical or potentially worst-case exposure conditions.

4.2.1 General Lighting Source (GLS) – Lamps and Luminaires. These are sources for which sound assumptions of typical use can be made. They are to be evaluated at either: 1) the assessment distance where the illuminance from the lamp is 500 lux; or 2) 20 cm from the outermost integral bulb surface of the lamp if the 500-lux distance is less than 20 cm.

Figure 4-1 shows the process of identifying GLS lamps.

Note 1: The 500-lux value does not represent illuminance values for spaces that GLS sources are intended to light (see **Section 3.11**); rather it is the reasonable, foreseeable worst-case, time-weighted-average lux value averaged over an 8-hour day. Increasing the duration past 8 hours does not increase the risk for any of the photobiological hazards considered. The UV irradiance at this 500-lux distance from a lamp correlates with the typically highest time-weighted-average values that can occur with conventional general lighting systems.

Note 2: The actual measurement distance is determined by the size and luminous intensity of the source. Typically, 20, 50, or 100 cm is used, as needed, for the measurement distance to obtain an acceptable signal-to-noise ratio but prevent the detector from heating up from the lamp radiation; lamp geometry may be considered in the case of directional lamps such as PAR lamps. The 500-lux distance is typically calculated from the ratio of the measured-to-500 lux values times the measurement distance.

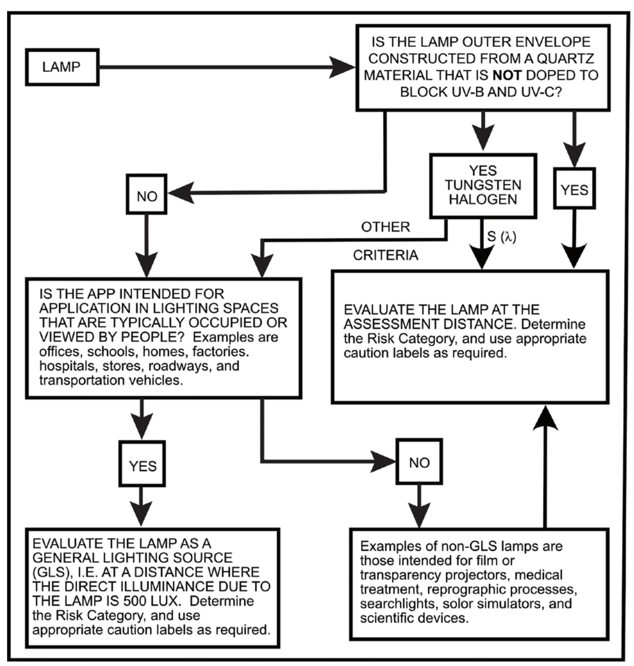

Figure 4-1. Process for identifying assessment distance of lamps in the general lighting source (GLS) lamp category.

4.2.2 Non-general Lighting Lamps. Lamps in this category are those for which assumptions about typical conditions of use are uncertain and those that are more likely to represent a significant potential hazard under a broad range of use conditions. These lamps are to be evaluated at a worst-case distance of 20 cm from the effective radiating surface. For known applications where the time-weighted-average exposure is not likely to occur at distances under 1 meter, evaluation at 1 meter is acceptable in the unusual case. Where the recommended use distance is less than 20 cm, then that distance shall be used as the assessment distance.

4.3 Continuous Wave Lamp Emission Limits

These equations relate to radiance and irradiance as applied directly to the continuous wave lamp (non-pulsed lamp) criteria. (See **Section 4.4** for pulsed lamp equations. Refer to **Annex A**, **Section A.1**, for relevant background information.) A limiting aperture no greater than 2.5 cm diameter shall be used with sources producing a uniform optical radiation pattern. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e., contain hot spots less than 2.5 cm (0.98 in.), a 7-mm (0.275-in.) aperture shall be used to assess the effect of hot spots. The measurement shall be made in that position of the beam giving the maximum reading.

4.3.1 S(λ)-Weighted Ultraviolet Irradiance, E_S. To evaluate the actinic ultraviolet hazard criterion by the spectrally weighted ultraviolet irradiance E_s, the following formula shall be used:

$$E_S = \sum_{200}^{400} E_\lambda \cdot S(\lambda) \cdot \Delta\lambda \,, \tag{4-1}$$

where:

E_s is the $S(\lambda)$-weighted ultraviolet irradiance in W/cm²

E_λ is the spectral irradiance in W/cm²·nm, measured with a field of view of 1.4 rad (80 degrees)

$S(\lambda)$ is the ultraviolet hazard weighting function (see **Table A-2** in **Annex A**)

$\Delta\lambda$ is the bandwidth in nm

The summation extends from 200 to 400 nm

4.3.2 Unweighted Ultraviolet Irradiance, E_{UV}. To evaluate the 315-nm to 400-nm ultraviolet hazard criterion by the unweighted ultraviolet irradiance E_{UV}, the following formula shall be used:

$$E_{UV} = \sum_{315}^{400} E_\lambda \cdot \Delta\lambda \,, \tag{4-2}$$

where:

E_{UV} is the unweighted ultraviolet irradiance in W/cm²

E_λ is the spectral irradiance in W/cm²·nm, measured with a field of view of 1.4 rad (80 degrees)

$\Delta\lambda$ is the bandwidth in nm

The summation extends from 315 to 400 nm

4.3.3 R(λ)-Weighted Radiance, L_R. To evaluate the retinal thermal hazard criterion by the spectrally weighted radiance L_R, the following formula shall be used:

$$L_R = \sum_{380}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda \,, \tag{4-3}$$

where:

L_R is the $R(\lambda)$-weighted retinal thermal radiance in W/cm²·sr

L_λ is the spectral irradiance in W/cm²·sr·nm, and shall be averaged over a right circular cone field of view of 0.011-radian including angle

$\Delta\lambda$ is the bandwidth in nm

$R(\lambda)$ is the retinal thermal hazard weighting function from **Table A-2** (see **Annex A**, **Section A.3**)

The summation extends from 380 to 1,400 nm

The angle α (see **Table 5-2** in **Section 5.3**) is needed in applying the L_R criterion, and α is the angular subtense of the source in radians defined by the 50% peak radiance points. For a non-circular projected source area, α is determined from the arithmetic mean of the shortest and the longest dimensions. For example, α for a 20 cm (7.8 in.) long by 3 cm (1.18 in.) diameter tubular source at a viewing distance of $r = 200$ cm (78.7 in.) in a direction normal to the lamp axis would be determined from the mean dimension, $l = (20+3)/2 = 11.5$ cm (4.5 in.).

$$\alpha = l/r = 11.5/200 = 0.058 \text{ radian}$$

If the calculated α exceeds 0.11 radian, $\alpha = 0.11$ should be used in conjunction with **Equation 4-3**.

If the calculated α is less than 0.011 radian, $\alpha = 0.011$ radian should be used in conjunction with **Equation 4-3**.

Note: In the case of multiple source elements that are not contiguous, this criterion applies to a single source element. It also applies to the source as a whole when the average radiance over the full source is used.

4.3.4 B(λ)-Weighted Radiance, L_B. To evaluate the retinal photochemical blue light hazard criterion by the spectrally weighted radiance L_B, the following formula shall be used:

$$L_B = \sum_{300}^{700} L_\lambda \cdot B(\lambda) \cdot \Delta\lambda \ , \qquad \textbf{(4-4)}$$

where:

L_B is the $B(\lambda)$-weighted radiance in W/cm²·sr

L_λ is the spectral radiance in W/cm²·sr·nm, and shall be averaged over a right circular cone field of view of 0.11-radian included angle for the Exempt Group (i.e., 10,000-second criterion); L_λ shall be averaged over a right circular cone field of view of 0.011-radian included angle for the RG-1 and RG-2 criteria

$\Delta\lambda$ is the bandwidth in nm

$B(\lambda)$ is the blue light hazard weighting function from **Table A-2** (see **Annex A**, **Section A.3**)

The summation extends from 300 to 700 nm

Note 1: The 0.11-radian field of view used herein is greater than the 0.011-radian field of view specified in **Section A.3.2**. In evaluating the potential of risk (see **Section 5.1**), it is assumed that an individual will not maintain a stable fixation on a very small light source for prolonged periods of time, and that a spatial averaging of the source image will occur over a reasonable retinal area.

Note 2: In the case of multiple source elements that are not contiguous, this criterion applies to a single-source element. Also, it applies to the source as a whole when the average radiance over the full source is used.

4.3.5 B(λ)-Weighted Irradiance – Small Source, E_B. To evaluate the retinal photochemical blue light hazard criterion for a source subtending an angle less than 0.011 radian (11 milliradians) by the spectrally weighted irradiance E_B, the following formula shall be used:

$$E_B = \sum_{300}^{700} E_\lambda \cdot B(\lambda) \cdot \Delta\lambda \ , \qquad \textbf{(4-5)}$$

where:

E_B is the $B(\lambda)$-weighted irradiance in W/cm²

E_λ is the spectral irradiance in W/cm²·nm

$\Delta\lambda$ is the bandwidth in nm

$B(\lambda)$ is the blue light hazard weighting function from **Table A-2** (see **Annex A**, **Section A.3**)

The summation extends from 300 nm to 700 nm

Note: This small source equation is equivalent to using **Section 4.3.4** (L_B) with the spatial averaging for small sources specified in Note 1 of that section.

4.3.6 Unweighted Infrared Irradiance, E_{IR}. To evaluate the ocular lens and corneal hazard criterion by the unweighted infrared irradiance, E_{IR}, the following formula shall be used:

$$E_{IR} = \sum_{770}^{3,000} E_\lambda \cdot \Delta\lambda \ , \qquad \textbf{(4-6)}$$

where:

E_{IR} is the unweighted infrared irradiance in W/cm²

E_λ is the spectral irradiance in W/cm²·nm

$\Delta\lambda$ is bandwidth in nm

The summation extends from 770 nm to 3,000 nm

4.3.7 R(λ)-Weighted Infrared Radiance, L_{IR}. To evaluate the retinal hazard criterion for near infrared sources where a strong visual stimulus is absent by the weighted infrared radiance, L_{IR}, the following formula shall be used:

$$L_{IR} = \sum_{770}^{1,400} R(\lambda) \cdot L_\lambda \cdot \Delta\lambda \ , \qquad \textbf{(4-7)}$$

where:

L_{IR} is the $R(\lambda)$-weighted infrared radiance in W/cm²·sr

$R(\lambda)$ is the retinal thermal hazard weighting function from **Table A-2** (see **Annex A**, **Section A.3**)

L_λ is the spectral radiance in W/cm²·sr·nm

$\Delta\lambda$ is bandwidth in nm

The summation extends from 770 nm to 1,400 nm

The angle α (in radians) (see **Table 6-2** in **Section 5.3**) as described in **Section 4.3.3** is needed (including issues of non-circular sources and averaging over a 0.011-radian field of view) when applying **Equation 4-7**.

A strong visual stimulus is defined herein as one whose maximum luminance (averaged over a circular field of view subtending 0.011 radian) is greater than 100 cd/m².

Note: L_λ shall be averaged over a right circular cone field of view of 0.011-radian included angle.

4.4 Pulsed Lamp Emission Limits

For lamps emitting pulsed optical radiation, the classification criteria shall apply to the most restrictive of the requirements for a single pulse, or to any group of pulses within the time criteria of the applicable hazard region given in **Table 5-1** in **Section 5.3**. Examples include stroboscopic flash lamps and pulse-modulated lamps.

The relevant weighted radiant exposure H or time-integrated weighted spatially averaged radiance (spatially averaged radiance dose) D for each pulse or for a combination of pulses (section of a pulse train) shall be obtained by integration of the weighted irradiance or radiance emitted from the source over the overall pulse or duration of a section of the pulse train, with the integration time of an individual pulse limited to a maximum of 0.25 s, and for a pulse train limited to a maximum of the time-base for the hazard assessment given in **Table 5-2** in **Section 5.3**. The weighted radiant exposure or weighted radiance dose shall be compared to the emission limits given in **Table 5-1** in **Section 5.3** for each of the photobiological hazards evaluated and for all integration durations spanning a single pulse up to the respective time base.

Note 1: Comparing a time-integrated radiant exposure with the limit of **Table 5-1** evaluated for the time base is equivalent to comparing the average irradiance with the irradiance limit specified in **Table 5-2**.

The nominal pulse duration Δt for pulsed lamp evaluation is determined by the time interval between the half-power points on the leading and trailing edges of the pulse. The energy integration time t is the full pulse width, with t limited to a maximum of 0.25 s.

Note 2: For pulse-modulated lamps, it is normally only necessary to assess the lamp or lamp system under CW conditions and further assessment is unnecessary if the peak power is not increased significantly when modulated.

4.4.1 S(λ)-Weighted Ultraviolet Radiant Exposure, H_S.

$$H_S = \int_t E_S \cdot \mathrm{d}t \,, \tag{4-8}$$

where:

H_S is the $S(\lambda)$-weighted ultraviolet radiant exposure in J/cm²

E_S is the weighted spectral irradiance in W·cm⁻² as defined in **Section 4.3.1**

t is time in seconds

A lamp is not in the Exempt Group if $H_S > 0.003$ J/cm².

4.4.2 Unweighted Ultraviolet Radiant Exposure, H_{UV}.

$$H_{UV} = \int_t E_{UV} \cdot \mathrm{d}t \,, \tag{4-9}$$

where:

H_{UV} is the unweighted ultraviolet radiant exposure in J/cm²

E_{UV} is the unweighted ultraviolet irradiance in W/cm² as defined in **Section 4.3.2**

t is time in seconds

A lamp is not in the Exempt Group if $H_{UV} > 1.0$ J/cm².

4.4.3 R(λ)-Weighted Radiance Dose, D_R.

$$D_R = L_R \cdot \mathrm{t} = \int_t L_R \cdot \mathrm{dt} \,, \tag{4-10}$$

where:

D_R is the $R(\lambda)$-weighted radiance dose in J/cm²·sr

L_R is the weighted spectral radiance in W/cm²·sr, as defined in **Section 4.3.3**

t is time in seconds

However, for pulsed-lamp evaluation when the 0.011-radian field of view is insufficiently small to measure hot spots, the spectral radiance L_λ shall be averaged over a right circular cone field of view of 0.00175-radian included angle.

A lamp is not in the Exempt Group if $L_R \cdot t > 5(\Delta t)^{0.75}/\alpha$.

Note: Angle α (in radians) is defined in **Section 4.3.3**, and Δt is defined in **Annex E**, **Section E.6**.

4.4.4 B(λ)-Weighted Radiance Dose, D_B.

$$D_B = L_B \cdot t = \int_t L_B \cdot \mathrm{d}t \,, \tag{4-11}$$

where:

D_B is the $B(\lambda)$-weighted radiance dose in J/cm²·sr

L_B is the weighted spectral radiance in W/cm²·sr as defined in **Section 4.3.4**, except that for pulsed-lamp evaluation the spectral radiance L_λ shall be averaged over a right circular cone field of view of 0.011-radian included angle

t is time in seconds

A lamp is not in the Exempt Group if $L_B \cdot t > 100$.

4.4.5 Unweighted Infrared Radiant Exposure, H_{IR}.

$$H_{IR} = \int_t E_{IR} \cdot \mathrm{d}t \,, \qquad \textbf{(4-12)}$$

where:

H_{IR} is the unweighted infrared radiant exposure in J/cm²

E_{IR} is unweighted infrared irradiance in W/cm² as defined in **Section 4.3.6**

t is time in seconds

A lamp is not in the Exempt Group if $H_{IR} > 1.8(\Delta t)^{0.25}$.

4.4.6 R(λ)-Weighted Infrared Radiance Dose, D_{IR}.

$$D_{IR} = L_{IR} \cdot t = \int_t L_{IR} \cdot \mathrm{d}t \,, \qquad \textbf{(4-13)}$$

where:

D_{IR} is the $R(\lambda)$-weighted infrared radiance dose in J/cm²·sr

L_{IR} is the $R(\lambda)$-weighted infrared radiance in W/cm²·sr as defined in **Section 4.3.7**

t is time in seconds

However, for pulsed-lamp evaluation when the 0.011-radian field of view is insufficiently small to measure hot spots, the spectral radiance L_λ shall be averaged over a right circular cone field of view of 0.00175-radian included angle.

A lamp is not in the exempt group if $L_{IR} \cdot t > 20(\Delta t)^{0.75}/\alpha$.

Note 1: Angle α is defined in **Section 4.3.3**, and Δt is defined in **Annex E**, **Section E.6**.

Note 2: This criterion, as does **Section 4.3.7**, applies to near-infrared sources where a strong visual stimulus is absent. A strong visual stimulus is defined herein as one whose maximum time-integrated luminance (averaged over a circular field of view subtending 0.00175 radian) is greater than 3 lm·s/m²·sr.

5.0 Risk Assessment and Lamp Safety Groups

5.1 General

This section is concerned with risk assessment of the emitted radiation from lamps and lamp systems, including luminaires. Emissions are measured at a convenient distance from the source to minimize uncertainty. The lamp's emissions can then be transferred through calculations to the assessment distance for Risk Group classification (see **Sections 4.2.1** and **4.2.2**) and the effective exposure distance for hazard evaluation. The received exposure varies with the effective exposure distance and the TWA exposure duration. The risk group time criteria below provide reasonably foreseeable worst-case scenarios for the time-weighted-average exposure for each hazard defined above. For example, in general illumination (GLS lamps), the worst-case risk assessment is done at a distance corresponding to 500 lux.

Lamps and lamp systems are categorized according to the degree of the hazard associated, in either an Exempt Group or one of three Risk Groups (see **Section 5.2** for details).

- Risk Group 1 includes Very Low-Risk Sources
- Risk Group 2 includes Low-Risk Sources
- Risk Group 3 includes High-Risk Sources

The Exempt Group consists of those lamps or lamp systems for which no risks have been identified under realistic use conditions. Furthermore, Risk Group 1 is normally not considered hazardous. Control measures and safety performance engineering requirements become more important and more stringent with increasing risk category.

5.2 Lamp Classifications

Since lamps and lamp systems may be hazardous for multiple reasons, a classification scheme is helpful. The class indicates only the potential risk. Depending upon use factors, time of exposure, and type of fixture, these potential hazards may or may not actually become real hazards. Other standards in this series are intended to deal with specific applications, e.g., copying machines, scientific instruments.

5.2.1 Exempt Group. The requirement for the Exempt Group classification is that the lamp does not pose any photobiological hazard for the end points in this Standard. This requirement is met by any lamp that does not pose an actinic (capable of producing a photochemical reaction) ultraviolet hazard (E_s) within 8 hours' exposure, nor a near-ultraviolet hazard (E_{UV}), nor an infrared cornea or lens hazard (E_{IR}) within 1,000 s, nor a retinal thermal hazard (L_R) within 10 s, nor a blue-light hazard (L_B) within 10,000 s (about 2.8 hours). Also, lamps that emit infrared radiation without a strong visual stimulus (i.e., less than 10 cd/m^2) and do not pose a near-infrared retinal hazard (L_{IR}) within 1,000 s are in the Exempt Group.

5.2.2 Risk Group 1 (RG-1, Very Low Risk). The requirement for this classification is that the lamp does not pose a hazard due to normal behavioral limitations on exposure. In other words, the performance of normal work tasks will limit the exposure duration so that the lamp does not pose a hazard. This requirement is met by any lamp that exceeds the limits for the Exempt Group but that does not pose an actinic ultraviolet hazard (E_s) within 10,000 s (about 2.8 hours), nor a near ultraviolet hazard (E_{UV}) within 300 s, nor a blue-light hazard (L_R), nor an infrared cornea or lens hazard (E_{IR}) within 100 s, nor a retinal thermal hazard (L_R) within 10 s. Also, lamps that emit infrared radiation without a strong visual stimulus (i.e., less than 10 cd/m^2) and do not pose a near-infrared retinal hazard (L_{IR}), within 100 s are in Risk Group 1 (Very Low Risk).

5.2.3 Risk Group 2 (RG-2, Low Risk). The requirement for Risk Group 2 (Low Risk) classification is that the lamp does not pose a hazard because the eye's aversion response to very bright light sources limits the exposure duration to a safe period, or due to thermal discomfort. This requirement is met by any lamp that exceeds the limits for Risk Group 1 (Low Risk), but that does not pose an actinic ultraviolet hazard (E_s) within 1,000 s of exposure, nor a near ultraviolet hazard (E_{UV}) within 100 s, nor an infrared cornea or lens hazard (E_{IR}) within 10 s, nor a retinal thermal hazard (L_R), nor a blue-light hazard (L_B) within 0.25 s (initiating an aversion response). Also, lamps that emit infrared radiation without a strong visual stimulus (i.e., less than 10 cd/m^2) and do not pose a near-infrared retinal hazard (L_{IR}) within 10 s are in Risk Group 2 (Low Risk).

5.2.4 Risk Group 3 (RG-3, High Risk). The requirement for this classification is that the lamp may pose a hazard even for momentary or brief exposure. Lamps that exceed the limits for Risk Group 2 (Low Risk) are in Risk Group 3 (High Risk).

5.3 Risk Group Evaluation for Continuous Wave Lamps

The following steps are to be used in assess the photobiological risk of continuous wave Lamps.

- A lamp shall be evaluated either as a general lighting source lamp (see **Section 4.2.1**) or as a non-general lighting source lamp (see **Section 4.2.2**).
- Emission Limits for determination of Risk Group classification are given in **Table 5-1**.

  Note 1: These emission limits are derived from the Risk Group time criteria of **Section 5.2**, as summarized in **Table 5-2**, together with the Exposure Limits defined in **Annex A**, **Section A.1**. Times greater than 0.25 s are time-weighted average exposures at the assessment distance.

  Note 2: Equations for lamp emission quantities and ancillary measurement requirements are given in **Section 4.3**.
- If the emissions of a lamp do not exceed any of the Emission Limits for the Exempt Group, then the lamp is in the Exempt Group.
- If a lamp is not in the Exempt Group and if its emissions do not exceed any of the Emission Limits for Risk Group 1, then the lamp is in Risk Group 1 (Very Low Risk).
- If a lamp is not in the Exempt Group or in Risk Group 1 and if its emissions do not exceed any of the Emission Limits for Risk Group 2, then the lamp is in Risk Group 2 (Low Risk).

Table 5-1. Emission Limits for Risk Groups of Continuous-Wave (Non-pulsed) Lamps

Risk	Metric	Formula in Section:	Notes	Exempt	RG-1 , Very Low Risk	RG-2, Low Risk	Units
Actinic UV, $S(\lambda)$	E_S	4.2.4.1		0.1	0.3	3.0	μW/cm^2
Near UV, 315 - 400 nm	E_{UV}	4.2.4.2		1.0	3.3	10.0	mW/cm^2
Retinal Thermal, $R(\lambda)$	L_R	4.2.4.3	Arc sources only	4.5/α	N.A.*	N.A.*	W/cm^2·sr
Blue Light, $B(\lambda)$	L_B	4.2.4.4		0.01	1	400	W/cm^2·sr
Blue Light, $B(\lambda)$	E_B	4.2.4.5	Small-source alternative	0.1	0.1	40	mW/cm^2
Cornea or Lens, IR	E_{IR}	4.2.4.6		10	57	320	mW/cm^2
Retinal IR	L_{IR}	4.2.4.7	Non-GLS only	0.6/α	3.2/($\alpha^{0.25}$)	N.A.	W/cm^2·sr

* The retinal thermal hazard emission limit does not change with time for durations longer than 0.25 s. A consequence of this is that if the emission limit is exceeded for up to 0.25 s, the source classified on the basis of the emission limit for L_R, is allocated to Risk Group 3.

Additional table notes:

1. The reader should refer to specific sections for averaging requirements on radiance and irradiance measurements and for definition of angle α.
2. L_{IR} applies only to sources where $L \leq 10$ cd/m^2.
3. General lighting source (GLS) lamps are evaluated at a distance where direct light from the lamp produces 500 lux but not at a distance less than 20 cm. (See **Section 4.3.1**.)
4. Non-general lighting source lamps are evaluated at a distance specified by the manufacturer if no vertical, application-specific standard exists unless, as in some medical applications, the TWA intended use distance is less.
5. *A lamp exceeding any Exposure Limit of RG-2 is in RG-3.*

Table 5-2. Risk Group Time Criteria for Continuous Wave (Non-pulsed) Lamps

Risk	Metric	Formula in Section:	Exempt	RG-1, Very Low Risk	RG-2, Low Risk
Actinic UV, $S(\lambda)$	E_S	4.2.4.1	8 hours	10,000 s	1,000 s
Near-UV, 315 – 400 nm	E_{UV}	4.2.4.2	1,000 s	300 s	100 s
Retinal Thermal, $R(\lambda)$	L_R	4.2.4.3	0.25 s	N.A.	N.A.
Blue Light, $B(\lambda)$	L_B	4.2.4.4	10,000 s	100 s	0.25 s
Blue Light, $B(\lambda)$	E_B	4.2.4.5	10,000 s	100 s	0.25 s
Cornea or Lens, IR	E_{IR}	4.2.4.6	1,000 s	100 s	10 s
Low Luminance, Retinal IR	L_{IR}	4.2.4.7	810 s	10 s	N.A.

NOTE: For the Exempt and RG 1 Groups, if the emission is kept at or below the limits shown in **Table 5-1**, the time given applies to all day; e.g., the weighted emission of less than 0.1 mW/cm^2 for blue light small source will not exceed the applicable exposure limit even for exposures lasting much longer than 100 seconds.

- If a lamp is not in the Exempt Group, nor in Risk Group 1, nor in Risk Group 2, then the lamp is in Risk Group 3 (High Risk).
- Continuous wave lamps shall be evaluated at the highest nominal power loading. This shall be established by the manufacturer's specified operating conditions.

5.4 Risk Group Evaluation of Pulsed Lamps

Pulsed-lamp criteria shall apply both to a single pulse and to any group of pulses within 0.25 s.

Note: Pulsed-lamp criteria do not apply to general lighting source (GLS) lamps.

- A lamp that passes the pulsed-lamp emission limits of **Section 4.4** shall be in the Exempt Group except as qualified in the next paragraph. A lamp that does not pass the pulsed lamp emission limits of **Section 4.4** shall be in Risk Group 3.
- Pulsed lamps that are repetitively pulsed and that fall in the Exempt Group per the criteria of **Section 4.4** shall further be evaluated by the continuous wave lamp criteria for Risk Group assessment. The criteria for continuous wave lamps shall be applied to the time averaged values of repeatedly pulsed lamps.
- A pulsed lamp shall be evaluated at its highest nominal energy loading. This shall be established by the manufacturer's specified operating conditions.

6.0 Measurement Techniques

6.1 Measurement Conditions

6.1.1 Lamp Seasoning. To maintain stable output during the measurement process and provide reproducible results, lamps shall be seasoned for an appropriate period of time. During the initial period of operation, a lamp will change as its components come to equilibrium. If measurements are taken of an unseasoned lamp, the variations within the measurement period and between measurements will be significant. As the output of a lamp generally decreases over life, the seasoning time should be short in order to result in conservative hazard evaluations.

Seasoning of lamps shall be done as stated in *ANSI/IES LM-54-20, Approved Method: IES Guide to Lamp Seasoning*. For the purposes of this Standard, the lamp output at the end of the seasoning period is the initial output. For lamps not covered by ANSI/IES LM-54-20, a study may be required to find the minimum time required to stabilize the operation of a source.

6.1.2 Test Environment. Measurements shall be made in a controlled environment. The operation of sources and measurement equipment is impacted by environmental factors. In addition, the formation of ozone in the measurement path may compromise accuracy and may present a safety hazard.

6.1.3 Temperature. The ambient temperature will significantly influence the output of certain light sources, such as fluorescent lamps. The ambient temperature in which measurements are taken shall be maintained in accordance with the appropriate IES LM documents noted in **Section 2.1**.

6.1.4 Air Movement. The characteristics of some light sources are significantly affected by air movement. For the applicable light source, the reader is referred to the appropriate IES LM standards listed in **Section 2.1**. Other than normal convection currents, air movement over the surface of test lamps should be reduced as much as possible consistent with safety considerations (e.g., ozone production). When the system under test provides interlocks that maintain air circulation, measurements shall be performed with circulation.

6.1.5 Extraneous Radiation. Careful checks should be made to ensure that extraneous sources of radiation and reflections do not add significantly to the measurement results. Visually black surfaces can be highly reflective of UV and IR radiation.

Radiation from hot baffles shall be considered in infrared measurements due to the large input angle subtended by baffles. Water cooled baffles and double baffles are two methods for addressing baffle heating.

6.1.6 Lamp and Lamp System Operation. The lamp or lamp system shall be operated under conditions that are standardized.

6.1.6.1 Lamp Operation. The input power to the test lamp shall be provided in accordance with the appropriate LM standard noted in **Section 2.1**. If no standard for the lamp type exists, the lamp manufacturer's recommendation for operation should be used.

6.1.6.2 Lamp System Operation. The power source for a complete lamp system shall be at the manufacturer's principal specified operating voltage for the device. If the specified condition is a voltage range, the highest nominal value shall be used. The supply voltage should not vary more than plus or minus one-half percent (±0.5%) during the test. For AC operation, the supply frequency should not vary more than plus or minus one-half percent, (±0.5%), and the root mean square (RMS) value summation of the harmonics components should not exceed 3% of the total.

6.1.6.3 Safety. The systems tested are unknown in their optical hazard and shall be treated as hazardous unless proven otherwise. The possible formation of ozone presents a health hazard and shall be addressed for all sources, with the exception of those sources with an envelope known to block ozone-forming UV.

The operation of some light sources in a laboratory environment may require compromises in operation. All compromises shall be detailed. No compromises shall be made that impact electrical safety.

The sources under test may pose additional hazards (e.g., burn, explosion) which shall be addressed with appropriate measures.

6.2 Measurement Instrumentation

6.2.1 Recommended Measurement Instrumentation – Double Monochromator. The measurement of a source for the purpose of hazard classification requires accuracy during calibration and testing. The detector's broad spectral response and high spectral resolution required to provide accurate weighting leads to stringent requirements for out-of-band stray light rejection. Calibration sources provide wide spectral output, which needs to be rejected out of the pass-band. The ratio of out-of-band energy to pass-band energy at 270 nm for tungsten-halogen incandescent calibration lamps (e.g., the ANSI standard double-ended 1,000-watt quartz halogen lamp, FEL) is greater than 10^4:1. The double monochromator is the only instrument that provides the needed selectivity, and it is recommended for hazard measurements involving UV and visible radiation. It is recognized that monochromator systems introduce limitations in convenience and speed.

6.2.1.1 Instrument Spectral Response. The shape of the spectral response and the ratio of the measurement interval to the bandwidth will determine whether the system is able to accurately measure signals with narrow spectral extent, for example, atomic emission lines. (See **Section 2.2 Informative References**, Kostkowski 1997, Chapter 5.)

A monochromator with perfect triangular spectral response used in a system that has a reporting interval that divides into the bandwidth integrally will accurately measure all signals regardless of their spectral shape. (See CIE 063, Section 1.8.4.2.1; or **Section 2.2 Informative References**, Kostkowski 1997, Section 5.9.) Deviations from this may lead to errors in measured energy. The spectral response of the system is determined by a spectral scan of a narrow isolated spectral line—e.g., filtered laser or atomic emission—using scan steps much smaller than the instrument's bandwidth. The resulting spectrum is a mirror image of the system's spectral response. The spectral response is what would be found by holding the instrument at a single wavelength and noting the response to a monochromatic source whose wavelength is varied around the wavelength. (See **Section 2.2 Informative References**, Kostkowski 1997, Section 4.9.) The system's ability to accurately measure the energy in a narrow band signal is the sum of the spectral responses at each reported wavelength. The variation across the summed spectrum is the potential error in total measured signal and shall be included in the uncertainty analysis.

The result of hazard evaluations will be influenced by the instrument's characteristics. The bandwidth of the monochromator will change the weighted results of any spectrum with varying levels. All finite-bandwidth instruments report signal at the wrong wavelength, leading to errors in a weighted sum. (See **Table 6-1** for recommended bandwidths for 2% upper bound of uncertainty in weighted sums.)

Table 6-1 lists the recommended bandwidth for 2% upper bound of uncertainty in weighted sums.

Table 6-1. Recommended Bandwidths

Range (nm)	Bandwidth (FWHM)*
200 ≤ λ < 400	≤ 4 nm
400 ≤ λ < 600	≤ 8 nm
600 ≤ λ < 1,050	≤ 20 nm
1,050 ≤ λ	No bandwidth limitation

** full-width, half-maximum*

A more complete analysis that takes the source spectrum into account may be used to relax the suggested bandwidth accuracy. The results of analysis shall be included in the stated uncertainty of the measurement.

Note: Systems that constantly integrate the signal during the spectral scan will not experience errors in total measured power from spectral response shape or from the ratio of bandwidth to reporting interval. (See CIE 063, Section 1.8.4.2.2.) Large bandwidths will still lead to errors in weighted results with this type of instrument.

6.2.1.2 Wavelength Accuracy. The wavelength accuracy of the instrument used to determine the spectral output of a source has a great impact on the weighted values. The photobiological weighting functions change at an extreme rate, e.g., 250% over 3 nm at 300 nm for $S(\lambda)$, the UV Hazard Weighting Function. If a reasonable limit of error is desired, then the measured energy must be assigned to its proper wavelength so that it is appropriately weighted.

Table 6-2 is an example showing the change in weighted results from a line when the line is moved by 0.1 nm. The measured values are calculated by assuming a spectroradiometer with a triangular response, 2-nm bandwidth, and 1-nm reporting interval. The sums of the measured values are equal as the line moves because of the principles described in **Section 6.2.4**. The weighted measurement changes by 2.6% for a wavelength change of 0.1 nm. The wavelength deviation in this example is 0.033%, causing an error in the weighted result of 2.6%. In the propagation of uncertainty, the percent wavelength uncertainty shall be multiplied by a sensitivity factor of 78.

The wavelength accuracy of the monochromator used for hazard testing should be sufficient to provide weighted results with an error of less than 2% arising from wavelength inaccuracy. The needed accuracy therefore depends on the region of the spectrum and the weighting function used. **Table 6-3** summarizes the suggested accuracy that will bound the error to approximately 2%.

Only unweighted power is used above 1,400 nm. Any bandwidth is permissible providing that it does not introduce an unacceptable error in the integrated power.

Table 6-3. Recommended Wavelength Accuracy

Range (nm)	Wavelength Accuracy
200 ≤ λ ≤ 300	0.2 nm
301 ≤ λ ≤ 325	0.1 nm
326 ≤ λ ≤ 600	0.2 nm
601 ≤ λ ≤ 1,050*	2.0 nm

** There is no spectrally varying weighting of power above 1,050 nm.*

Table 6-2. Example of Error in Weighted Value for Wavelength Error

nm	S(λ)*	305 nm		305.1 nm		Ratio of Sums
		Measured	Weighted	Measured	Weighted	
304	0.08485	0.25000	0.02121	0.22500	0.01909	
305	0.06000	0.50000	0.03000	0.47500	0.02850	
306	0.04540	0.25000	0.01135	0.27500	0.01249	
307	0.03436	0.00000	0.00000	0.02500	0.00086	
Sums		**1.00000**		**1.00000**		**100%**
Sums			**0.06256**		**0.06094**	**97.4%**

** S(λ) is the UV Hazard Weighting Function.*

A more complete analysis that takes the source spectrum into account may be used to relax the suggested wavelength accuracy. (See **Annex E**, **Section E.4**.) The results of analysis shall be included in the stated uncertainty of the measurement.

6.2.1.3 Stray Radiant Power. The absolute calibration of spectroradiometers requires the use of sources with broad spectral output and high energy. If the spectral rejection is insufficient, additional energies from other parts of the spectrum will be included in the calibration. The result of this type of error is to under-calibrate the spectoradiometer and leads to lower readings of the potential hazard.

A single monochrometer typically rejects 99.99% of the out-of-band radiation; i.e., 0.01 % is passed through as in-band radiation. However, in the UV region the actual radiation that is important for hazard evaluation is often on the order of 0.01% of the light in the visible and/or infrared regions. Thus, a rejection of out-of-band radiation needs to be 100 times better than that of most single monochrometers in order to obtain an accuracy of 1% in calibration for the UV region. Hence, there is a need for double monochrometers.

Diode- or CCD-array spectrometers have, up to now, been unsuitable for use in the UV due to stray radiant power. Recent improvements in modeling that allow mathematical removal of the stray radiant power, coupled with the use of blocking filters, have made such instruments more acceptable for use, particularly in the UV-A region.

Methods to check for out-of-band stray light, and remedies, are detailed in **Annex C – Stray Radiant Power**.

6.2.1.4 Input Optics Recommendation: Integrating Sphere. A properly designed and suitably coated integrating sphere will address the three issues detailed below. The integrating sphere will be the most convenient method for irradiance measurements.

The random reflectance of the coating depolarizes the incoming light, and proper design can closely match a cosine response. The multiple reflections within the integrating sphere consistently fill the radiometer's input.

6.2.1.4.1 Polarization. The calibration source and the device tested may not have the same polarization. The internal optics of a monochromator will generally be selective to polarization. Selectively rejecting some of the calibration or test signal can lead to an erroneous determination. Input optics for the double monochromator shall be selected to depolarize the input light to maintain a valid calibration.

The random reflections in an integrating sphere will depolarize the incoming light.

6.2.1.4.2 Angular Response. The calibration of the system for irradiance is typically accomplished at a moderate distance with a small source, leading to a small input angle. Measurements may then be performed on sources that subtend a greater angle. Irradiance is the measure of flux on a surface element. For both calibration and test source to be comparable, the angular response of the input optics should conform to a cosine function, the response varying as the cosine of the angle to the normal.

Proper design of an integrating sphere can closely match a cosine response.

6.2.1.4.3 Monochromator Input. The throughput of a monochromator will vary with position and input angle. The response of the detector will be affected by the position of the flux on the detector. For a calibration to be valid the calibration and the test signal must have the same angular extent.

The multiple reflections within the integrating sphere will negate the issues of position and input angle by consistently filling the spectroradiometer's input.

6.2.1.5 Calibration Sources. The suggested sources for calibration are the deuterium discharge lamp and the calibration grade tungsten-halogen lamp (e.g., the FEL) with NIST type base. The deuterium lamp may vary in output level while maintaining its spectral shape. Therefore, the calibration of the system in the region of 200 nm to 250 nm using the deuterium lamp shall be adjusted by comparison to the calibration level from the FEL between 250 nm and 350 nm. The wavelength below which the deuterium shape is used shall be at as short a wavelength as practical considering the noise in the FEL calibration.

6.2.2 Broadband Detectors. When faced with short-duration pulsed or low intensity sources, it is convenient to use broadband detectors. Broadband hazard sensors attempt to match the weighting spectra by the use of filters. The matching is never exact and leads to some amount of error. If the source spectrum is unknown, then the point of largest percentage deviation between the detector and the action spectrum must be assumed as the uncertainty. When both the detector's response and source's spectrum are known, straightforward calculations can generate a correction factor. Used with an appropriate correction factor, the broadband detector provides a valid method for measurements under this Standard. It is incumbent upon the radiometrist to show that the correction factor is valid in each specific case. Variations that lead, or may lead, to changes in spectra require redetermination of the correction factor.

6.2.2.1 An Example of Combined Instrument Testing. The combination of weighted broadband radiometers and a double monochromator can improve the measurement process in many instances. A survey of spatial, temporal, or item to item variations of a source type can be rapidly accomplished using the filtered detector. Representative samples—mean, high, and low—can then be measured using the double monochromator system. The results are analyzed to determine the correction factor and the factor's variation. Further tests can then be accomplished using the filtered detector with the improved knowledge of the correction factor.

6.2.3 Instrument Limitations. All instruments have limits below and above which measurements are not possible. The instrument used for hazard testing shall be characterized to determine the measurement limit level or levels. When the instrument reading is zero, the corresponding measurement limit will be reported. On the other hand, if these instrument limitations can affect the classification of sources, further evaluation is necessary. Such evaluation is likely needed in the UV-C region and the mid-IR region, e.g., near the ends of the wavelength limits for hazard evaluation.

The measurement limit of a spectroradiometer is usually dominated by the noise limit of the detector. The noise limit of the detector can be taken as the root-mean-square (RMS) variation in the detector dark signal. The detector's noise limit is multiplied by the calibration spectrum for the system to give a spectrum with the same dimensions as the measurements, referred to as the noise equivalent input (NEI). Normally the reported spectral measurement is the higher value at each wavelength of the measured spectrum and the NEI.

6.2.3.1 An Example of Detector Limitations for a Spectroradiometer System. This example will consider the calibration of a photomultiplier tube in the UV-C region. Due to decreasing signal strength of the deuterium calibration lamp and decreasing sensitivity of the detector as the wavelength decreases between 250 and 200 nm, the calibration factor increases rapidly with decreasing wavelength. Thus, the NEI is usually found to be about one order of magnitude larger near at 200 nm than it is at 250 nm. If this noise power is recorded as it is, it may, depending on the source, add significantly to the total power of the UV reported, and in some cases change the Risk Group categorization. One method to reduce the NEI when it is believed that no actual signal exists is to make a measurement of the source with the power on and then with the power off. If this shows no change in the NEI over a given wavelength region, say 200 to 270 nm, it is permissible to report the NEI power at 10% of the detector response.

6.2.4 Linearity. The individual lamp or device under test must be assumed to have a different radiometric magnitude than the source used to calibrate the testing system. For the calibration to be useful, the linearity of the system must be known, and the test measurements shall be performed within the range that is linear. Non-linearity within the system may be corrected using a calibration function to bring the system to linearity. It is important to note that this adjustment shall be applied to both the calibration and the measurements.

6.2.5 Source Size Measurement. The determination of α, the angle subtended by a source, requires determining 50% of the maximum radiance locations on the projected image of the source in the direction of view. Common methods using photography or solid-state cameras should be used only after verifying that the spectral uniformity is sufficient to warrant the use of visible radiation as an analog for the IR or UV radiation. Changes in spectra across a source can lead to different sizes in different regions of the spectra. (See **Section 2.2 Informative References**, Sliney and Wolbarsht, Section 12.6.6.)

6.2.6 Pulse Width Measurement. The determination of Δt, the nominal pulse duration of a source, requires the determination of time between the 50% of maximum values on the rise and decay of the pulse. Common methods using a photocell and an oscilloscope should be used only after verifying that the spectral uniformity is sufficient to warrant the use of visible radiation as an analog for the IR or UV radiation. Changes in the spectrum during a pulse may lead to different pulse widths in different regions of the spectrum and complicate weighted pulse width measurements. (See **Annex E**, **Section E.4**.)

6.3 Measurement Procedure

To clarify the intent of the measurements in **Section 4.3**, ideal instruments will be described that make the measurements in a direct fashion. In few cases will it be practical to construct physical instruments that sufficiently approximate the ideal instruments. Frequently, it is more practical to use indirect measurements and computational procedures to determine the desired quantities. The ideal instruments shown would directly measure the needed quantities and therefore provide examples that may be used as starting points for measurement design.

6.3.1 Irradiance Measurement.

6.3.1.1 Ideal Instrument. A plane, circular area detector of diameter D, where values for D are specified. (Refer to **Figure 6-1**.)

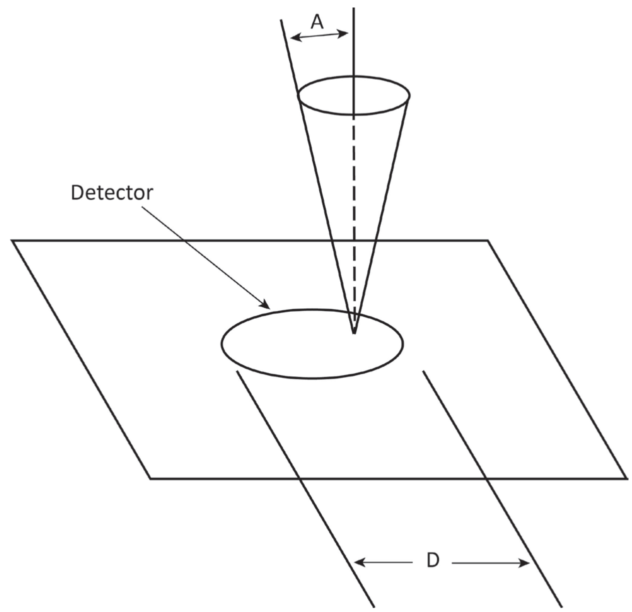

Figure 6-1. Fixed-bandwidth radiometer.

6.3.1.1.2 Circular Area Detector Characteristics.

- Accepts radiation within a right circular cone whose centerline is normal to the area and of specified half angle A. If A is not specified, then it is 90° for spatially total incident power.
- Has an angular spatial response varying as the cosine from a normal to the area.
- Has the same magnitude response.
- Has a spectral response that is constant with wavelength within a specified band from λ_1 to λ_2 and is zero elsewhere.

6.3.1.1.3 Calibration. Calibrated to read in absolute incident radiant power per unit receiving area.

6.3.1.2 Conceptual Realization.

- A fixed-bandwidth radiometer with a cosine corrected detector.
- Providing that a radiant source does not extend beyond the given angle A for each point on the receiving area, the measurement can be made with the cosine corrected receptor.
- If a radiant source extends beyond a specified angle A, then the cosine receptor field of view can be limited by an aperture at some distance from the receptor that is large with respect to the receptor diameter. (See **Figure 6-2**.)

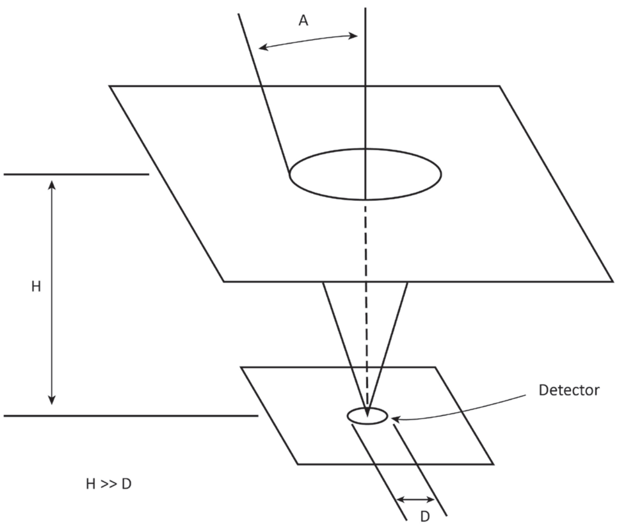

Figure 6-2. Limited cosine receptor field of view.

6.3.1.3 Application. Irradiance measurements apply to the following criteria identified by reference to **Section 4.3 Continuous Wave Lamp Emission Limits**:

- 315 nm to 400 nm Eye Exposure Limit (**Section 4.3.2**)
- Infrared Radiation Hazard Exposure Limit (**Section 4.3.6**)

6.3.2 Spectral Irradiance Measurement.

6.3.2.1 Geometry. Spectral irradiance instruments have the same geometric characteristics as irradiance instruments described in **Section 6.3.1**.

6.3.2.2 Detector. The radiometric detector of **Section 6.3.1** is replaced by a spectroradiometric detector having the characteristics described in **Section 6.2**.

6.3.2.3 Application. Spectral irradiance measurements can be used for irradiance measurements in **Section 4.3** by integrating over the appropriate spectral band. Spectral irradiance measurements apply to the following criteria identified by reference to sections in this document:

- 200 nm to 400 nm Skin and Eye Exposure Limit (**Section 4.3.1**)
- Retinal Blue Light Hazard Exposure Limit: Small Source (**Section 4.3.5**)

6.3.3 Radiance Measurement.

6.3.3.1 Ideal Instrument. The following three sections refer to **Figure 6-3**.

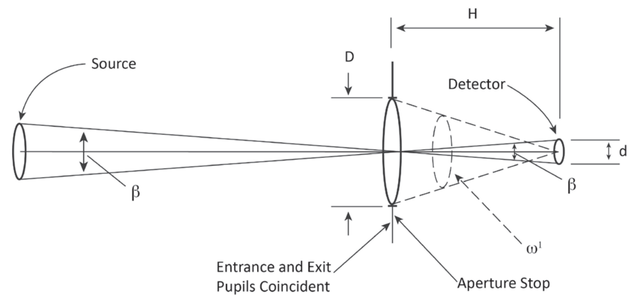

Figure 6-3. An optical system that images the projected area of a radiant source onto a detector.

6.3.3.1.1 Optical System Characteristics.

- Images the projected area of a radiant source onto a detector.
- Has a circular area detector that acts as a field stop to establish the angular extent of the field of view (β) at 0.011 radian or other angular extents e.g. 0.00175 radians for pulsed lamps (see **Section 4.4.6**).
- Has a circular entrance pupil equal to the assumed ocular pupil diameter.

  Note: An Ideal radiance instrument with a field of view $\beta = 0.011$ radians (see **Figure 7-3**) will properly measure a Retinal Blue Light Hazard (see **Section 4.3.4**) whether the source is small (subtending less than 0.011 radians) or large (subtending more than 0.011 radians). The alternative method using irradiance, Retinal Blue Light Hazard: Small Source (**Section 4.3.5**), is equivalent to the radiance method when the source is small. For LED sources consisting of multiple light-emitting elements, only the radiance method should be used.

6.3.3.1.2 Circular Area Detector Characteristics.

- Accepts radiation from within the solid angle ω' defined by the exit pupil.
- Has an angular spatial response varying as the cosine from a normal to the plane of the detector area.
- Has the same magnitude response.
- Has a spectral response that is constant with wavelength within a specified band from λ_1 to λ_2 and is zero elsewhere.

6.3.3.1.3 Calibration. Calibrated to read in absolute radiant power per unit area and per unit solid angle emitted at the source and averaged over the field of view of the instrument.

6.3.3.2 Conceptual Realization. The following three sections refer to **Figure 6-3**.

6.3.3.2.1 Lens Characteristics.

- Images a radiant source onto a detector.
- Has a transmittance t between λ_1 to λ_2.
- Has an appropriate diameter (here, the entrance pupil). It should be noted that in most cases, the change of radiance with direction at a point on the source is sufficiently small over small angles that the entrance pupil can be greater than the assumed ocular pupil diameter.

6.3.3.2.2 Detector Characteristics.

- Diameter $d = 0.011H$
- Spectrally flat response between λ_1 and λ_2

6.3.3.2.3 Calibration. Calibration is established by the relation between the source radiance L and the detector irradiance E:

$$E = \tau L\omega' = L[\tau(\pi D^2/4)/H^2]$$

Notes:

1) ω' is solid angle in steradians.

2) Equation is small angle approximation based on $D \ll H$.

6.3.3.3 Application. Radiance measurements apply to the criterion Infrared Radiation Hazard Exposure Limit: Weak Visual Stimulus (**Section 4.3.7**).

6.3.4 Spectral Radiance Measurement.

6.3.4.1 Geometry. Spectral radiance instruments have the same geometric considerations as radiance instruments described in **Section 6.3.3**.

6.3.4.2 Detector. The radiometric detector of **Section 6.3.3** is replaced by a spectroradiometric detector having characteristics as described in **Section 5.2**.

6.3.4.3 Application. Spectral radiance measurements can be used for radiance measurements in **Section 6.3.3** by integrating over the appropriate spectral band. Spectral radiance measurements apply to the following criteria identified by reference to sections of this document.

- Retinal Thermal Hazard Exposure Limit (**Section A.3.1**)
- Retinal Blue Light Hazard Exposure Limit (**Section A.3.2**)

6.3.5 Subtended Angle of the Source. At the point of hazard evaluation, angle α subtended by a radiant source is required for Retinal Thermal Hazard evaluation (see **Section 4.3.3**). Angle α is defined as the full-width half-maximum (FWHM) angle. It is the angle between two lines in a plane with: 1) the vertex at the point of measurement; and 2) the lines intersecting with points on each side of the source at 50% of the maximum radiance.

Note: Angle α is not the instrument field of view (angle β in **Figure 6-3**, **Section 6.3.3.2**). Angle β determines the projected area on a source over which the measured radiance is averaged and is independent of source size or distance.

6.3.5.1 Circular Uniform Radiance Source. For a source with a sharply defined edge, a dimension d normal to the direction of the measurement point, and a distance r from the measurement point, angle α is given by $\alpha = d/r$ radians.

6.3.5.2 Non-circular Uniform Radiance Source. Angle α for the Retinal Thermal Hazard evaluation is the average of the two physical angles calculated using the longest and the shortest dimensions d of the source.

6.3.5.3 Non-uniform Radiance Source. The dimension d in **Sections 6.3.5.1** and **6.3.5.2** is measured to FWHM radiance points on the source. This determination can be made by scanning a projected image of the source or by various photographic or imaging techniques.

6.4 Analysis of Results

6.4.1 Weighting Curve Interpolations. The weighting curves defined in **Section 4.3** are not sufficiently resolved to perform the hazard calculations. The functions are reasonably linear in any local region on semi-log coordinates. Therefore, to standardize interpolated values, linear interpolation on the log of given values should be used to obtain intermediate points at one nanometer intervals. Anti-logarithm of the interpolated values results in the weighting factors that shall be used.

Table 6-4 shows the method that should be used to find intermediate values for the photobiological hazard weighting functions. (*Note:* **Annex F** provides a table with the interpolations already performed.)

6.4.2 Calculations. The calculation of hazard values shall be performed by weighting the spectral scan by the appropriate function and calculating the total weighted energy. To provide a repeatable method, this Standard suggests interpolation or summing to one nanometer (1 nm) for the spectra below 400 nm. Weighting and summations are then performed at this (1 nm) resolution. From 400 nm to 780 nm, a 2-nm spacing should be used. Above 780 nm, a spacing of 5 nm is recommended.

6.4.3 Measurement Uncertainty. The quality of all measurement results shall be quantified by an analysis of the uncertainty. All calculated results shall be paired with uncertainty values that conform to the guidance in the normative references. The uncertainty of each result will be reported as combined standard uncertainty, u_c, as defined in the ISO guide listed in **Section 2.1**. The values of uncertainty shall be propagated from the calibration uncertainties through the calculations and shall include all sources as described in **Annex D – Uncertainty Analysis**.

7.0 Specific Labeling Requirements

7.1 Labeling Equivalency

Caution labels are intended to inform and instruct both original equipment manufacturers (OEMs) and end users regarding pertinent safety related information that should be communicated for lamp systems that are classified by the manufacturer under Risk Groups 1, 2 or 3.

A caution label shall provide these elements of information in an abbreviated or simplified manner, and variations in specific language shall still convey the same basic information elements to be considered "equivalent." The elements are:

- A statement of “caution” or “warning,” sometimes called the *signal word*.
- A statement of the potential hazard. This should simply advise as to what might be expected to occur (i.e., skin irritation or eye injury).
- A list of what precautions could or should be taken to avoid the risk. Examples include information on shielding or a warning against staring at a particular source.
- An abbreviated statement that specifies the Risk Group Classification for Risk Groups 1, 2 or 3 (i.e., ANSI RG1, ANSI RG2, ANSI RG3, or RG-1, RG-2, RG-3).
- Any federal-, state-, or province-mandated labeling requirements take precedence over requirements included in this recommended practice consensus document.

7.2 Evaluation Categories

These labels are only examples. If by the very nature of the lamp system (e.g., limited emission duration) an

Table 6-4. Examples of Weighting Function Interpolation

#1	#2	#3	#4	#5
nm	*S*(λ)	Logarithm: ln(#2)	Linear interpolation of #3	Anti-logarithm: $e^{\#4}$
300	0.300	-1.204		
301			-1.509	0.221
302			-1.815	0.163
303	0.120	-2.120		

individual cannot be exposed above the Exposure Limit as defined in **Section 6.1,** the labelling is not required.

7.3 Evaluation Categories

These labels are only examples. If by the very nature of the lamp system (e.g., limited emission duration) an individual cannot be exposed above the Exposure Limit as defined in **Annex A**, **Section A.1**, the labelling is not required.

7.4 Product Identification and Labeling

Each RG-1, RG-2, or RG-3 lamp system with unique or distinctive spectral emissions that would influence its photochemical effectiveness at wavelengths less than 400 nm shall be identified with a corresponding unique coding, designation, or other identification in lamp product information, catalogs, lamp packaging, and in other information provided to the user or purchaser.

7.5 Labeling Examples

The following labels (or their equivalent) shall appear on packaging, as applicable:

7.5.1 Ultraviolet Hazard, E_S and E_{UV}.

Exempt Group: Not required.

RG-1: *CAUTION. UV emitted from this lamp. Minimize exposure. Shielding lamps via glass or plastic normally provides adequate protection.*

RG-1 alternative: *CAUTION. UV emitted from this lamp. Skin or eye irritation is possible. Minimize exposure.*

RG-2: *CAUTION. UV emitted from this lamp. Possible skin or eye irritation can result from exposures exceeding 15 minutes in a day. Use appropriate shielding.*

RG-3: *WARNING. UV emitted from this lamp. Skin or eye injury could result. Avoid exposure of eyes and skin to unshielded lamp.*

7.5.2 Retinal Thermal and Blue Light Hazards, L_R and L_B, or E_B.

Exempt Group: Not required.

RG-1: Not required.

RG-2: *CAUTION. Do not stare at exposed lamp in operation. May be harmful to the eyes.*

RG-3: *WARNING. Do not look at exposed lamp in operation. Eye injury can result.*

7.5.3 Cornea or Lens Infrared Hazard, E_{IR}.

Exempt Group: Not required.

RG-1: *CAUTION. IR emitted from this lamp. Appropriate eye protection should be used when daily exposure at short distances is greater than 15 minutes.*

RG-2: *CAUTION. IR emitted from this lamp. Appropriate eye protection should be used when working in the vicinity of this lamp.*

RG-3: *WARNING. IR emitted from this lamp. Avoid eye exposure without appropriate protection.*

7.5.4 Retinal Thermal, Low Visible Light, L_{IR}.

Exempt Group: Not required

RG-1: *CAUTION. IR emitted from this lamp. Do not stare at exposed lamp.*

RG-2: *CAUTION. IR emitted from this lamp. Do not stare at exposed lamp.*

RG-3: *WARNING. IR emitted from this lamp. Do not look at this lamp from distances less than ______meters (or feet).*

7.6 Technical Information

A lamp manufacturer shall provide, or cause to be provided upon request, representative spectral distribution data for the lamp system it manufactures in the form of 1) spectral radiant power, or 2) spectral radiance, or 3) spectral intensity, or 4) spectral irradiance; and the lumen-to-radiant-power conversion factor. The data shall be provided over the wavelength range 300 nm to 800 nm. If there is potentially hazardous emission in the range 200 nm to 1,400 nm, the data range shall be extended to include it.

The manufacturer should provide, upon request, available radiometric information relating to the potential hazards associated with the lamp system it manufactures (e.g., effective ultraviolet irradiance at a reference distance or, alternatively, for general lighting sources at a reference illuminance level of 500 lux, the lamp's luminance and radiance). This may also include engineering design suggestions for protective enclosures, globes, fixtures, and luminaires.

A lamp manufacturer should provide (or cause to be provided upon request) representative spectral irradiance in SI units at reference conditions (as specified in the first paragraph of this section) for lamps it manufactures, to clearly specify the spectral emissions of the lamps over the wavelength range from 200 nm to 1,400 nm. Because the output of some lamps degrades with use over time, the manufacturer should, upon request, provide the lamp's date of manufacture.

Annex A – Exposure Limits

A.1 General

Personnel working with or in the vicinity of lamps and lamp systems should not be exposed to levels exceeding the criteria based on recommendations in Threshold Limit Values (TLVs®) of the American Conference of Governmental Industrial Hygienists and recommendations in publications of the International Commission on Non-ionizing Radiation Protection (ICNIRP) (see **Annex B**, **Section B.7 – General**). The specific techniques of applying these criteria to lamps and lamp systems are given in **Section 4**.

The limiting aperture dimensions for irradiance measurements given in this document are for general application. However, they may not be appropriate for certain specific conditions and devices.

A.2 Ultraviolet Exposure Limits

A.2.1 General. The ultraviolet exposure limits represent conditions under which it is believed that nearly all individuals in the general population may be repeatedly exposed without adverse health effects. However, they do not apply to photosensitive individuals or to individuals concomitantly exposed to photosensitizing agents (see **Section B.6.10 – Key References** for Fitzpatrick et al. reference). Such individuals, in general, are more susceptible to adverse health effects from optical radiation than individuals who are not photosensitive or concomitantly exposed to photosensitizing agents. The susceptibility of photosensitive individuals varies greatly, and it is not possible to set exposure limits for this portion of the population. These exposure limits shall be used as guides in the control of exposure to continuous sources where the exposure duration is not less than 0.1 ms. These values shall be used as guides in the control of exposure and shall not be regarded as a fine line between safe and dangerous levels.

A.2.2 Skin and Eye Exposure Limits for 200 nm to 400 nm. The limits for human exposure to ultraviolet radiation incident upon the unprotected skin or eye where irradiance values are known and exposure time is controlled are as follows:

- The effective irradiance exposure for the unprotected skin or eye should not exceed the values given in **Table A-1** within an 8-hour period.
- To determine the effective ultraviolet irradiance E_S of a broad-band source, the following weighting formula shall be used:

$$E_S = \sum_{200}^{400} E_\lambda \cdot S(\lambda) \cdot \Delta\lambda \,, \qquad \textbf{(A-1a)}$$

where:

E_S is the effective ultraviolet irradiance in W/cm²

E_λ is the spectral irradiance in W/cm²·nm

$S(\lambda)$ is the ultraviolet hazard weighting function (see **Table A-2**)

$\Delta\lambda$ is the calculation interval in nm

The summation extends from 200 nm to 400 nm

The permissible exposure time for exposure to ultraviolet radiation incident upon the unprotected eye or skin shall be computed by:

$$t_{max} = (0.003)/E_S \,, \qquad \textbf{(A-1b)}$$

where:

t_{max} is the permissible exposure time in seconds

E_s is the effective ultraviolet irradiance in W/cm²

The value "0.003" is in J/cm²

The exposure time may also be determined using **Table A-1**, which provides exposure times corresponding to an effective irradiance in µW/cm^2.

Note 1: All the preceding exposure limits for ultraviolet radiation apply to sources that subtend an angle of less than 80° (1.4 radians) i.e., sources within 40° of the normal to the irradiance area. Sources that subtend a greater angle need to be measured only over an angle of 80°.

Note 2: A limiting aperture no greater than 2.5-cm diameter shall be used with sources producing a uniform optical radiation pattern. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e., contain hot spots less than 2.5 cm), a 7-mm aperture shall be used. The measurement shall be made in that position of the beam giving the maximum reading.

A.2.3 Eye Exposure Limit for 315 nm to 400 nm. For the spectral region 315 nm to 400 nm (within the UV-A), the total ultraviolet irradiance for the unprotected eye, E_{UV}, should not exceed 1 mW/cm^2 for periods greater than 1,000 s (approximately 16 minutes). For exposure times less than 1000 s, the total radiant exposure to the eye should not exceed 1.0 J/cm^2.

Note: All of the preceding exposure levels for ultraviolet radiation apply to sources that subtend an angle less

Table A-1. Permissible Ultraviolet Exposure Duration

Duration of Exposure Per Day		TWA Effective Ultraviolet Irradiance E_S (µW/cm^2)
8	hours	0.1
4	hours	0.2
2	hours	0.4
1	hour	0.8
30	minutes	1.7
15	minutes	3.3
10	minutes	5
5	minutes	10
1	minute	50
30	seconds	100
10	seconds	300
1	second	3,000
0.5	second	6,000
0.1	second	30,000

Table A-2. Spectral Weighting Function for Assessing Ultraviolet Hazards

Wavelength* λ, nm	Relative Spectral Efficiency, *S*(λ)	Wavelength* λ, nm	Relative Spectral Efficiency, *S*(λ)
200	0.030	313**	0.006
205	0.051	315	0.003
210	0.075	316	0.0024
215	0.095	317	0.0020
220	0.120	318	0.0016
225	0.150	319	0.0012
230	0.190	320	0.0010
235	0.240	322	0.00067
240	0.300	323	0.00054
245	0.360	325	0.00050
250	0.430	328	0.00044
254**	0.500	330	0.00041
255	0.520	335	0.00034
260	0.650	340	0.00028
265	0.810	345	0.00024
270	1.000	350	0.00020
275	0.960	355	0.00016
280*	0.880	360	0.00013
285	0.770	365**	0.00011
290	0.640	370	0.000093
295	0.540	375	0.000077
297**	0.460	380	0.000064
300	0.300	385	0.000053
303	0.120	390	0.000044
305	0.060	395	0.000036
308	0.026	400	0.000030
310	0.015		

Table notes:

* Wavelengths chosen are representative; other values should be interpolated at intermediate wavelengths.

** Emission lines of a mercury-discharge spectrum, rounded to the nearest integer.

than 80°, i.e., sources within 40°of the normal to the irradiance area. Sources that subtend a greater angle need be measured only over an angle of 80°.

This can be expressed as:

$E_{uv} \leq 0.001$ (for $t \geq 10^3$) , **(A-2a)**

$t_{\max} \leq 1.0/E_{uv}$ (for $t \geq 10^3$) , **(A-2b)**

where:

t is time in seconds

$t_{\max}$ is time in seconds

E_{uv} is the total ultraviolet irradiance between 315 nm and 400 nm on the unprotected eye in W/cm^2; the value 1.0 is in J/cm^2

Notes: (Refer to Notes for **Section A.2.2**.)

A.3 Light and Near-Infrared Radiation Exposure Limits

The limits for exposure to broad-band light and IR-A radiation for the eye apply to exposure within an 8-hour period and require knowledge of the spectral radiance, L_λ, and total irradiance, E, of the source as measured at the position(s) of the eyes of the exposed person. Such detailed spectral data of a white light source is generally required only if the luminance of the source exceeds 10^4 cd/m^2. At a luminance less than this value, the exposure limit should not be exceeded. The exposure limits are given in **Sections A.3.1** through **A.3.6**.

The light and near-infrared limits represent conditions under which it is believed that nearly all individuals in the general population may be repeatedly exposed without adverse health effects. However, they do not apply to photosensitive individuals or to individuals concomitantly exposed to photosensitizing agents. These exposure limits shall be used as guides in the control of exposure and shall not be regarded as a fine line between safe and dangerous levels.

Spectral weighting functions for aphakic hazards, blue-light hazards, and retinal thermal hazards are shown in **Table A-3**.

A.3.1 Retinal Thermal Hazard Exposure Limit. To protect against retinal thermal injury, the integrated spectral radiance of the light source L_R weighted by the retinal thermal hazard weighting function $R(\lambda)$ (from **Table A-2**), should not exceed the levels defined by:

$L_R = \sum_{380}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda \leq 640/\left(\propto \cdot t^{0.25}\right)$ for $1\ \mu s < t \leq 0.63$ ms, all α , **(A-3a)**

$L_R = \sum_{380}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda \leq 3.2/\left(\propto \cdot t^{0.25}\right)$ for $0.63 \text{ ms} < t \leq 0.25$ s, $\alpha < 0.1$, **(A-3b)**

$L_R = \sum_{380}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda \leq 16/t^{0.75}$ for $0.63 \text{ ms} < t \leq 0.25$ s, $\alpha > 0.1$, **(A-3c)**

$L_R = \sum_{380}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda \leq 45$ for $t > 0.25$ s, $\alpha > 0.1$, **(A-3d)**

$L_R \leq 4.5\,\alpha^{-1}$ for $t > 0.25$ s, $\alpha < 0.1$, **(A-3e)**

where:

L_R is the retinal thermal hazard weighted radiance of the light source in W/cm^2·sr·nm

L_λ is the spectral radiance in W/cm^2·sr·nm

$R(\lambda)$ is the retinal thermal hazard weighting function

t is the viewing duration (or pulse duration if the lamp is pulsed) expressed in seconds but restricted to durations of 1 μs to 0.25 s

$\Delta\lambda$ is the calculation interval in nm

α is the visual angle of the source or projected source in radians defined by the 50% peak radiance points

The summation extends from 400 nm to 1,400 nm

Note: **Equations A-3a** and **A-3b** are empirical and are not dimensionally correct unless a dimensional correction factor $K_1 = (1\text{W}\cdot\text{rad}\cdot\text{s}^{0.25})/(\text{cm}^2\cdot\text{sr})$ is inserted in the right-hand numerator.

For a non-circular projected source area, α is determined from the arithmetic mean of the shortest and the longest dimensions. For example, α for a 20-cm long by 3-cm diameter tubular source at a viewing distance of $r = 200$ cm in a direction normal to the lamp axis would be determined from the mean dimension of $l = (20 + 3)/2 = 11.5$ cm: $\alpha = l/r = 11.5/200 = 0.058$ radian.

If the calculated α exceeds 0.1 radian, then $\alpha = 0.1$ in **Equations A-3a** and **A-3c** should be used. If the calculated α is less than 0.0017 radian, and the time duration is less than 0.25 sec, then $\alpha = 0.0017$ radian

Table A-3. Spectral Weighting Function for Assessing Retinal Hazards from Broadband Optical Sources

Wavelength, λ (nm)	Aphakic Hazard Function, $A(\lambda)$	Blue-Light Hazard Function, $B(\lambda)$	Retinal Thermal Hazard Function, $R(\lambda)$
300	N.A.	0.010	N.A.
305	6.00	0.010	N.A.
310	6.00	0.010	N.A.
315	6.00	0.010	N.A.
320	6.00	0.010	N.A.
325	6.00	0.010	N.A.
330	6.00	0.010	N.A.
335	6.00	0.010	N.A.
340	5.88	0.010	N.A.
345	5.71	0.010	N.A.
350	5.46	0.010	N.A.
355	5.22	0.010	N.A.
360	4.62	0.010	N.A.
365	4.29	0.010	N.A.
370	3.75	0.010	N.A.
375	3.56	0.010	N.A.
380	3.19	0.010	0.01
385	2.31	0.013	0.01
390	1.88	0.025	0.03
395	1.58	0.050	0.05
400	1.43	0.100	0.10
405	1.30	0.200	0.20
410	1.25	0.400	0.40
415	1.20	0.800	0.80
420	1.15	0.900	0.90
425	1.11	0.950	0.95
430	1.07	0.980	0.98
435	1.03	1.000	1.00
440	1.00	1.000	1.00
445	0.97	0.970	1.00
450	0.94	0.940	1.00
455	0.90	0.900	1.00
460	0.80	0.800	1.00
465	0.70	0.700	1.00
470	0.62	0.620	1.00
475	0.55	0.550	1.00
480	0.45	0.450	1.00
485	0.40	0.400	1.00
490	0.22	0.220	1.00
495	0.16	0.160	1.00
500 to <600	$10^{(450-\lambda)/50}$	$10^{(450-\lambda)/50}$	1.00
600 to <700	0.001	0.001	1.00
700 to <1,050	N.A.	N.A.	$10^{(700-\lambda)/500}$
1,050 – 1,150	N.A.	N.A.	0.2
1,150 – 1,200	N.A.	N.A.	$0.2 \times 10^{[0.02\,(1,150-\lambda)]}$
1,200 – 1,400	N.A.	N.A.	0.02

N.A. = not applicable

in **Equations A-3a** and **A-3b** should be used. If the calculated α is less than 0.011 radian and the time duration is greater than 0.25 sec, then α = 0.011 radian in **Equations A-3a** and **A-3b** should be used. No criteria for thermal hazards are relevant for exposure durations longer than 10 s, because while thermal injury is the dominant injury mechanism to 10 s, photochemical mechanisms predominate for exposure durations longer than 10 s.

Note 1: For these exposure limits, a 7-mm pupil diameter is assumed for viewing durations less than approximately one second.

Note 2: L_λ shall be averaged over a right circular cone field of view of 0.011-radian included angle.

Note 3: This limit should only be assessed for very high-brightness arc lamps (e.g., xenon arc lamps) and super-luminescent diodes (SLDs, highly specialized LEDs with a very small source size) that can be viewed through magnifying optics (e.g., arc-lamp projector). Less-bright visible lamps would not be expected to pose a hazard for retinal thermal injury (e.g., tungsten filament lamps and common white light emitting diodes used in flashlights).

Note 4: In the case of multiple-source elements that are not contiguous, this criterion applies to a single-source element. It also applies to the source as a whole when the average radiance over the full source is used.

A.3.2 Retinal Blue Light Hazard Exposure Limit. To protect against retinal photochemical injury from chronic blue-light exposure, the integrated spectral radiance of the light source weighted against the blue-light hazard function $B(\lambda)$ (from **Table A-2**, **Section A.3**) should not exceed the levels defined by:

$$L_B \cdot t = \sum_{300}^{700} L_\lambda \cdot B(\lambda) \cdot t \cdot \Delta\lambda \le 100 \ \left(\text{for } t \le 10^4\right), \quad \textbf{(A-4a)}$$

and

$$L_B \cdot t = \sum_{300}^{700} L_\lambda \cdot B(\lambda) \cdot \Delta\lambda \le 0.01 \ \left(\text{for } t \le 10^4\right), \quad \textbf{(A-4b)}$$

where:

$(L_B \cdot t)$ is in J/cm²·sr

L_B is the blue light hazard weighted radiance in W/cm²·sr

L_λ is the spectral radiance in W/cm²·sr·nm

$B(\lambda)$ the blue-light hazard weighting function

$\Delta\lambda$ is the calculation interval in nm

t is the exposure duration in seconds

The summations extend from 300 nm to 700 nm

For a weighted source radiance L_B exceeding 10 mW/cm²·sr, the maximum permissible exposure duration t_{max} is:

$$t_{max} = 100/L_B \ (\text{for } t \le 10^4), \quad \textbf{(A-4c)}$$

where:

t_{max} is the maximum permissible exposure duration in seconds

t is exposure time in seconds

L_B is the blue-light hazard weighted radiance in W/cm²·sr

The value 100 is in J/cm²·sr

Note 1: For these exposure limits, a 3-mm pupil diameter is assumed.

Note 2: The spectral radiance L_λ shall be averaged over a right circular cone field of view of 0.011-radian included angle.

Note 3: In the case of multiple-source elements that are not contiguous, this criterion applies to a single-source element. It also applies to the source as a whole when the average radiance over the full source is used.

A.3.3 Retinal Blue Light Hazard Exposure Limit: Small Source. For a light source subtending an angle less than 0.011 radian (11 milliradians), the limits of **Section A.3.2** are relaxed such that the spectral irradiance at the eye E_λ weighted against the blue-light hazard function $B(\lambda)$ should not exceed the levels defined by:

$$E_B \cdot t = \sum_{300}^{700} E_\lambda \cdot t \cdot B(\lambda) \cdot \Delta\lambda \le 0.01 \ (\text{for } t \le 100), \quad \textbf{(A-5a)}$$

$$E_B = \sum_{300}^{700} E_\lambda \cdot B(\lambda) \cdot \Delta\lambda \le 10^{-4} \ (\text{for } t \le 100), \quad \textbf{(A-5b)}$$

where:

$(E_B \cdot t)$ is in J/cm²

E_B is the blue-light hazard weighted irradiance in W/cm²

E_λ is the spectral irradiance at the eye in $W/cm^2 \cdot nm$

B_λ is the blue-light hazard weighting function

$\Delta\lambda$ is the calculation interval in nm

t is the exposure duration in seconds

The summations extend from 300 nm to 700 nm

For a source where the blue-light weighted irradiance E_B exceeds 1 $\mu W/cm^2$, the maximum permissible exposure duration is:

$$t(\max) \leq 0.01/E_B \text{ (for } t \leq 10^4\text{)}, \qquad \textbf{(A-5c)}$$

where:

$t(\max)$ is the maximum permissible exposure duration in seconds

t is exposure time in seconds

E_B is the blue-light weighted irradiance in W/cm^2

The value 0.01 is in J/cm^2

Note 1: A limiting aperture no greater than 2.5-cm diameter shall be used with sources producing a uniform optical radiation pattern. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e., contain hot spots less than 2.5 cm), a 7-mm aperture shall be used. The measurement shall be made in that position of the beam giving the maximum reading.

Note 2: These small source equations are equivalent to using **A.3.2** with the spatial averaging for small sources specified in Note 2 of that Section.

A.3.4 The Aphakic Eye Hazard Exposure Limit. To find the exposure values for individuals who have had cataract surgery (or otherwise lack a normal ocular lens) when an ultraviolet blocking intraocular lens has not been implanted, the aphakic retinal hazard weighting function $A(\lambda)$ (from **Table A-2**, **Section A.3**) shall be used in **Equations A-4a** and **A-4b** through **Equation A-5c** in place of $B(\lambda)$. The summation shall be from 305 nm to 700 nm. Except for sources that have a strong UV-A signal, evaluations using the $B(\lambda)$ weighting function can normally ignore the summation between 305 and 380 nm.

A.3.5 Infrared Radiation Hazard Exposure Limit. To avoid thermal injury of the cornea and possible delayed effects upon the lens of the eye (e.g., cataractogenesis), ocular exposure to infrared radiation E_{IR}, over the wavelength range 770 nm to 3,000 nm should be limited to 0.01 W/cm^2 for periods exceeding 1,000 seconds. This limit can be expressed as:

$$E_{IR} = \sum_{770}^{3{,}000} E_\lambda \cdot \Delta\lambda \leq 0.01 \text{ (for } t \leq 1{,}000\text{)}, \qquad \textbf{(A-6)}$$

For exposures of less than 1,000 seconds, the irradiance limit should be:

$$E_{IR} = \sum_{770}^{3{,}000} E_\lambda \cdot \Delta\lambda \leq 1.8\, t^{-0.75} \text{ (for } t \leq 1{,}000\text{)}, \qquad \textbf{(A-7)}$$

where:

E_{IR} is the ocular exposure to infrared radiation in W/cm^2

E_λ is the spectral irradiance in $W/cm^2 \cdot nm$

$\Delta\lambda$ is the calculation interval in nm

t is the exposure time in seconds

The summations extend from 770 nm to 3,000 nm

Note 1: For certain optical instruments, an averaging aperture greater than 3 mm but less than 7 mm may be required in updated versions of this Standard.

Note 2: A limiting aperture no greater than 2.5-cm (1-in.) diameter shall be used with sources producing a uniform optical radiation pattern. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e., contain hot spots less than 2.5 cm [1 in.]), a 7-mm (0.28-in.) aperture shall be used. The measurement shall be made in that position of the beam giving the maximum reading.

A.3.6 Infrared Radiation Hazard Exposure Limit: Weak Visual Stimulus. A weak visual stimulus is defined herein as one whose maximum luminance (averaged over a circular field of view subtending 0.011 radian) is less than 100 cd/m^2.

For an infrared heat lamp or any near-infrared source where a weak visual stimulus is inadequate to activate the aversion response, the near infrared (770 nm to 1,400 nm) radiance L_{IR}, as viewed by the eye for exposure times greater than 10 s, should be limited to:

$$L_{IR} = \sum_{770}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda < 3.2/\alpha t^{0.25} \quad \text{for } t \leq 810$$

(A-8a)

$$L_{IR} = \sum_{770}^{1,400} L_\lambda \cdot R(\lambda) \cdot \Delta\lambda < 0.6/\alpha \quad \text{for } t \leq 810$$

(A-8b)

where:

L_{IR} is the near-infrared radiance in W/cm²·sr

L_λ is the spectral radiance in W/cm²·sr·nm

$R(\lambda)$ is the retinal thermal hazard weighting function

$\Delta\lambda$ is the calculated interval in nm

t is exposure time in seconds

α is the angle (in radians) described in **Section A.3.1** (including the provisions for non-circular sources, and maximum and minimum limits)

The summations extend from 770 nm to 1,400 nm

The limit expressed in **Equation A-8** is based on a 7-mm (0.28-in.) pupil diameter.

Note 1: **Equation A-8** is empirical and is not dimensionally correct unless a dimensional correction factor (K_2 = 1 W·rad/cm²·sr) is inserted in the right-hand numerator.

Note 2: L_λ shall be averaged over a right circular cone field of view of 0.011-radian included angle.

A.3.7 Skin: Thermal Hazard Exposure Limit. Visible and infrared radiant exposure (400 nm to 3,000 nm) of the skin should be limited to:

$$H = \sum_{400}^{3,000} E_\lambda \cdot t \cdot \Delta\lambda \leq 2 \text{ (for } 1 < t \leq 10\text{)},$$ **(A-9a)**

$$H = \sum_{400}^{3,000} E_\lambda \cdot t \cdot \Delta\lambda \leq 2t^{0.25} \text{ (for } 1 < t \leq 10\text{)},$$

(A-9b)

where:

H is the radiant exposure in J/cm²

E_λ is the spectral irradiance in W/cm²·nm

$\Delta\lambda$ is the calculated interval in nm

t is exposure time in seconds

The summations extend from 400 nm to 3,000 nm

Note 1: A limiting aperture no greater than 2.5 cm in diameter shall be used with sources producing a uniform optical radiation pattern. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e., contain hot spots less than 2.5 cm), a 7-mm aperture shall be used. The measurement shall be made in that position of the beam giving the maximum reading.

Note 2: This exposure limit is based on skin injury due to a rise in tissue temperature and applies only to small-area irradiation. Exposure limits for periods greater than 10 s are not provided. Severe pain occurs below the skin temperature for skin injury, and an individual's exposure normally will be limited for comfort. Large body-area irradiation and heat stress are not evaluated since this involves consideration of heat exchange between the individual and the environment, physical activity, and various other factors.

Note 3: **Equation A-9b** is empirical* and is not dimensionally correct unless a dimensional correction factor ($K_3 = s^{-0.35}$) is inserted in the right-hand numerator.

Annex B – Summary of Biological Effects

Note: References are listed at the end of each Bioeffect section. General References are found in **Section B.7**.

B.1 Bioeffect: Infrared Cataract

Also known as "industrial heat cataract," "furnace man's cataract," or "glassblower's cataract."

B.1.1 Organ/Site: Eye: crystalline lens.

B.1.2 Spectral Range: 700 – 800 nm to 1,400 nm and possibly to 3,000 nm.

B.1.3 Peak of Action Spectrum: Not known; probably between 900 nm and 1,000 nm.

B.1.4 State of Knowledge: Limited threshold data available for acute cataract for rabbit at 1,064 nm (Wolbarsht) and IR-A region (Pitts and Cullen); no data for man. Degree of additivity and action spectrum

*Based on *Bioastronautics Data Book*, NTIS No. N65¬400 15594; p.7-15. 1964.

unknown. Good epidemiological evidence (Lydahl 1984).

B.1.5 Time Course: Noticeable clouding of the lens generally following years of chronic high-level exposure, the elapsed time depending upon how much difference between exposure and threshold, with heavy exposures producing reaction in shortest time.

B.1.6 Mechanism: Generally presumed to be thermal, although recent evidence suggests possible photochemical reaction; details not understood. The lens may be heated either from direct irradiation (Vogt) or by conductive heating from the heated iris (Goldmann).

B.1.7 Symptoms: Clouding of vision.

B.1.8 Needed Information: Action spectrum, if existent, for acute exposure and for effects of concomitant ultraviolet radiation exposure; additivity of multiple exposures; and the possibility of delayed effects from recurrent exposures.

B.1.9 Experience with Lamps: Accidental injury is not known, even from exposure to heat lamps. Limited population exposed.

B.1.10 Key References:

Goldmann H. Experimentelle untersuchungen uber die Genese des Feuerstars. III Mitteilung. Die Physik des Feuerstars I. Teil. Arch fur Ophthalmol. 1983;130:93130.

Lydahl E. Infrared radiation and cataract. Acta Ophthalmologica; Suppl. 1984;166:1-63.

Pitts DG, Cullen AP. Determination of infrared radiation levels for acute ocular cataractogenesis. von Graefes Arch Ophthal. 1981;217:285-297.

Sliney DH, Wolbarsht ML. Safety with Lasers and Other Optical Sources. New York: Plenum Press; 1980.

Vogt A. Experimentelle Erzeugung von Katarakt durch isoliertes kurtzwelliges Ultrarot, dem Rot beigemischt ist. Klin Mbl Augenheilk. 1919;63:230-231.

Wolbarsht ML. Cataract from infrared lasers: Evidence for photochemical mechanisms. Lasers and Light Ophthalmology. 1992;4:91-96.

B.2 Bioeffect: Photokeratitis

B2.1 Organ/Site: Eye: cornea.

B2.2 Spectral Range: 180 – 200 nm to 400 – 420 nm; principally 200 to 320 nm.

B.2.3 Peak of Action Spectrum: Approximately 270 nm (Pitts); approximately 288 nm (Cogan and Kinsey).

B.2.4 State of Knowledge: Good acute threshold data available for rabbit (200 nm to 400 nm); for monkey (200 nm to 320 nm); for human (200 nm to 300 nm). Data from different laboratories generally in good agreement.

B.2.5 Time Course: Noticeable reaction generally delayed by 4 to 12 hours following the exposure, the elapsed time depending upon how much difference in exposure and threshold, with heavy exposures producing reaction in shortest time; clearing in 24 to 48 hours, except for extremely severe exposures.

B.2.6 Mechanism: Photochemical reaction initiates chain of biological reactions; details not understood.

B.2.7 Symptoms: "Sand in the eye," blepharospasm (sudden, violent, involuntary contraction of the muscles of the eyelid), some clouding of vision; reaction in the palpebral fissure (opening between the upper and lower eyelids).

B.2.8 Needed Information: Higher resolution of thresholds in 305- to 320-nm range; possibility of delayed effects due to recurrent exposures.

B.2.9 Experience with Lamps: Not uncommon accidental exposure from germicidal lamps and mercury and xenon-arc lamps, but only in special applications. Limited population exposed.

B.2.10 Key References:

Cogan DG, Kinsey VE. Action spectrum of keratitis [roduced by ultraviolet radiation. Arch. Ophthalmol. 1946;35:670-617.

Hedblom EE. Snowscape eye protection. Arch Environ Health. 1961;2:685-704..

Leach, WM. Biological Aspects of Ultraviolet Radiation: A Review of Hazards; BRH/DBE 70-3. Rockville, Maryland: U.S. Public Health Service, Bureau of Radiological Health; Sept 1970.

MacKeen D, Fine S, Aaron A, Fine BS. Preventable hazards at UV wavelengths. Laser Focus. 1971;7(4):29.

Pitts DG, Tredici TJ. The effects of ultraviolet on the eye. Amer Ind Hyg Assoc J. 1971;32(4):235-246.

B.3 Bioeffect: Photoretinitis

Also known as "blue-light retinal injury."

B.3.1 Organ/Site: Eye: retina.

B.3.2 Spectral Range: 400 nm to 700 nm (principally 400 to 500 nm) in phakic eye (crystalline lens intact); 310 nm to 700 nm in aphakic (crystalline lens removed) eye (principally 310 nm to 500 nm).

B.3.3 Peak of Action Spectrum: Approximately 445 nm (Ham); approximately 310 nm in aphakic (Ham) in rhesus monkey.

B.3.4 State of Knowledge: Good acute threshold data available for monkey and some corroborative data for human at medically used laser wavelengths, and from accidental viewing of the sun or welding arcs.

B.3.5 Time Course: This mechanism of injury is dominant over thermal injury only for lengthy (greater than 10 seconds) exposures. Noticeable reaction generally delayed by more than 12 hours following the exposure, the elapsed time depending upon how much difference in exposure and threshold, with heavy exposures producing reaction in shortest time; greatest reaction generally noted at 48 hours. Some recovery is noted in human accidental exposures to arcs and sun-gazing.

B.3.6 Mechanism: Photochemical reaction initiates chain of biological reactions, apparently centered in retinal pigment epithelium; details not understood.

B.3.7 Symptoms: "Blind spot," or scotoma, where the bright arc was imaged on the retina. A retinal visible lesion (normally depigmented from blue light or hyper-pigmented from some ultraviolet wavelengths) is seen under ophthalmic examination within 48 hours post-exposure. Loss of vision may be permanent, although recovery is noted in mild cases.

B.3.8 Needed Information: More knowledge of injury mechanism; data at 400 nm to 450 nm for exposure durations less than 10 seconds; data on additivity of multiple exposures and the possibility of delayed effects from to recurrent exposures at levels below the acute threshold.

B.3.9 Experience with Lamps: Extremely rare or largely unreported injuries due to excessive exposure from staring at lamps. The natural aversion response normally limits exposure to preclude photoretinitis. Limited population potentially exposed.

B.3.10 Key References:

Ham WT Jr, Mueller HA, Sliney DH. Retinal sensitivity to damage by short-wavelength light. Nature. 1976;260(5547):153-155.

Ham WT Jr, Ruffolo JJ Jr, Mueller HA, Guerry D III. The nature of retinal radiation damage: Dependence on wavelength, power level and exposure time. Vision Res. 1980;20(12):1105-1111.

Lund DJ, Stuck BE, Edsell P. Retinal injury thresholds for blue wavelength lasers. Health Physics. 2006;90(5).

Mainster MA. Spectral transmission of intraocular lenses and retinal damage from intense light sources. Am. J Ophthalmol. 1978;85:167-170.

Marshal, J. Light damage and the practice of ophthalmology; in: Intraocular Lens Implantation; Rosen E, Arnott E, Haining W (eds). London: Moseby-Yearbook, Ltd.; 1983.

Pitts DG. The human ultraviolet action spectrum. American Journal Optom. Physiol. Opt. 1974;51:946-960.

Sliney DH. Eye protective techniques for bright light. Ophthalmology.1983; 90(8):937-944.

Sliney DH, Wolbarsht ML. Safety with Lasers and Other Optical Sources. New York: Plenum; 1980.

Sperling HG (ed). Intense Light Hazards in Ophthalmic Diagnosis and Treatment, Proceedings of a Symposium, Vision Res, 20(12), 1033-1203 (1980).

Varma SD, Lerman S (eds). Proceedings of the First International Symposium on Light and Oxygen Effects on the Eye. Oxford: IRL Press; 1984. [also published as Current Eye Res 3(1)].

Waxler M, Hitchens V (eds). Optical Radiation and Visual Health. Boca Raton: CRC Press; 1986.

Williams TB, Baker BN (eds). The Effects of Constant Light on the Visual System. New York: Plenum Press; 1980.

Young RW. A theory of central retinal disease; in: New Directions in Ophthalmic Research. Sears ML (ed). New Haven: Yale University Press; pp 237-270; 1981.

B.4 Bioeffect: Retinal Thermal Injury

B.4.1 Organ/Site: Eye: retina and choroid.

B.4.2 Spectral Range: 400 nm to 1,400 nm (principally 400 nm to 1,100 nm).

B.4.3 Peak of Action Spectrum: Approximately 500 nm (Ham).

B.4.4 State of Knowledge: Good acute threshold data available for rabbit and monkey, and limited data for man. Data from different laboratories generally in good agreement.

B.4.5 Time Course: This mechanism of injury is dominant over photochemical retinal injury for short (less than 10 seconds) exposures or at wavelengths greater than 700 nm. Noticeable reaction is normally immediate (or within 5 minutes) following the exposure. Recovery is limited or nonexistent.

B.4.6 Mechanism: Thermochemical reaction denatures proteins and other key biological components of cells with destruction of biological tissue. Light absorption and initial injury centered in retinal pigment epithelium and choroid.

B.4.7 Symptoms: "Blind spot," or scotoma, where the bright source was imaged on the retina. A retinal lesion visible (normally depigmented) is seen under ophthalmic examination—normally within 5 minutes, and certainly within 24 hours post-exposure. Loss of vision will be greatest just after the exposure, and some limited recovery may occur within 14 days.

B.4.8 Needed Information: More data on large image size (greater than 1 mm) exposures.

B.4.9 Experience with Lamps: Virtually no lamp is capable of causing this type of injury. A xenon-arc focused into the eye can cause the effect as shown clinically. Hence, the incidence may be extremely rare or largely unreported injuries from staring at magnified xenon arc. Natural aversion response normally limits exposure to preclude injury.

B.4.10 Key References:

Allen RA. Retinal thermal injury. Proc ACGIH Topical Symposium. 26-28 November 1979. Cincinnati, Ohio: ACGIH; 1980.

Ham WT Jr, Ruffolo JJ Jr, Mueller HA, Guerry D. III. The nature of retinal radiation damage: Dependence on wavelength, power level and exposure time. Vision Res. 1980;20(12):1105-1111.

Ham WT Jr, Williams RC, Geeraets WJ, Mueller HA, Guerry D, Clarke AM, Geeraets WJ. Effects of laser radiation on the mammalian eye. Trans NY Acad Sci. 1966;28:517-526.

Lund DJ, Stuck BE. Retinal injury thresholds for blue wavelength lasers. International Laser Safety Conference Proceedings; pp 50-66. Jacksonville, FL; 2003.

Schulmeister K. Averaging apertures and field-of-views for hazard measurements. Measurements of Optical Radiation Hazards. Sliney D H, Matthes R, et al. (eds). CIE Publication x016-1998 and ICNIRP 6/98, Geneva; 1998.

Sliney DH, Matthes R, DiDomenico S, Murray P, Phillips R, Wengraitis S. Measurements of Optical Radiation Hazards. CIE Publication 016-1998 and ICNIRP 6/98. Geneva; 1998.

Sliney DH, Wolbarsht ML. Safety with Lasers and Other Optical Sources. New York: Plenum Press; 1980.

B.5 Bioeffect: Ultraviolet Cataract

B5.1 Organ/Site: Eye: crystalline lens

B5.2 Spectral Range: 290 nm to 325 nm; possibly to 400 nm.

B.5.3 Peak of Action Spectrum: Approximately 305 nm (Pitts) for acute cataract; no action spectrum available for effect at wavelengths greater than 325 nm (Lerman, Zigman).

B.5.4 State of Knowledge: Good acute threshold data available for rabbit and monkey (295 nm to 325 nm); no data for human for acute cataract, but epidemiological evidence exists for chronic exposure to UV-B radiation (Taylor).

B.5.5 Time Course: Noticeable clouding of the lens generally delayed by 4 or more hours following the exposure, the elapsed time depending upon how much difference in exposure and threshold, with heavy exposures producing reaction in shortest time; some clearing within days only near threshold; otherwise, permanent opacification of the lens.

B.5.6 Mechanism: Photochemical reaction; details not understood.

B.5.7 Symptoms: Clouding of vision.

B.5.8 Needed Information: Action spectrum, if existent, for acute exposure and effects for UV-A exposure; additivity of multiple exposures and the possibility of delayed effects from recurrent exposures.

B.5.9 Experience with Lamps: Accidental injury is known, even from exposure to xenon-arc lamps. Limited population exposed.

B.5.10 Key References:

Lerman S. Radiant Energy and the Eye. New York: MacMillian, Inc.; 1980.

Parrish JA, Anderson RR., Urbach F, Pitts D. UV-A: Biological Effects of Ultraviolet Radiation with Emphasis on Human Responses to Longwave Ultraviolet. New York: Plenum Press; 1978.

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Pitts, DG, Cullen AP, Hacker PD. Ocular ultraviolet effects from 295-100 nm. Invest Ophthalmol Vis Sci. 1977;16(10):932-939.

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Taylor HR, West SK, Rosenthal FS, Munoz B, Newland HS, Abbey H, Emmett EA. Effect of ultraviolet radiation on cataract formation. New England Journal of Medicine. 1988;319:1429.

Waxler M, Hitchens V (eds). Optical Radiation and Visual Health. Boca Raton: CRC Press; 1986.

Zigman S, Datiles M, Torczynski E. Sunlight and human cataracts. Invest Ophthalmol Vis Sci. 1979;18(5):462-467.

Zuclich JA, Connolly JS. Ocular damage induced by near ultraviolet laser radiation. Invest Ophthalmol. 1976;15:760-764.

B.6 Bioeffect: Ultraviolet Erythema

B6.1 Organ/Site: Skin.

B6.2 Spectral Range: 180 – 200 nm to 400 – 420 nm; principally 200 to 320 nm.

B.6.3 Peak of Action Spectrum: Approximately 295 nm (Urbach, Parrish).

B.6.4 State of Knowledge: Good acute threshold data available for human (254 to 400 nm). Data from different laboratories generally in good agreement if two action spectra are taken into account—one for 4 to 8 hours and another for 24 to 48 hours.

B.6.5 Time Course: Noticeable reaction generally delayed by 4 to 12 hours following the exposure, the elapsed time depending upon how much difference in exposure and threshold, with heavy exposures producing reaction in shortest time; clearing in 24 to 48 hours, except for extremely severe exposures.

B.6.6 Mechanism: Photochemical reaction initiates chain of biological reactions; details not well understood (van der Leun).

B.6.7 Symptoms: "Sunburn," reddening of the skin, at sites of ultraviolet radiation exposure.

B.6.8 Needed Information: Higher resolution of thresholds in the 305- to 320-nm range; possibility of delayed effects due to recurrent exposures.

B.6.9 Experience with Lamps: Not uncommon accidental exposure from germicidal lamps and mercury and xenon-arc lamps, but only in special applications. Intentional exposure from sunlamp products.

B.6.10 Key References:

Anders A, Altheide H, Knalmann M, Tronnier H. Action spectrum for erythema in humans investigated with dye lasers. Photochemistry and Photobiology. 1995;61:200.

Fitzpatrick TB, Pathak MA, Harber LC, Sieji M, and Kukita A (eds). Sunlight and Man. Tokyo: Tokyo University Press; 1974.

Freeman RG, Owens DW, Knox JM, Hudson HT. Relative energy requirements for an erythemal response of skin to monochromatic wavelengths of ultraviolet present in the solar spectrum. J Invest Dermat. 1966;64:586-592.

Hausser KW, Vahle W. Sunburn and suntanning. Wissenchaftliche Veroffnungen des Siemens Konzern. 1927;6(1):101-120. Translated in the Biologic Effects of Ultraviolet Radiation (F. Urbach, ed). New York: Pergamon Press; 1969.

Schmidt K. On the skin erythema effect of UV flashes. Strahlentherapie. May 1964;124:127-136.

Urbach F (ed). The Biologic Effects of Ultraviolet Radiation; pp. 83-39, 327-436, 541-654. New York: Pergamon Press; 1968.

van der Leun JC. Theory of ultraviole erythema. Photochemistry and Photobiology. 1965;4:453.

World Health Organization. Environmental Health Criteria 14: Ultraviolet Radiation. Geneva: WHO; 1979.

B.7 General References

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Sliney DH, Matthes R, et al. (eds). Measurements of Optical Radiation Hazards . CIE Publication x016-1998 and ICNIRP 6/98. Geneva; 1998.

Annex C– Stray Radiant Power

Stray radiant power (SRP), often called "stray light," is any radiation that adds to the response of the photodetector but that: 1) has traversed a different geometrical path from the radiant source (lamp) to the instrument input than the designed optical path; or 2) is of a wavelength outside of the pass-band of the spectroradiometer.

Out-of-path SRP should be prevented by measurement design and implementation. Out-of-path SRP is not addressed in this annex, though its impact shall be considered in the uncertainty analysis.

Out-of-band SRP is further categorized as "near" or "remote." For purposes of this standard, *near SRP* is from wavelengths within 5 spectral bandwidths, i.e., 5 full-width half-maximum (FWHM) bandwidths, of the wavelength setting of the spectroradiometer. Near SRP may be ignored. *Remote SRP* is from wavelengths beyond 5 spectral bandwidths from the wavelength setting. Such radiation can introduce significant errors in a spectroradiometric determination. Hereafter in this annex, *SRP* refers only to remote out-of-band SRP.

C.1 Causes and Implications

The ability of gratings and filters to reject light of unwanted wavelengths is never perfect. This rejected radiation may be scattered, thus arriving at the detector through other paths. Absorbed radiation may change wavelength through heating or fluorescence and continue along the original optical path.

Spectroradiometers with photodetectors that are sensitive over a wide range of wavelengths, when measuring radiation from broad-band sources, have the greatest risk of SRP-generated errors. This risk can be realized if the spectral distribution of radiation from a lamp under test differs appreciably from that of the calibration source. If, at a test wavelength, the SRP from the calibration source is a greater fraction of the in-band radiation than is the case for the test lamp, the spectral radiant emission from the test lamp will be underestimated. The converse, too, is realizable, and underestimation should not be assumed without testing or giving due consideration to relative source distribution and the detector spectral sensitivity function.

C.2 Tests for SRP

In general, a spectroradiometer that has a good, properly aligned double monochromator and appropriate order-sorting and SRP-suppression filters will have negligible SRP. However, over time, internal scattering usually increases, and SRP can become significant. Periodic testing for SRP should be done.

C.3 Test during Calibration

Periodic system checks for SRP can be made conveniently during calibration with standard lamps; both a calibrated tungsten-halogen incandescent source (TH lamp) and a calibrated deuterium-arc source are used for this test. A comparison should be made of the graphs of the instrument's spectral response determined using the two lamps over the wavelength range where both have sufficient power for good measurements (250 nm to 350 nm). If the ratio of the curves is essentially constant, SRP in this region of the spectrum is negligible during calibration. This region has the greatest potential for SRP during incandescent calibration.

C.4 Full Test

When a critical test for SRP is indicated, the ASTM Filter Method referenced in **Section 2.1** is used. The filters used (see **Table C-1**) are sharp-cut, long-pass filters. A filter should be selected that has a transmittance less than 0.01 percent at and below the wavelength being tested. A known source with large long-wavelength energy content, such as the TH calibration lamp, should be aligned.

The response of the spectroradiometer with the SRP-blocking filter in place and the wavelength set below the cut-on wavelength will include the SRP. The filter's transmission and the TH lamp's output will determine the expected signal, and any additional radiation detected is SRP.

The above use of the method is the most sensitive because all wavelengths above the filter's cut-on will be passed to the instrument and their total contribution measured with little confounding from signals at the pass wavelength.

The above method will also work with well-characterized band pass filters. The addition of a band pass filter to the optical path should only change the reported signal by the filter's transmittance at the monochromator's wavelength. If the signal falls an additional amount, it will be due to the filter's blocking of out-of-band power. This method may be required for the testing of stray light from shorter wavelengths because of the lack of useful short-pass filters. Due to the presence of the signal at the pass wavelength, this version of the method is less sensitive than using opaque filters.

The SRP test filter is inserted ahead of the entrance port of the input optics but not near a plane that is conjugate to the entrance slit of the monochromator. Additional methods may be devised, such as introducing a strong monochromatic signal into the system and measuring the indicated spectrum. This method requires a source with negligible off-wavelength output—i.e., a filtered laser—and a system with wide dynamic range. Other methods exist and can be found in Kostkowski (see **Section 2.2**).

C.5 Outcome of Test Results

The SRP of an instrument shall be considered a flaw to be eliminated or reported. Corrections to the spectrum for SRP should not be introduced for the purpose of this test procedure. The level of SRP shall be included in the uncertainty analysis of all test results.

Table C-1. Sharp-Cut Long-Pass Glass Transmission Filter for Stray Radiant Power (SRP) Ratio Test

Spectral Range, nm[a]	Cut-Off Wavelength, nm[b]	Filter Identification[c,d]
185 – 215	215	Kopp (Corning) CS9-54 (7910 glass), 2.0 mm Schott WG-295 1.0 mm
210 – 230	230	Schott WG-295, 2.0 mm Kopp (Corning) CS 0-56 (7058 glass), 1.6 mm Hoya UV-28, 2.5 mm
225 – 240	240	Schott WG-295, 3.0 mm
240 - 260	260	Schott WG-305, 1.0 mm
250 – 270	270	Schott WG-305, 2.0 mm Hoya UV-30, 2.5 mm Kopp (Corning) CS 0-53 (7740 glass), 2.0 mm
260 – 280	280	Schott WG-305, 3.0 mm
270 – 295	295	Hoya UV-32, 2.5 mm Schott WG-320, 3.0 mm Kopp (Corning) CS 0-54 (0160 glass), 2.0 mm
280 – 300	300	Schott WG-320, 3.0 mm
300 – 318	318	Hoya UV-34, 2.5 mm
310 – 325	325	Hoya UV-36, 2.5 mm
320 – 335	335	Kopp (Corning) CS 0-52 (7380 glass), 2.0 mm
330 – 350	350	Kopp (Corning) CS 0-51 (3850 glass), 4.0 mm
Infrared SRP Test[e]	–	Schott RG-665, 3 mm Kopp (Corning) CS 2-58 (2403 glass), 3 mm Hoya R-66, 2.5 mm

Table notes:

a. The useful spectral range is nominal. The long wavelength limit of the range is the cut-off wavelength of the SRP ratio test filter actually used.

b. The cut-off wavelength of the SRP ratio test filter is the wavelength at which its transmission is $1x10^{-4}$ (0.01%). It is determined by measurement of the filter being used. The cut-off wavelength for the listed glasses, for the listed thickness, may deviate from the nominal values listed here by as much as 8 nm. The listed cut-off wavelengths are nominal values estimated from graphs in manufacturers' catalogs.

c. The listed filter thicknesses are nominal for stock filters.

d. Sources of the filters:
- Kopp (Corning): Kopp Glass, Inc., Pittsburgh, PA
- Schott: Schott Glass Technologies, Duryea, PA
- Hoya: Hoya Corp. USA, San Jose, CA

e. For an auxiliary test for near-infrared SRP, the reader is referred to the ASTM standard (see **Section 2.1**).

C.6 Methods for Reduction

Any action taken to reduce SRP shall be implemented for both the calibration and the test lamp measurements. After determining the critical wavelength range over which measurements are to be made to evaluate the biological hazard, two modes of action are available:

- Prevent as much of the out-of-band radiation as possible from entering the spectroradiometer system; and/or
- Install a detector that is not sensitive to out-of-band radiation, particularly to the wavelengths over which the calibration source or the test lamp radiates strongly.

The second method is likely to be the more effective, but less convenient to implement. For measurements in the 200-nm to 330-nm range, a photomultiplier with a CsTe cathode, known as a *solar blind*, is helpful.

The wavelength range over which pre-filtering can be used (the first method above) is limited to 250 nm to 380 nm because suitable filters are not available for other ranges. A Schott UG-11 or UG-5 blocking filter is used. The UG-11 has a higher SRP-rejection but lower throughput in the measurement range. The filters pass some near-infrared radiation. If the photodetector is sensitive to radiation with wavelengths greater than 700 nm, and the SRP is still indicated with the blocking filter, a sharp-cut red filter may be additionally inserted (see **Table C-1**); if the indicated SRP remains large, it is near-infrared radiation.

Annex D– Uncertainty Analysis

The analysis of uncertainty requires that all sources of uncertainty be quantitatively evaluated. The first step in uncertainty analysis is identifying the various uncertainty sources. Below is a list that can be the start of the evaluation of uncertainty for measurements used in classifying lamps and lamp systems.

- Instrument
  - Wavelength
  - Bandwidth
  - Response to varying spectra
  - Linearity
  - Stability
  - Stray Light
    - Out-of-Path
    - Out-of-Band
- Calibration
  - Standards (assigned uncertainty)
  - Distance
  - Alignment
  - Electrical Operation
- Measurement
  - Alignment
  - Distance
  - Source Under Test
    - Stability
    - Temperature
    - Size
    - Electrical Operation

The individual uncertainty factors shall be found or estimated. Each factor shall then be propagated through the measurement to find the impact on the weighted values used in hazard classification. As seen in **Section 6.4**, the percent impact on the weighted values may be different from the percent uncertainty of the individual factors.

The normal way to propagate the uncertainty is to find the ratio of the final value's change to the initial factors' change. The ratio of the impact to the input change is called *sensitivity*. According to the *Guide to the Expression of Uncertainty in Measurement* (GUM), the normal way to obtain estimates of the uncertainty of a given output is through the use of sensitivity coefficients.[*] Essentially, sensitivity coefficients show how the variables in an equation or function are related to the desired output. First, an equation describing the situation is required. When the value of a variable in this equation is changed, it will have an effect on the magnitude of the result. Often, the relationship between the input variable and the output result is linear, but not always. For example, for an incandescent or tungsten-halogen lamp, a change in lamp current will have a non-linear (exponential) effect on the spectral flux. This is because an increase in, for example, lamp current, induces an increase in the temperature of the tungsten wire. This increased temperature causes a shift in the spectral power distribution; the magnitude of the shift depends on what wavelength is used to measure the flux. The net result is that for small changes in current, the spectral flux changes by a factor of five to ten times the change in current in the near-UV and visible range; the smaller the wavelength, the larger the spectral power shift. **Table D-1** shows an example of the calculated change in spectral power at 300 nm due to a small change in current.

Table D-1. Example of Uncertainty Propagation

	Lamp Current	Signal at 300 nm
Setting 1	8.2000	8,451
Setting 2	8.2011	8,461
Change	0.0134%	0.118%

Table note: Sensitivity is 0.118/0.0134 = 9

Each of the uncertainties shall be carried through to the final value and expressed as a percentage. The full set of uncertainties is combined in quadrature, and this combined uncertainty (expressed as a percentage) is reported with the measured value.

Annex E – Application Notes and Examples

Generally, spectral data on lamps and lamp systems will be available in some form, e.g., as spatially total spectral power or spectral irradiance. Such data may be absolute or normalized to luminous values. Alternatively, the data may be relative spectral power where the scaling factor is arbitrary. It is desirable to use the available data whenever possible rather than require additional measurements for analysis. With appropriate calculations, it is usually possible to use data in almost any available form for the purposes of this Standard. Several examples are given to indicate possible calculation and measurement techniques.* It should be noted that there generally will be more than one approach to a problem. The analyst's choice will depend on the techniques that are familiar.

E.1 Example 1

Figure E-1 shows a relative spectral distribution for an incandescent lamp. Convert this to a spectral power distribution that is normalized to a per-lumen basis; i.e., determine new ordinate values such that the ordinate will be in units of W/nm·lm.

Assume that the relative curve is a spectral power distribution with the ordinate in watts per nanometer. The results of the weighting the curve by the spectral luminous efficiency function, $V(\lambda)$, integrating the resulting product curve, and multiplying by 683 would be the luminous flux in lumens represented by that curve. Clearly, this is best done using a numeric tabulation for the curve.

$$\Phi_V = 683 \sum \Phi_{\theta,\lambda}\, V(\lambda)\mathrm{d}\lambda = 32{,}160 \text{ lm} \qquad \textbf{(E-1)}$$

Dividing the ordinate values as shown by 32,160 would decrease the luminous flux to unity. The ordinate units are then watts per nanometer per lumen.

E.2 Example 2

Figure E-2 shows a normalized spectral power distribution curve for a floodlight utilizing a 1,000-watt tungsten-halogen incandescent lamp, an aluminum reflector behind the lamp, and a 20-cm diameter lenticular cover lens. Evaluate the retinal thermal hazard exposure against the criterion found in **Annex A.3.1**, at a distance from this lamp system where the illuminance on the centerline is 500 lux.

The measured distance along the beam centerline to the location of 500 lx is measured as 6.90 m. The retinal thermal hazard weighted radiance, L_R, is to be averaged through a right circular cone field of view of 0.011-radian included angle (see Note 2 in **Annex**

* International Bureau of Weights and Measures (BIPM). JCGM 100:2008, Evaluation of Measurement Data – Guide to the Expression of Uncertainty in Measurement. Paris: BIPM; 2008.

* More significant figures may be used than are mathematically and physically justifiable in order to clarify calculations and permit easy tracking of quantities.

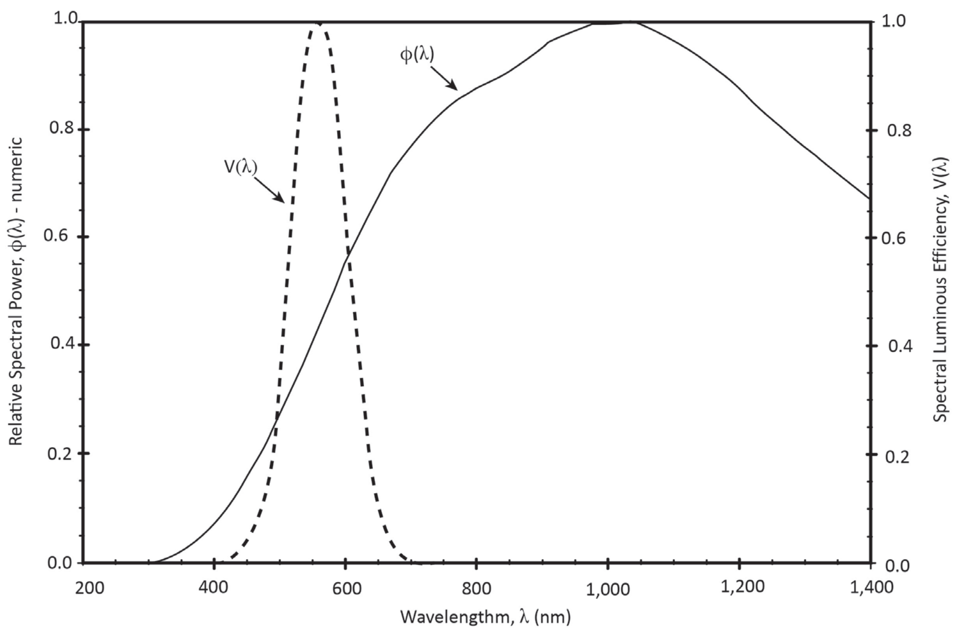

Figure E-1. Relative spectral power and spectral luminous efficiency as a function of wavelength.

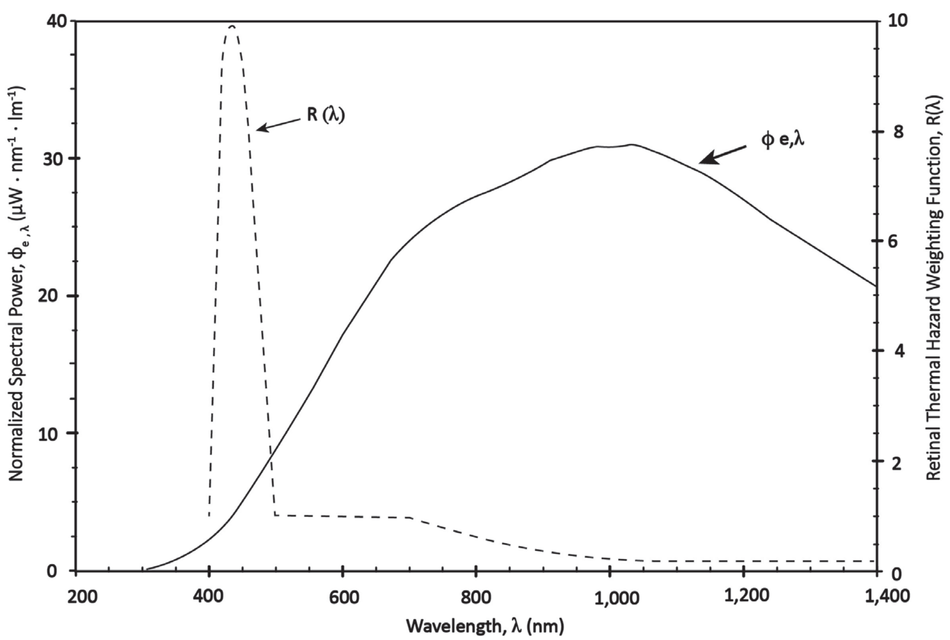

Figure E-2. Normalized spectral power and retinal thermal hazard weighting function as a function of wavelength.

A.3.1). A circular mask of 0.011 x 6.9 = 0.076 m diameter was placed in front of the lens. With this mask in place, luminous flux reaches an illuminance meter at the 6.9-meter distance through the specified solid angle, and the measured illuminance correlates with the average luminance within this angle. The mask was moved over the surface of the lens while observing the illuminance variation to locate the maximum illuminance reading. The measured maximum reading of 148 lx corresponds to the maximum source luminance averaged over the defined solid angle. Using the inverse-square law, the luminous intensity is 148 x 6.9^2 = 7,046 cd. The maximum average luminance is:

$$L_V = I_V/A = 7{,}046/[\pi\ (0.076/2)^2] = 1.5^5 \text{ x } 10^6\ \text{cd/m}^2$$

(E-2)

As an alternative, using square centimeters instead of square meters, the result is 155 cd/cm^2 or 155 lm/cm^2·sr. Weighting the spectral power curve of **Figure**

E-2 by the retinal thermal hazard weighting function and integrating the product curve produces the relation between $R(\lambda)$ weighted power and luminous flux. Data often are not available in consistent units. Here, the microwatts of **Figure E-2** are converted to watts for the summation.

$$\Phi_r/\Phi_v = \sum \Phi_{\theta,\lambda} \cdot R(\lambda)\mathrm{d}\lambda \qquad \textbf{(E-3)}$$

Multiply the luminance by this factor to obtain the retinal thermal hazard weighted radiance:

$$L_R = (155\ \mathrm{lm/cm^2{\cdot}sr})(0.0131\ \mathrm{W/lm}) = 2.03\ \mathrm{W/cm^2{\cdot}sr} \qquad \textbf{(E-4)}$$

The angle α subtended by the source from the viewing position is required for the exposure limit evaluation (see **Annex A.3.1**). An examination of the lens through an optically dense viewing glass shows a well-defined region of high luminance (9.5 cm x 13 cm) surrounded by a region of much lower luminance. Practically, this high luminance gradient defines the 50% luminance value used to establish α. The mean of the length and width dimensions, $(9.5 + 13)/2 = 11.2$ cm, is used to calculate α. Thus, $\alpha = (0.112\ \mathrm{m})/(6.9\ \mathrm{m}) = 0.016$ rad.

The test for the exposure limit (see **Annex A.3.1** and **Equation E-4**) expresses L_R in W/cm²·sr and requires that:

$$L_R \leq 5/\alpha t^{0.25}\ , \qquad \textbf{(E-5)}$$

where t is the exposure time but restricted to a maximum of 10 s.

Evaluating the right-hand side of **Equation E-5** at the maximum time of 10 s:

$$5/[(0.016)(100.25)] = 176\ \mathrm{W/cm^2{\cdot}sr} \qquad \textbf{(E-6)}$$

Continuous exposure to this lamp system will not exceed the criterion because L_R is less than the value in **Equation E-6** at the maximum applicable exposure time. In this example, α was measured in terms of a luminous quantity, luminance, rather than in terms of the 400-nm to 1,400-nm radiance for which α is defined. This was possible because the spectral distribution is essentially consistent over the lens of the lamp system. The 50%-of-maximum radiance and the 50%-of-maximum luminance coincide. In situations where the ratio of radiance to luminance is not constant, an appropriate technique for the specific situation shall be found.

E.3 Example 3

Figure E-3 shows the spectral power distribution for **a 40-watt**, 4-foot (122-cm), T12 (12/8 inch = 3.8 cm diameter), black light fluorescent lamp with a dark blue, UV-transmitting bulb. A long continuous row of the black light lamps is mounted on a ceiling to excite fluorescence of various displays. An individual may access locations such that the unobstructed distance

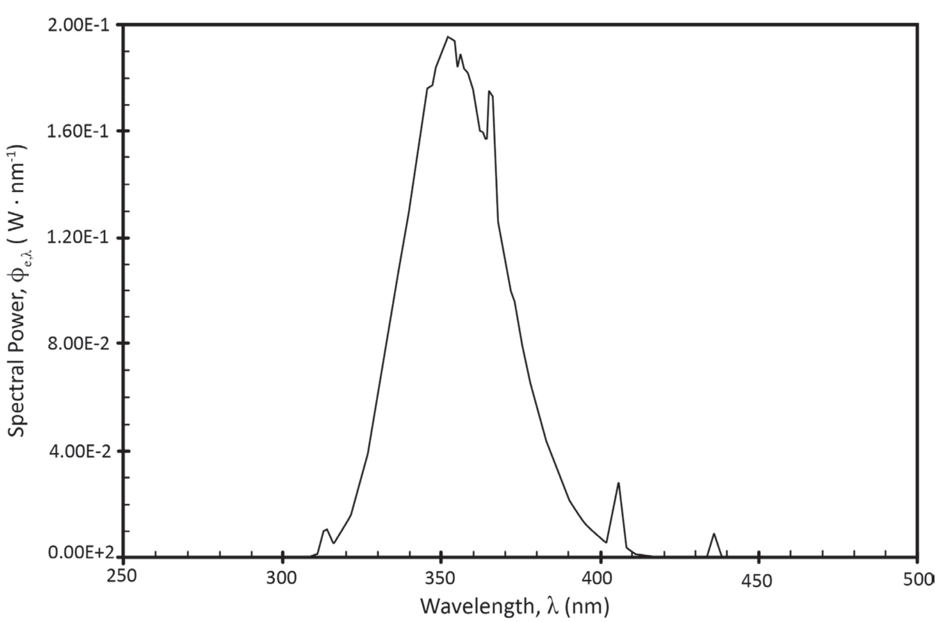

Figure E-3. Spectral power as a function of wavelength.

between his or her head and the row of lamps is no less than one meter. Evaluate whether a maximum exposure time limit needs to be established.

There are two relevant criteria: that for eye exposure to radiant power in the 320-nm to 400-nm bands (**Annex A**, **Section A.2.3**), and that for skin and eye exposure to $S(\lambda)$-weighted radiant power in the 200-nm to 400-nm band (**Annex A**, **Section A.2.2**). Evaluation of both criteria requires the irradiance at the point of exposure, and both are based on sources within 40° of a normal to the plane of the irradiated area. The maximum irradiance will be on a surface element normal to a line that is perpendicular to the line of lamps. This can be calculated for that length of the lamp row subtending 80° at the point of exposure.

The fluorescent lamp can be approximated as a cylindrical Lambertian radiator, and this approximation is quite good except in directions nearly parallel to the lamp axis. On this basis, the irradiance can be calculated using the configuration factor between a point and a finite Lambertian cylindrical radiating source. **Figure E-4** shows the standard geometry for the configuration factor C of element ΔA on a normal at one end of finite-length Lambertian cylinder S. Here, angle θ from the normal line is 40°, the half-angle subtended by the source length applicable to the UV hazard analysis. C is defined as the ratio of irradiance on ΔA to the radiant exitance of Lambertian cylinder S. For a distance of 100 cm and

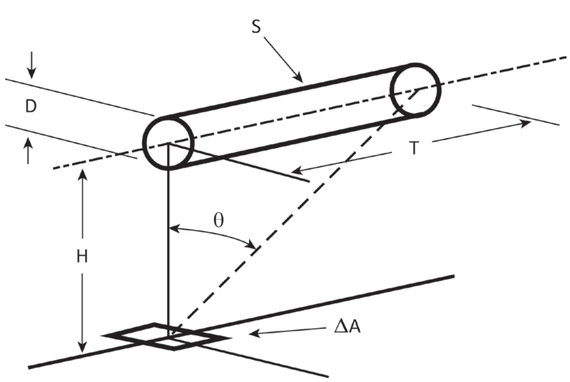

Figure E-4. Standard configuration factor geometry.

[*] Sparrow E, Cess R. Radiation Heat Transfer. Belmont, CA: Brooks/Cole Publishing Co.;1970. Appendix A.
Rohsenow W, Hartnett J. Handbook of Heat Transfer. New York: McGraw-Hill Book Company; 1973. Chapter 15.
Hamilton D, Morgan W. Radiant-Interchange Configuration Factors. NACA Technical Note 2836. Washington, DC: NACA; 1952.

a lamp of diameter of 3.8 cm, and using the effective length of 100 x tan40° = 83.9 cm, C is calculated to be 0.0072. Equations for this and other useful configuration factors can be found in many references.[*]

The irradiance in this example is due to a lamp row extending 40° on each side of the line from ΔA, and this doubles the irradiance and therefore doubles C of the standard case. Thus, the effective configuration factor is 2 x 0.0072 = 0.0144. The small variation of radiant exitance along the lamp length has negligible effect for this application, and its average value will be used. It is the radiant power divided by the lamp surface area A_S. AS = 3.8π x 122 = 1,450 cm^2.

The relevant radiant power for the eye criterion is that between 320 nm and 400 nm.

$$\Phi_\theta = \sum {}_{\Phi\theta,\lambda}\, \mathrm{d}_\lambda = 7.7\ \mathrm{W} \quad \textbf{(E-7)}$$

Then radiant exitance is:

$$M_\theta = \Phi_\theta / A_s = 7.7\ \mathrm{W}/1{,}450\ \mathrm{cm}^2 = 5.3\ \mathrm{mW/cm}^2 \quad \textbf{(E-8)}$$

Multiplying this exitance by the configuration factor, the irradiance is (0.0144)(5.3) = 0.076 mW/cm^2. The maximum irradiance for continuous exposure under **Annex A**, **Section A.2.3** is 1.0 mW/cm^2. Therefore, no time limit is required for exposure under this criterion.

The skin and eye criterion require weighting the radiant power by the $S(\lambda)$ function (see **Annex G**) shown in **Figure E-5**. The spectral power distribution of the lamp also is plotted in **Figure E-5** on a log ordinate, clearly showing power below 310 nm that is not evident on the linear ordinate of **Figure E-3**.

The weighted power is:

$$\Phi_{\theta,S(\lambda)} = \sum_{200}^{400} \Phi_{\theta,\lambda} \cdot S(\lambda)\mathrm{d}\lambda \ = \ 0.00192\ \mathrm{W} \quad \textbf{(E-9)}$$

And the radiant exitance is:

$$M_{\theta,S(\lambda)} = \Phi_{\theta,S(\lambda)} / A_S = 0.00192/1{,}450 = 1.32 \times 10^{-6}\ \mathrm{W/cm}^2 \quad \textbf{(E-10)}$$

Multiplying by the configuration factor, the irradiance is (0.0144)(1.32 x 10^{-6}) = 1.9 x 10-8 W/cm^2. The maximum exposure in an 8-hour period under **Annex A**, **Section**

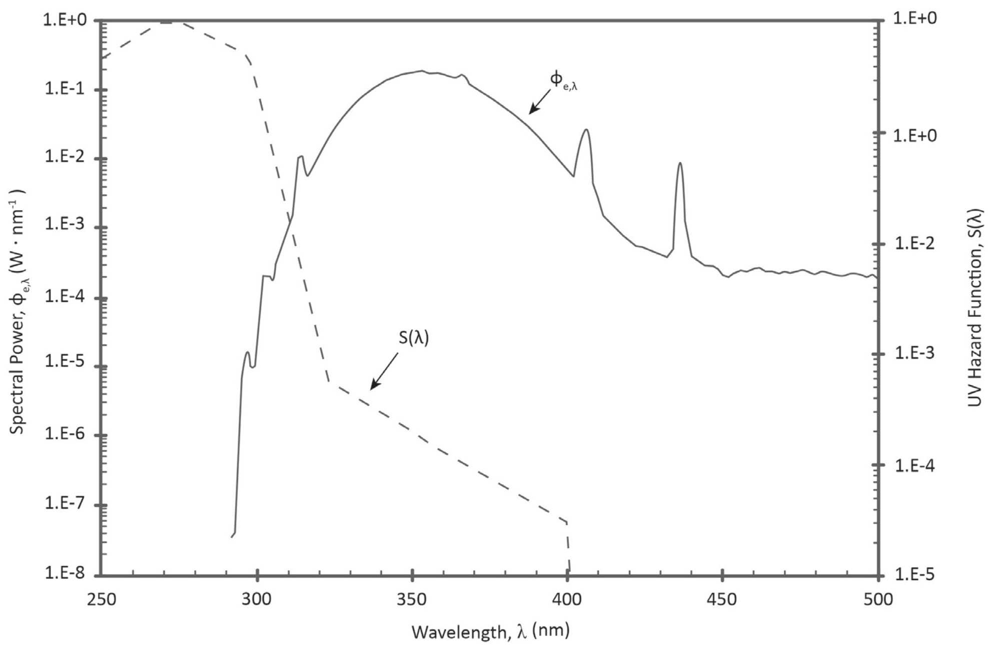

Figure E-5. Log spectral power and log UV hazard function *S*(λ) as a function of wavelength.

A.2.2 is 0.003 J/cm². The actual 8-hour exposure would be 1.9 x 10^{-8} x 60 x 60 = 0.00055 J/cm². Consequently, no time limit is required for exposure under this criterion.

Several observations should be made about this example:

- $S(\lambda)$ changes rapidly in the vicinity of 290 nm to 320 nm, increasing an order of magnitude for each 10-nm decrease. Therefore, a spectrum that is decreasing as the wavelength decreases in the vicinity of 300 nm may have spectral power that makes an important contribution to the $S(\lambda)$ summation.
- Spectral power plotted on a linear scale, as in **Figure E-3**, does not necessarily show where UV radiation becomes negligible. A log ordinate, as in **Figure E-5**, permits better evaluation.
- Many exposure conditions will be well below critical criteria values. In this example, they were about an order of magnitude lower. If a back-of-the-envelope analysis shows such a relation, then there is no need to carry out detailed calculations or require extensive measurements. However, such decisions can be made only if the analyst has a thorough knowledge of lamp and lamp system characteristics, together with experience in radiometric analysis.

E.4 Example 4

A strobe source is evaluated for weighted radiance dose ($L_R{\cdot}t$) (see **Section E.8**). It should be noted that more significant figures than are mathematically justified will be used at times to aid in tracking values.

Three values are needed to conduct this evaluation:

- **Nominal pulse duration, $\Delta\tau$:** Evaluation of $\Delta\tau$ shall be performed using $R(\lambda)$ weighted spectrum.
- **Weighted radiance dose, ($L_R{\cdot}t$):** This is the maximum weighted radiance dose found across the source and averaged using a selected field of view. The weighted radiance dose for sources that are not uniform will depend on the angle of the field of view used to average.
- **Angular subtense of the source, α:** For purposes of calculating α, peak radiance (see **Section 4.4.3**.) is the maximum weighted radiance dose ($L_R{\cdot}t$) for the particular averaging field of view selected. The 50% peak radiance points are determined from the spatial distribution of the source appropriately weighted to give the same units as ($L_R{\cdot}t$).

The fact that α is a function of ($L_R{\cdot}t$) means that the criterion, $5(\Delta t)^{0.75}/\alpha$, changes as the weighted radiance dose changes. As the field of view is increased, the criterion and the weighted radiance dose both decrease. To be classified as exempt, the source's weighted radiance dose shall be below the criterion for all averaging fields of view from 0.00175 radian to 0.100 radian.

E.5 R(λ)-Weighted Spectrum

The spectrum of the source shall be determined so that $R(\lambda)$ weighting can be performed. The spectroradiometer used is calibrated for irradiance. Consequently, the measured spectrum is in units of W·cm²/nm. Irradiance is measured at 20 cm, the defined evaluation distance for non-general lighting source (non-GLS) lamps.

The spectroradiometer is configured to step at increments of 1 nm and has FWHM bandwidth of 4 nm. It should be noted that this is the instrument dispersion bandwidth, not the summation bandwidth $\Delta\lambda$ of **Section 4.4.3**. The spectroradiometer operates by stopping at each step and averaging for a selected amount of time. The strobe repeats at 2 Hz. The spectroradiometer's averaging time is set to 0.49 second to ensure that there will be no more than one pulse per averaging period. A missed pulse is identifiable as a zero value in the spectrum.

The spectroradiometer's detector system may include non-linearities that compromise the spectroradiometer's ability to accurately measure short pulses. The level of the measured spectrum is confirmed using an integrating pulse illuminance meter that is accurate for pulses with durations of 10^{-7} second and longer. Using the meter, the integrated luminous energy density from ten pulses is measured as 3.5 x 10^{-2} lm·s/cm². The illuminance is calculated from the spectrum as 7.008 x 10^{-3} m/cm², or 3.504 x 10^{-3} lm·s/cm² for a single pulse. Good agreement between the two luminance values confirms the spectral level.

The energy in joules in a single pulse is calculated from the power in watts by multiplying the spectrum by 0.49 because the integration is over 0.49 second.

Figure E-6 is a graph of the spectral irradiance dose per pulse ($E_\lambda \cdot t$) at 20 cm in units of µJ/(cm²·nm).

Figure E-7 shows the $R(\lambda)$ retinal thermal hazard weighting function action spectrum used in this hazard analysis.

Figure E-8 is a graph of the weighted spectral irradiance dose per pulse ($E_{R,\lambda} \cdot t$) at 20 cm in units of µJ/(cm²·nm).

Summing the weighted irradiance dose spectrum from 400 nm to 1,400 nm determines the weighted irradiance dose per pulse ($E_R \cdot t$) to be 5.6 x 10^{-5} J/cm².

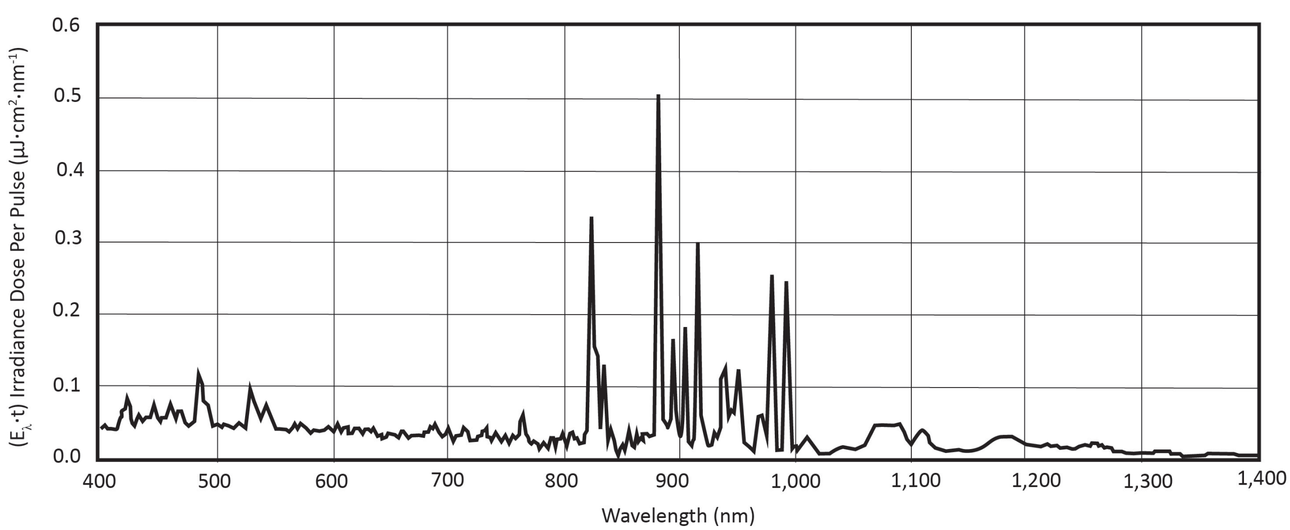

Figure E-6. Spectral irradiance dose per pulse at a distance of 20 cm.

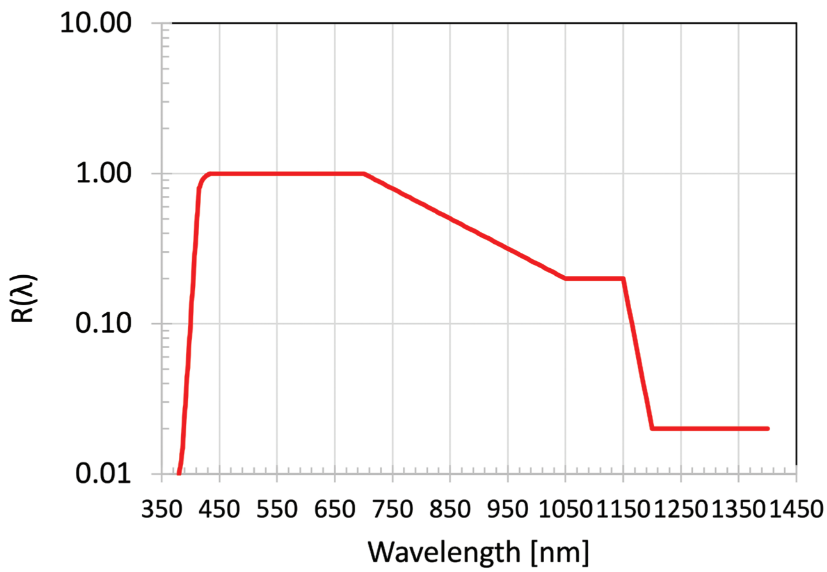

Figure E-7. Retinal thermal hazard weighting function action spectrum. (Graph courtesy of Rolf Bergman)

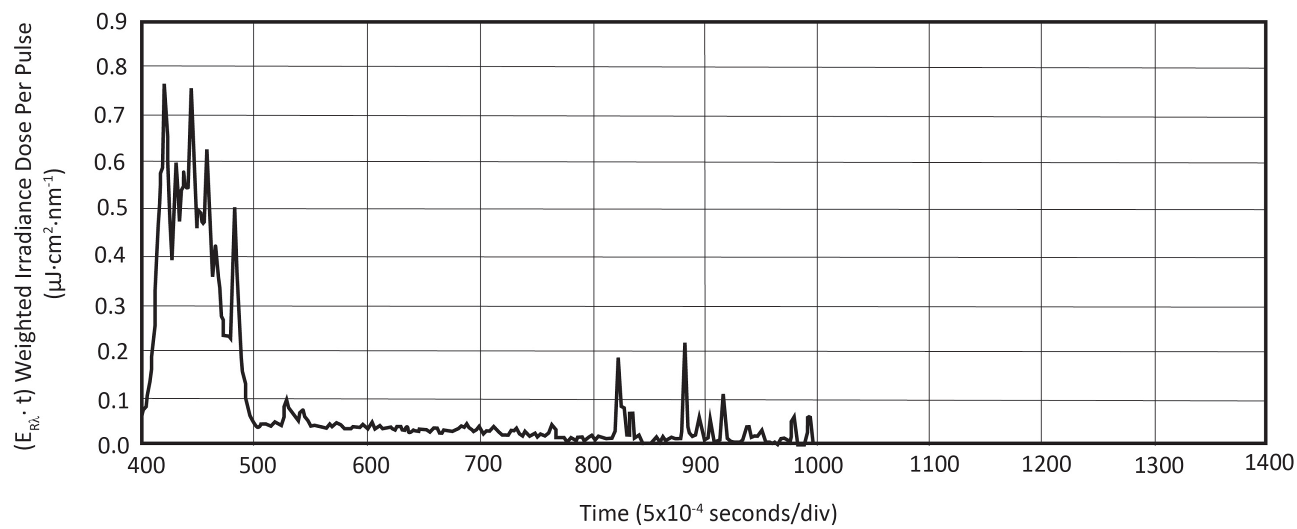

Figure E-8. Weighted irradiance dose per pulse at a distance of 20 cm.

E.6 Nominal Pulse Duration, Δt

A silicon detector is used for measurements of pulse width because there is insignificant contribution above 1,000 nm, as shown in **Figure E-8.** If contributions were seen at wavelengths above the spectral response of silicon detectors, additional measurements, equipment, and techniques would be required.

The temporal behavior at the different wavelengths is scaled by the proportion of weighted irradiance in the corresponding spectral band to duplicate the response of a $R(\lambda)$-weighted high speed detector. Bands of radiation from 400 nm to 750 nm in steps of 50 nm are selected using band pass filters. A long-pass filter is used to select irradiance above 800 nm. The filtered irradiance is sensed by a silicon detector system whose electrical output is captured on a digitizing oscilloscope. The measurements of irradiance vs. time show different pulse widths and shapes for different wavelengths. **Figure E-9** shows the time behavior of the irradiance found using four of the filters.

Each recorded time trace is then normalized for energy, i.e., adjusted to unit area under the time trace. Each normalized time trace was multiplied by the percentage of energy in the corresponding part of the $R(\lambda)$-weighted spectrum. Because each trace is from a different pulse with different triggering, the time on each trace was shifted to align the peaks of each time trace. This maximizes the peak, shortens the pulse duration, and gives a lower criterion, $[5(\Delta t)^{0.75}/\alpha]$, which leads to a conservative hazard evaluation. The scaled pulse traces are summed to give a temporal behavior, which would be measured by a high speed $R(\lambda)$-weighted sensor. **Figure E-10** shows the composite trace.

The nominal pulse duration, Δt, is calculated from the data shown in **Figure E-10** to be 14.7 x 10^{-6} second.

E.7 Uncertainty of Δ*t*

The uncertainty of the nominal pulse duration is estimated from the uncertainties of the detector system's temporal response and spectral weighting. The detector's response is checked using an LED driven by a square-wave oscillator as a fast pulse source. The rise and fall time of the detector system is less than or equal to 2 x 10^{-7} second. No adjustment by deconvolution for instrument rise and fall time behavior is made to the time traces, because the rise and fall times measured may be a limitation of the LED pulse source used to check the detector system. The rise time error will cause

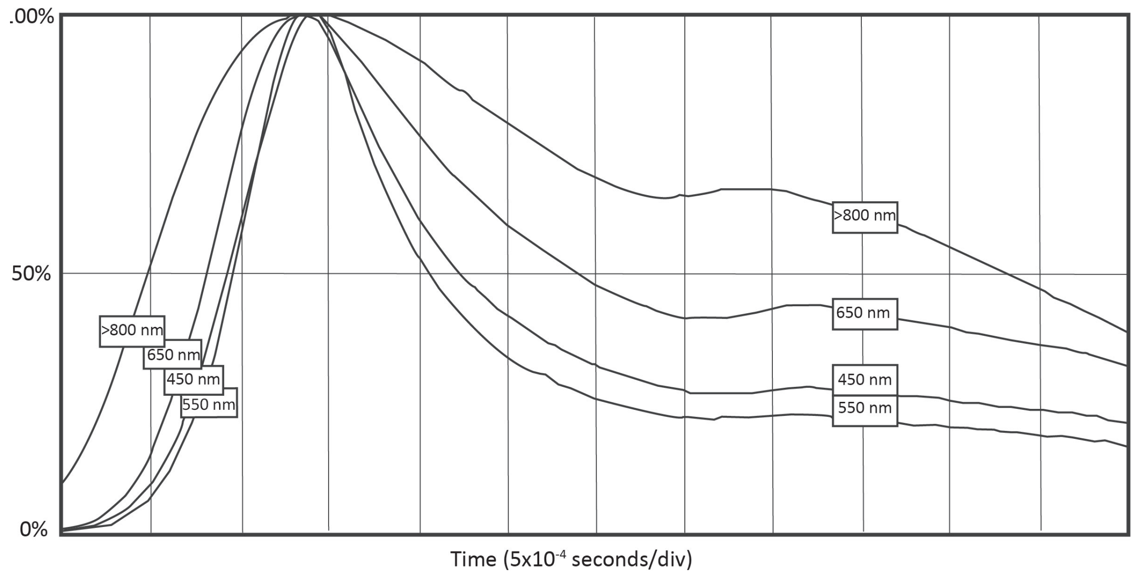

Figure E-9. Time behavior of irradiance with four filters.

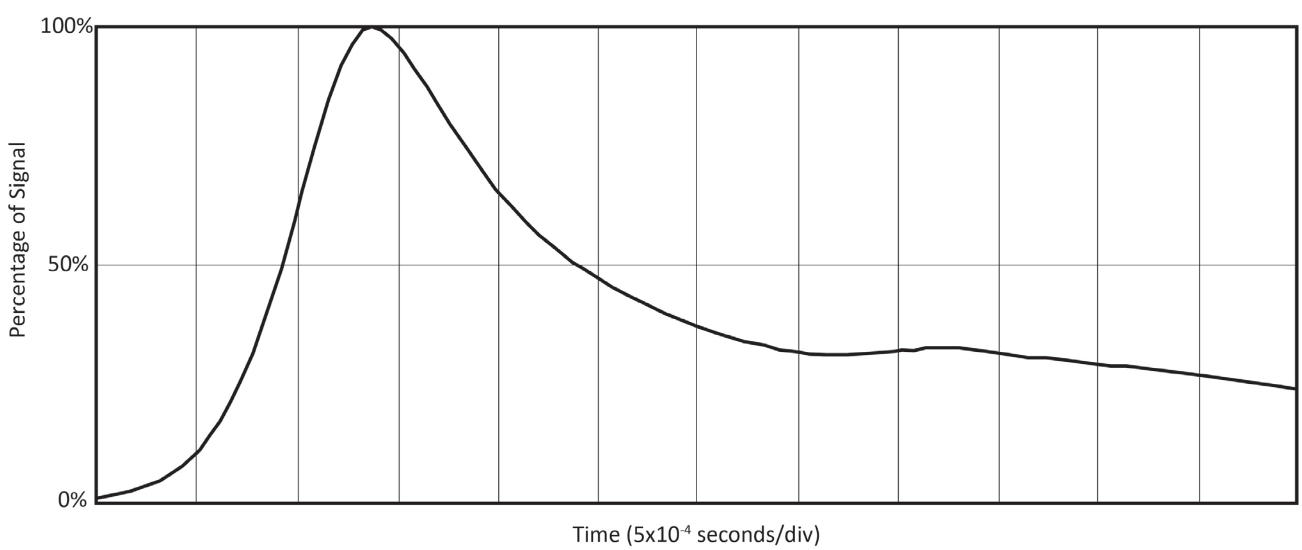

Figure E-10. Composite of summed scaled pulse traces.

a small change in the conservative direction and will be ignored. Only the fall time will increase the measured pulse width and underestimate the hazard. The fall time may increase the measured 50% pulse width by one-half the fall time. The potential error in pulse width determined at 50% maximum pulse height is taken as the uncertainty:

$$(0.5 \text{ x } 2 \text{ x } 10^{-7}) \,/\, 14.7 \text{ x } 10^{-6} = 0.680\% \qquad \textbf{(E-11)}$$

A careful reader may note that the detector's response could smooth small structures, lowering the peak height. Because these small temporal structures would contain little energy, this is ignored.

The oscilloscope time base has an uncertainty of 0.005% in the range used. The uncertainty of the level of absolute spectral irradiance measurement for the instrument used is known to be 1.10%, and the uncertainty in wavelength for this instrument is 0.1 nm. The known spectrum is shifted by one nm, weighted, and compared with the un-shifted weighted irradiance to find the sensitivity to wavelength error. A change in one nm leads to a change of 0.4%. The uncertainty of the weighted irradiance is:

$$[(1.10\%)^2 + (0.1 \times 0.4\%)^2]^{0.5} = 1.10\% \qquad \textbf{(E-12)}$$

As the pulse durations at different wavelengths are of the same order, a change in weighting would lead to a smaller relative change in nominal pulse duration. Changing the weighting used for the 450-nm range by 1.10% in the calculations changes the nominal pulse duration, Δt, by 0.080%. The combined uncertainty for the nominal pulse duration, $u_c(\Delta t)$, is equal to:

$$[(0.685\%)^2 + (0.005\%)^2 + (0.080\%)^2]^{0.5} = 0.685\% \qquad \textbf{(E-13)}$$

E.8 Weighted Radiance Dose ($L_R \cdot t$)

Note: To avoid confusion, the term weighted radiance dose levels is used to indicate the radiance dose pattern of the source weighted by $R(\lambda)$. The term weighted radiance dose refers to ($L_R{\cdot}t$), the maximum of the weighted radiance dose levels when averaged using a right circular cone field of view.

As shown in **Figure E-8**, photographic techniques may be used for the measurements of spatial radiance distribution because there are only insignificant contributions above 800 nm. If a major proportion of weighted energy was seen at wavelengths above the spectral response of photographic film, then additional measurements, equipment, and techniques would be required.

The determination of ($L_R{\cdot}t$) is achieved by capturing an image of the source that can be calibrated for weighted radiance dose level and linear spatial dimensions. A single, thin lens of 2.5 cm diameter is placed 20 cm from the source, at the position where the irradiance measurement was performed. An image is projected onto a white card at a distance of 145 cm from the lens, producing a magnification of 7.25.

A spot check across the image indicates that the spectral energy distribution is consistent over the image.

Therefore, the density in a black and white photograph will be proportional to the $R(\lambda)$ weighted radiance dose. A black and white instant-film camera with adjustable exposure time and aperture is focused on the projected image on the white card and fixed in place. The camera is set to a ½-second exposure time to photographically record the projected images of single flashes. Photographs are taken at each aperture setting to provide an image of the source without over- or under-exposure.

The lens is removed, and the strobe is configured with diffuse reflectors to give a spatially uniform irradiance over the white card. This is checked with a luminance meter. A grayscale card, a reflective standard with 6 levels of known reflectance, is placed at the position of the white card to provide a scale of radiance. Illuminating this card with the source under test takes into account the possible short-time reciprocity failure of the film. Photographs are taken at each aperture setting.

The photographs are scanned into a computer image file using an office flatbed scanner; then the images are translated into tables of pixel values for analysis as a spreadsheet. The dimensions of the gray steps on the card are 5.22 cm by 5.22 cm. The corresponding spatial dimension at the source is:

$$5.22/7.25 = 0.720 \text{ cm} \qquad \textbf{(E-14)}$$

The digitized image shows these proportions as 36 x 36 pixels; therefore, the system has a resolution in both directions of:

$$0.720/36 = 0.020 \text{ cm per pixel} \quad \textbf{(E-15)}$$

This conversion factor will be referred to as p.

The uncertainty of the conversion factor p is calculated from the uncertainty of each value used in its determination. The distances from the thin lens are measured to an uncertainty of 0.5 cm. The relative uncertainty is therefore 0.5/20 and 0.5/145, or 2.500% and 0.345%, respectively. The measurement of the gray step card has an uncertainty of 0.5 mm, leading to an uncertainty of 0.05/5.22, or 0.958%. The *U.S. Guide to the Expression of Uncertainty in Measurements* (ANSI Z540-2-1997; Section F.2.2.1) gives the relative uncertainty for digitized data as 0.29/(number of counts). The measurement of pixels is quantized, giving an uncertainty of 0.29/36, or 0.806%. The combined standard uncertainty for the conversion factor for pixels to cm, $u_c(p)$, is:

$$[2.500\%^2 + 0.345\%^2 + 0.958\%^2 + 0.806\%^2]^{0.5} = 2.817\% \quad \textbf{(E-16)}$$

One image of the gray scale card is chosen that shows a range of values for the levels on the card that are not over- or under-exposed. This digitized image of the gray scale card shows that the system of film and scanner is not linear under this illumination. A table is formed that transforms the digitized values to accurate relative radiance levels. The lower portion of the scale, with values from 0 to 90, is linear, and a straight line is used to interpolate the values for this portion of the table. Values above 90 are well fit with a logarithmic function, and the interpolated values are calculated from this function. **Figure E-11** shows the correction for level non-linearity, with measured data as circles and the interpolated values as the line.

A photograph of the source that has values in the range from 0 to 160 counts is selected. **Figure E-12** shows the strobe source as projected, photographed, and scanned.

The two images used, gray-scale card and source, are captured under the same conditions except the aperture setting. The aperture size does not affect the linearity.

For each pixel in the image, the value is replaced by the appropriate value as found in the look-up table described above. The resulting image is linear in radiance, but the levels are uncalibrated.

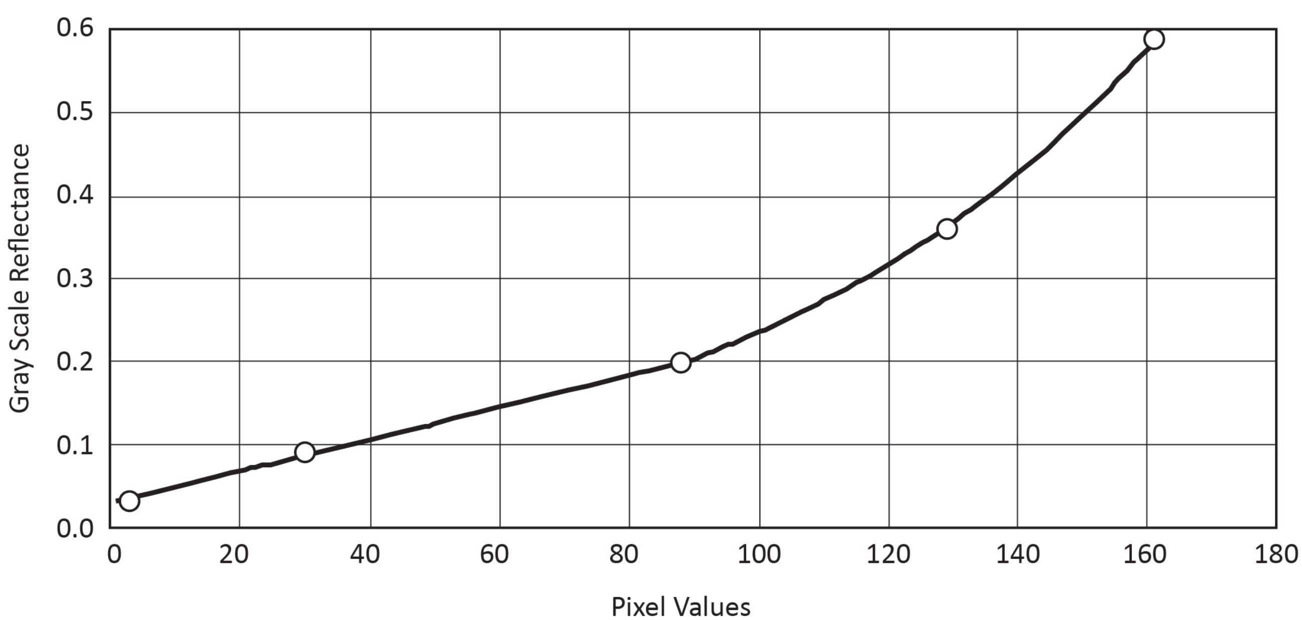

Figure E-11. Correction of digitized grayscale card values to radiance levels, adjusting for non-linearity.

The solid angle, $\Delta\omega$, of a single pixel at the source seen from 20 cm is:

$$(0.02 \text{ x } 0.02)/20^2 = 1 \text{ x } 10^{-6} \text{ sr} \quad \textbf{(E-17)}$$

The weighted radiance dose level per pulse is calculated from:

$$(L_R \cdot t) = (E_R \cdot t)/\Delta\omega = (E_R \cdot t)(d^2/p^2) \quad \textbf{(E-18)}$$

where:

d is the distance from the point of irradiance dose measurement

p is the linear dimension of a pixel

If the entire weighted irradiance dose per pulse, 5.6 x 10^{-5} J/cm^2, came from a single pixel, the weighted radiance dose level per pulse of that pixel would be:

$$(5.6 \text{ x } 10^{-5})/(1 \text{ x } 10^{-6}) = 56 \text{ (J/cm}^2\text{·sr)} \quad \textbf{(E-19)}$$

The sum of all pixel values, after linearity correction, is 500. The image is converted to weighted radiance dose levels by multiplying each measured value by:

$$56/500 = 0.112 \quad \textbf{(E-20)}$$

Figure E-13 is a digital grayscale image of the strobe source in **Figure E-12**. Its grayscale levels correspond to $R(\lambda)$-weighted radiance dose levels from 0 to 0.05 effective J/cm^2·sr.

E.9 Uncertainty of (L_R • t)

The weighted radiance dose levels are determined from the weighted irradiance dose, the measurement distance, and the pixel size, as indicated in **Equation E-18**. The uncertainty of the irradiance measurement distance is the same as the uncertainty of the distance from the source to the lens, developed above. The uncertainty of these quantities as previously determined are 1.100%, 2.500%, and 2.817%, respectively. Because

Figure E-12. Strobe source.

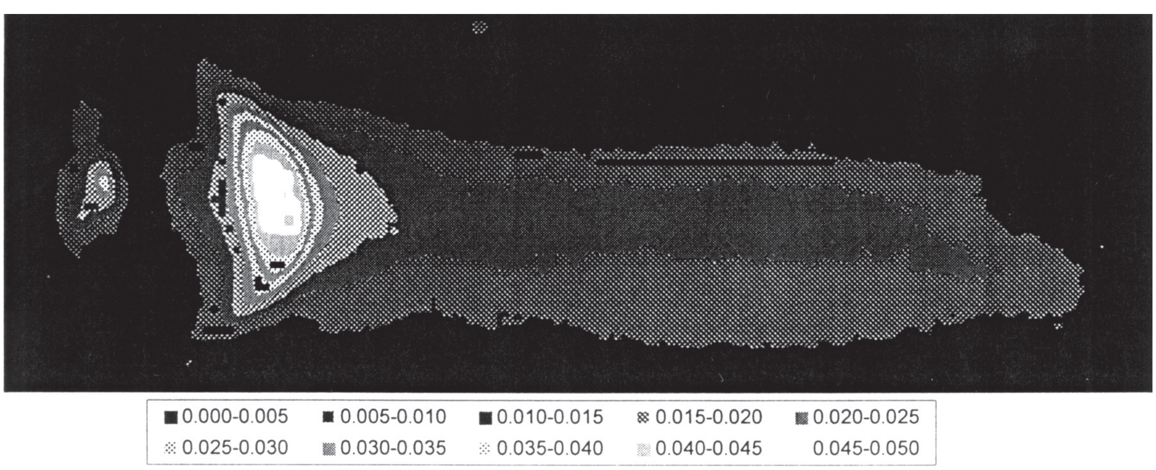

Figure E-13. *R*(λ)-weighted radiance dose levels, J/cm²· sr.

the linear dimensions are raised to the second power in the equation, the partial derivative with respect to each factor is used as coefficient for that factor in combining the uncertainties. The combined relative uncertainty of the weighted radiance dose levels is:

$$[(1.100\%)^2 + (2 \cdot 2.500\%)^2 + (-2 \cdot 2.817\%)^2]^{0.5} = 7.613\% \qquad \textbf{(E-21)}$$

The weighted radiance dose levels are not uniform; therefore, the smallest angle, 0.00175 radian, will be used to start the analysis.

A right circular cone of 0.00175 radian projects an area of 0.000962 cm^2 at a distance of 20 cm. As each pixel has an area of 0.0004 cm^2, it is not practical to average this image over a right circular cone of 0.00175 radian. A new image is constructed with four times as many pixels by dividing each original pixel into four square pixels with the same weighted radiance dose values as the original and an area of 0.0001 cm^2. An average of 9 cells in a three-by-three pattern approximates the right circular cone of 0.00175 radian in this image. This average is computed for all points of the image, and the highest average is noted. These average weighted radiance dose levels are slightly higher than what would be measured with a weighted radiance dose meter that averages over a right-circular-cone field of view of 0.00175-radian included angle, and therefore conservative in the hazard evaluation.

The highest value found, 0.050 J/cm^2·sr, is the weighted radiance dose ($L_R \cdot t$) for the averaging field of view of 0.00175-radian included angle.

E.10 Angular Subtense of the Source, α

With the weighted radiance dose level determined, the 50% profile of the source is found using this weighted radiance dose as 100%.

Figure E-14 shows the image from **Figure E-13**, adjusted so that only the pixels with effective radiance dose within 50% of the maximum (i.e., greater than 0.025 effective J/cm^2·sr) are shown. The remaining pixels in **Figure E 14** are a maximum of 25 pixels high and 13 pixels wide. The angular subtense of the source, α, is:

$$(0.5)(0.020)(25 + 13)/20 = 0.019 \text{ radian} \qquad \textbf{(E-22)}$$

E.11 Uncertainty of α

The uncertainty of α is found by quadrature sum of the relative uncertainty from each factor used to calculate α. Because pixels are quantized and have only integer values, a change in weighted radiance dose will have different effects on width plus height, depending on the starting weighted radiance dose. The level of weighted radiance dose is varied upwards until a change in width plus height occurs. It is then varied downward until a change in width plus height occurs. The weighted radiance dose difference between the two transitions is used to determine the sensitivity of α to a change in weighted radiance dose. A change in weighted radiance dose of -5.140% is equivalent to one pixel in the width plus height, which occasions a change in α of 2.564%. The sensitivity of α to weighted radiance dose, $\partial\alpha/\partial(L_R \cdot t)$, is:

$$2.564\%/-5.140\% = -0.499 \qquad \textbf{(E-23)}$$

Figure E-14. Profile of source from Figure E-13, adjusted to show only the pixels whose weighted radiance dose is within 50% of the maximum.

This sensitivity is multiplied by the uncertainty in weighted radiance dose to get the contribution of weighted radiance dose to the uncertainty of α. Therefore, the uncertainty of α arising from the uncertainty in weighted radiance dose is:

$$(7.613\%)\ (-0.499) = -3.798\% \quad \textbf{(E-24)}$$

As seen in **Section 4.4.3 Weighted Radiance Dose (L_R·t),** the uncertainty of the conversion factor p is 2.817%. The uncertainty of the height plus width of the source is estimated, as shown above, as 0.29/(number of counts). As the width plus height is a sum of terms rather than a product of factors, the quadrature sum is taken of the absolute uncertainties rather than from the relative uncertainties. The ratio of the combined uncertainties to the width plus height is taken to get the relative uncertainty of the factor:

$$[(0.29)^2 + (0.29)^2]^{0.5}/(13 + 25) = 1.079\% \quad \textbf{(E-25)}$$

The 20-cm distance is a statutory distance that is not measured and therefore has no uncertainty.

The last term shown in the combined uncertainty of α is a covariance term because the two factors, $(L_R \cdot t)$ and p, are not independent. The covariance is twice the product of the two contributions of the uncertainty of p to the uncertainty of α:

$$(2)\{[\partial a/\partial(L_R \cdot t)][\partial(L_R \cdot t)/\partial p][u_c(p)]\}[(\partial a/\partial p)\cdot u_c(p)] \quad \textbf{(E-26)}$$

Numerically evaluated:

$$(2)[(-0.499)(-2)(2.817\%)][(1)(2.817\%)] = 0.001584 \quad \textbf{(E-27)}$$

The combined standard uncertainty of angular subtense of the source $u_c(\alpha)$ is:

$$[(-3.798\%)^2 + (2.817\%)^2 + (1.079\%)^2 + 0.001584]^{0.5} = 6.273\% \quad \textbf{(E-28)}$$

E.12 Comparison of Weighted Radiance Dose to Criteria

The criterion, $5(\Delta t)^{0.75}/\alpha$, for an averaging angle of 0.00175 radian is:

$$5(14.7\times10^{-6})^{0.75}/0.019 = 0.0625 \quad \textbf{(E-29)}$$

As the weighted radiance dose, 0.050, is below 0.0625, the source is exempt as measured using a 0.00175-radian included angle averaging field of view.

E.13 Uncertainty of Comparison

For purposes of uncertainty evaluation, a parameter, H, is defined as the ratio of the weighted radiance dose to the criteria, and this value has to be below 1 to be exempt.

$$H = (L_R \cdot t)/[5(\Delta t)^{0.75}/\alpha] = \alpha\ (L_R \cdot t)/5(\Delta t)^{0.75} \quad \textbf{(E-30)}$$

The uncertainty of H is the quadrature sum of the uncertainty of the weighted radiance dose plus the nominal pulse duration plus the angular subtense of the source. The value of Δt is raised to a power of -0.75. Therefore, $\partial H/\partial(\Delta t) = -0.75$ is included as a coefficient to the uncertainty, $u_c(\Delta t)$.

The individual terms are not independent, leading to two covariance terms: the first because α is a function of weighted radiance dose, and the second because α and $(L_R \cdot t)$ are a function of p, the cm/pixel factor.

The first covariance term is:

$$(2)(\partial H/\partial\alpha)[\partial\alpha/\partial(L_R \cdot t)][uc(L_R \cdot t)][\partial H/\partial(L_R \cdot t)][u_c(L_R \cdot t)] \quad \textbf{(E-31)}$$

Numerically evaluated:

$$(2)(1)(-0.499)(7.613\%)(1)(7.613\%) = -0.005784 \quad \textbf{(E-32)}$$

The second covariance term is:

$$(2)[\partial H/\partial(L_R \cdot t)][\partial(L_R \cdot t)/\partial p][u_c(p)](\partial H/\partial\alpha)(\partial\alpha/\partial p)[u_c(p)] \quad \textbf{(E-33)}$$

Numerically evaluated as:

$$(2)(1)(-2)(2.817\%)(1)(1)(2.817\%) = -0.003174 \quad \textbf{(E-34)}$$

When performed with a measurement-averaging field of view of 0.00175-radian included angle, the combined standard uncertainty of the evaluation for $R(\lambda)$-weighted radiance dose, $u_c(H)$, is calculated as:

$$[(7.613\%)^2 + (-0.75 \cdot 0.685\%)^2 + (6.2730\%)^2 - 0.005784 - 0.003174]^{1/2} = 2.827\% \quad \textbf{(E-35)}$$

Note: The method of uncertainty analysis used above was chosen to demonstrate the propagation of uncertainty and illustrate various techniques. A more direct method is to form the equation for H in terms of directly measured parameters. Multiple occurrences of factors are combined through simplification of the equation. The uncertainty of H can then be calculated from uncertainty of the directly measured parameters without the need for covariance terms.

E.14 All Averaging Angles

The source has to pass the criteria for all averaging-cone fields of view between the limits of 0.00175 radian and 0.100 radian to be classified as exempt. Using calculations as shown above for various cone angles, the source can be evaluated in the range of angles. A simpler method is used for the completion of this example.

The criterion is a function of α, which is a function of the weighted radiance dose. The angle of averaging will only affect the criterion through a change of the weighted radiance dose. The highest weighted radiance dose is seen with the smallest averaging angle; all larger angles will lead to lower weighted radiance doses. Rather than choosing larger angles for evaluation, lower levels of weighted radiance dose may be used directly in the calculations. Results are shown in **Table E-1**.

The fourth column in **Table E-1** shows a marked increase in the 50% width. As shown in this column, the pixels with weighted radiance dose greater than 0.020 effective J/cm^2·sr are from an area of 33 pixels by 109 pixels. This corresponds to a source with an angular subtense of 0.071 rad (see **Figure E 15**) and an Exempt Risk Group criterion for of 0.0170 effective J/cm^2·sr. Because the effective radiance doses were greater than 0.020 effective J/cm^2·sr, the source is not Exempt.

It should be noted that the source fails exempt classification for an intermediate value of weighted radiance dose, and checking both extremes is not sufficiently thorough. As these calculations are done on a computer, all values should be checked.

This example demonstrates the effort that may be needed to achieve small uncertainties. Less complexity, time and effort may be traded for higher uncertainties.

High confidence in classification may be achieved with large uncertainties if the measurement result is far from the classification limit.

Table E-1. Results of Averaging Field-of-View Angles

Weighted Radiance Dose	0.050	0.040	0.030	0.020	0.020
50% Length	25	28	30	33	39
50% Width	13	16	26	109	124
α	0.019	0.022	0.028	0.071	0.082
Criterion	0.062	0.054	0.042	0.017	0.015

Figure E-15. Profile of the strobe source from Figure E-13, adjusted in order to investigate the lamp's Risk Group classification.

Annex F – Table of Weighting Functions, Interpolated

Table F-1 shows the interpolated values in 1-nm increments (refer to **Section 6.4.1** in main text) for these functions:

$S(\lambda)$: UV Hazard Weighting Function

$A(\lambda)$: Aphakic Hazard Weighting Function

$B(\lambda)$: Retinal Blue-Light UV Hazard Weighting Function

$R(\lambda)$: Retinal Burn Hazard Weighting Function

Table F-1. Table of Weighting Functions

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(λ)	Aphakic *A*(λ)	Blue Light *B*(λ)	Retinal Thermal *R*(λ)
200	0.030000			
201	0.033359			
202	0.037094			
203	0.041247			
204	0.045865			
205	0.051000			
206	0.055089			
207	0.059507			
208	0.064278			
209	0.069433			
210	0.075000			
211	0.078631			
212	0.082438			
213	0.086429			
214	0.090613			
215	0.095000			
216	0.099544			
217	0.104305			
218	0.109294			
219	0.114522			
220	0.120000			
221	0.125477			
222	0.131203			
223	0.137192			
224	0.143453			
225	0.150000			
226	0.157262			
227	0.164876			
228	0.172858			
229	0.181226			
230	0.190000			
231	0.199088			
232	0.208611			
233	0.218589			
234	0.229044			
235	0.240000			
236	0.250953			
237	0.262407			
238	0.274383			
239	0.286906			
240	0.300000			
241	0.311141			
242	0.322696			
243	0.334680			
244	0.347109			
245	0.360000			
246	0.373023			
247	0.386517			
248	0.400500			
249	0.414988			
250	0.430000			

Table F-1. Table of Weighting Functions *(continued)*

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(λ)	Aphakic *A*(λ)	Blue Light *B*(λ)	Retinal Thermal *R*(λ)
251	0.446523			
252	0.463681			
253	0.481498			
254	0.500000			
255	0.520000			
256	0.543733			
257	0.568548			
258	0.594497			
259	0.621629			
260	0.650000			
261	0.679247			
262	0.709810			
263	0.741748			
264	0.775123			
265	0.810000			
266	0.844866			
267	0.881234			
268	0.919166			
269	0.958732			
270	1.000000			
271	0.991869			
272	0.983804			
273	0.975804			
274	0.967870			
275	0.960000			
276	0.943438			
277	0.927162			
278	0.911167			
279	0.895448			
280	0.880000			
281	0.856810			
282	0.834230			
283	0.812246			
284	0.790841			
285	0.770000			
286	0.742042			
287	0.715099			
288	0.689135			
289	0.664113			
290	0.640000			
291	0.618618			
292	0.597951			
293	0.577974			
294	0.558664			
295	0.540000			
296	0.498397			
297	0.460000			
298	0.398914			
299	0.345939			
300	0.300000			
301	0.221042			

* *Table note:*

Interpolated to 1 nm between 200 and 700 nm, to 5 nm between 705 and 1,200nm, and to 20 nm above 1,200 nm.

(continued on next page)

Table F-1. Table of Weighting Functions

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(λ)	Aphakic *A*(λ)	Blue Light *B*(λ)	Retinal Thermal *R*(λ)
302	0.162865			
303	0.120000			
304	0.084853			
305	0.060000	6.000	0.010	
306	0.045404	6.000	0.010	
307	0.034358	6.000	0.010	
308	0.026000	6.000	0.010	
309	0.019748	6.000	0.010	
310	0.015000	6.000	0.010	
311	0.011052	6.000	0.010	
312	0.008143	6.000	0.010	
313	0.006000	6.000	0.010	
314	0.004243	6.000	0.010	
315	0.003000	6.000	0.010	
316	0.002400	6.000	0.010	
317	0.002000	6.000	0.010	
318	0.001600	6.000	0.010	
319	0.001200	6.000	0.010	
320	0.001000	6.000	0.010	
321	0.000819	6.000	0.010	
322	0.000670	6.000	0.010	
323	0.000540	6.000	0.010	
324	0.000520	6.000	0.010	
325	0.000500	6.000	0.010	
326	0.000479	6.000	0.010	
327	0.000459	6.000	0.010	
328	0.000440	6.000	0.010	
329	0.000425	6.000	0.010	
330	0.000410	6.000	0.010	
331	0.000396	6.000	0.010	
332	0.000383	6.000	0.010	
333	0.000370	6.000	0.010	
334	0.000355	6.000	0.010	
335	0.000340	6.000	0.010	
336	0.000327	5.990	0.010	
337	0.000315	5.970	0.010	
338	0.000303	5.940	0.010	
339	0.000291	5.910	0.010	
340	0.000280	5.880	0.010	
341	0.000271	5.850	0.010	
342	0.000263	5.820	0.010	
343	0.000255	5.790	0.010	
344	0.000248	5.750	0.010	
345	0.000240	5.710	0.010	
346	0.000231	5.660	0.010	
347	0.000223	5.610	0.010	
348	0.000215	5.560	0.010	
349	0.000207	5.510	0.010	
350	0.000200	5.460	0.010	
351	0.000191	5.420	0.010	
352	0.000183	5.370	0.010	
353	0.000175	5.320	0.010	
354	0.000167	5.270	0.010	
355	0.000160	5.220	0.010	
356	0.000153	5.150	0.010	
357	0.000147	5.060	0.010	
358	0.000141	4.960	0.010	
359	0.000136	4.860	0.010	
360	0.000130	4.750	0.010	
361	0.000126	4.650	0.010	
362	0.000122	4.550	0.010	
363	0.000118	4.450	0.010	
364	0.000114	4.350	0.010	
365	0.000110	4.250	0.010	
366	0.000106	4.150	0.010	
367	0.000103	4.050	0.010	

Table F-1. Table of Weighting Functions *(continued)*

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(λ)	Aphakic *A*(λ)	Blue Light *B*(λ)	Retinal Thermal *R*(λ)
368	0.000099	3.950	0.010	
369	0.000096	3.850	0.010	
370	0.000093	3.750	0.010	
371	0.000090	3.650	0.010	
372	0.000086	3.550	0.010	
373	0.000083	3.450	0.010	
374	0.000080	3.350	0.010	
375	0.000077	3.250	0.010	
376	0.000074	3.150	0.010	
377	0.000072	3.050	0.010	
378	0.000069	2.950	0.010	
379	0.000066	2.850	0.010	
380	0.000064	2.760	0.010	0.010
381	0.000062	2.670	0.011	0.011
382	0.000059	2.580	0.011	0.011
383	0.000057	2.490	0.012	0.012
384	0.000055	2.400	0.012	0.012
385	0.000053	2.310	0.014	0.014
386	0.000051	2.220	0.015	0.015
387	0.000049	2.130	0.017	0.017
388	0.000047	2.050	0.019	0.019
389	0.000046	1.960	0.022	0.022
390	0.000044	1.880	0.025	0.025
391	0.000042	1.810	0.029	0.029
392	0.000041	1.750	0.033	0.033
393	0.000039	1.690	0.039	0.039
394	0.000037	1.630	0.044	0.044
395	0.000036	1.580	0.051	0.051
396	0.000035	1.540	0.057	0.057
397	0.000033	1.510	0.067	0.067
398	0.000032	1.470	0.076	0.076
399	0.000031	1.440	0.088	0.088
400	0.000030	1.410	0.100	0.100
401		1.380	0.120	0.120
402		1.350	0.140	0.140
403		1.330	0.160	0.160
404		1.310	0.180	0.180
405		1.300	0.200	0.200
406		1.290	0.240	0.240
407		1.280	0.280	0.280
408		1.270	0.320	0.320
409		1.260	0.360	0.360
410		1.250	0.400	0.400
411		1.240	0.480	0.480
412		1.230	0.560	0.560
413		1.220	0.640	0.640
414		1.210	0.720	0.720
415		1.200	0.800	0.800
416		1.190	0.820	0.820
417		1.180	0.840	0.840
418		1.170	0.860	0.860
419		1.160	0.880	0.880
420		1.150	0.900	0.900
421		1.142	0.910	0.910
422		1.134	0.920	0.920
423		1.126	0.930	0.930
424		1.118	0.940	0.940
425		1.110	0.950	0.950
426		1.102	0.956	0.956
427		1.094	0.962	0.962
428		1.086	0.968	0.968
429		1.078	0.974	0.974
430		1.070	0.980	0.980
431		1.062	0.984	0.984
432		1.054	0.988	0.988
433		1.046	0.992	0.992

* *Table note:*

Interpolated to 1 nm between 200 and 700 nm, to 5 nm between 705 and 1,200nm, and to 20 nm above 1,200 nm.

(continued on next page)

Table F-1. Table of Weighting Functions

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(λ)	Aphakic *A*(λ)	Blue Light *B*(λ)	Retinal Thermal *R*(λ)
434		1.038	0.996	0.996
435		1.030	1.000	1.000
436		1.023	1.000	1.000
437		1.018	1.000	1.000
438		1.011	1.000	1.000
439		1.005	1.000	1.000
440		1.000	1.000	1.000
441		0.994	0.994	1.000
442		0.988	0.988	1.000
443		0.982	0.982	1.000
444		0.976	0.976	1.000
445		0.970	0.970	1.000
446		0.964	0.964	1.000
447		0.958	0.958	1.000
448		0.952	0.952	1.000
449		0.946	0.946	1.000
450		0.940	0.940	1.000
451		0.932	0.932	1.000
452		0.924	0.924	1.000
453		0.916	0.916	1.000
454		0.908	0.908	1.000
455		0.900	0.900	1.000
456		0.880	0.880	1.000
457		0.860	0.860	1.000
458		0.840	0.840	1.000
459		0.820	0.820	1.000
460		0.800	0.800	1.000
461		0.780	0.780	1.000
462		0.760	0.760	1.000
463		0.740	0.740	1.000
464		0.720	0.720	1.000
465		0.705	0.700	1.000
466		0.684	0.684	1.000
467		0.668	0.668	1.000
468		0.652	0.652	1.000
469		0.636	0.636	1.000
470		0.620	0.620	1.000
471		0.606	0.606	1.000
472		0.592	0.592	1.000
473		0.578	0.578	1.000
474		0.564	0.564	1.000
475		0.550	0.550	1.000
476		0.530	0.530	1.000
477		0.510	0.510	1.000
478		0.490	0.490	1.000
479		0.470	0.470	1.000
480		0.450	0.450	1.000
481		0.440	0.440	1.000
482		0.430	0.430	1.000
483		0.420	0.420	1.000
484		0.410	0.410	1.000
485		0.400	0.400	1.000
486		0.364	0.364	1.000
487		0.328	0.328	1.000
488		0.292	0.292	1.000
489		0.256	0.256	1.000
490		0.220	0.220	1.000
491		0.208	0.208	1.000
492		0.196	0.196	1.000
493		0.184	0.184	1.000
494		0.172	0.172	1.000
495		0.160	0.160	1.000
496		0.148	0.148	1.000
497		0.136	0.136	1.000
498		0.124	0.124	1.000
499		0.112	0.112	1.000

Table F-1. Table of Weighting Functions *(continued)*

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(λ)	Aphakic *A*(λ)	Blue Light *B*(λ)	Retinal Thermal *R*(λ)
500		0.100	0.100	1.000
501		0.095	0.095	1.000
502		0.091	0.091	1.000
503		0.087	0.087	1.000
504		0.083	0.083	1.000
505		0.079	0.079	1.000
506		0.076	0.076	1.000
507		0.072	0.072	1.000
508		0.069	0.069	1.000
509		0.066	0.066	1.000
510		0.063	0.063	1.000
511		0.060	0.060	1.000
512		0.058	0.058	1.000
513		0.055	0.055	1.000
514		0.052	0.052	1.000
515		0.050	0.050	1.000
516		0.048	0.048	1.000
517		0.046	0.046	1.000
518		0.044	0.044	1.000
519		0.042	0.042	1.000
520		0.040	0.040	1.000
521		0.038	0.038	1.000
522		0.036	0.036	1.000
523		0.035	0.035	1.000
524		0.033	0.033	1.000
525		0.032	0.032	1.000
526		0.030	0.030	1.000
527		0.029	0.029	1.000
528		0.028	0.028	1.000
529		0.026	0.026	1.000
530		0.025	0.025	1.000
531		0.024	0.024	1.000
532		0.023	0.023	1.000
533		0.022	0.022	1.000
534		0.021	0.021	1.000
535		0.020	0.020	1.000
536		0.019	0.019	1.000
537		0.018	0.018	1.000
538		0.017	0.017	1.000
539		0.017	0.017	1.000
540		0.016	0.016	1.000
541		0.015	0.015	1.000
542		0.014	0.014	1.000
543		0.014	0.014	1.000
544		0.013	0.013	1.000
545		0.013	0.013	1.000
546		0.012	0.012	1.000
547		0.011	0.011	1.000
548		0.011	0.011	1.000
549		0.010	0.010	1.000
550		0.010	0.010	1.000
551		0.010	0.010	1.000
552		0.009	0.009	1.000
553		0.009	0.009	1.000
554		0.008	0.008	1.000
555		0.008	0.008	1.000
556		0.008	0.008	1.000
557		0.007	0.007	1.000
558		0.007	0.007	1.000
559		0.007	0.007	1.000
560		0.006	0.006	1.000
561		0.006	0.006	1.000
562		0.006	0.006	1.000
563		0.005	0.005	1.000
564		0.005	0.005	1.000
565		0.005	0.005	1.000

* *Table note:*

Interpolated to 1 nm between 200 and 700 nm, to 5 nm between 705 and 1,200nm, and to 20 nm above 1,200 nm.

(continued on next page)

Table F-1. Table of Weighting Functions

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV $S(\lambda)$	Aphakic $A(\lambda)$	Blue Light $B(\lambda)$	Retinal Thermal $R(\lambda)$
566		0.005	0.005	1.000
567		0.005	0.005	1.000
568		0.004	0.004	1.000
569		0.004	0.004	1.000
570		0.004	0.004	1.000
571		0.004	0.004	1.000
572		0.004	0.004	1.000
573		0.003	0.003	1.000
574		0.003	0.003	1.000
575		0.003	0.003	1.000
576		0.003	0.003	1.000
577		0.003	0.003	1.000
578		0.003	0.003	1.000
579		0.003	0.003	1.000
580		0.003	0.003	1.000
581		0.002	0.002	1.000
582		0.002	0.002	1.000
583		0.002	0.002	1.000
584		0.002	0.002	1.000
585		0.002	0.002	1.000
586		0.002	0.002	1.000
587		0.002	0.002	1.000
588		0.002	0.002	1.000
589		0.002	0.002	1.000
590		0.002	0.002	1.000
591		0.002	0.002	1.000
592		0.001	0.001	1.000
593		0.001	0.001	1.000
594		0.001	0.001	1.000
595		0.001	0.001	1.000
596		0.001	0.001	1.000
597		0.001	0.001	1.000
598		0.001	0.001	1.000
599		0.001	0.001	1.000
600		0.001	0.001	1.000
601		0.001	0.001	1.000
602		0.001	0.001	1.000
603		0.001	0.001	1.000
604		0.001	0.001	1.000
605		0.001	0.001	1.000
606		0.001	0.001	1.000
607		0.001	0.001	1.000
608		0.001	0.001	1.000
609		0.001	0.001	1.000
610		0.001	0.001	1.000
611		0.001	0.001	1.000
612		0.001	0.001	1.000
613		0.001	0.001	1.000
614		0.001	0.001	1.000
615		0.001	0.001	1.000
616		0.001	0.001	1.000
617		0.001	0.001	1.000
618		0.001	0.001	1.000
619		0.001	0.001	1.000
620		0.001	0.001	1.000
621		0.001	0.001	1.000
622		0.001	0.001	1.000
623		0.001	0.001	1.000
624		0.001	0.001	1.000
625		0.001	0.001	1.000
626		0.001	0.001	1.000
627		0.001	0.001	1.000
628		0.001	0.001	1.000
629		0.001	0.001	1.000
630		0.001	0.001	1.000
631		0.001	0.001	1.000

Table F-1. Table of Weighting Functions *(continued)*

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV $S(\lambda)$	Aphakic $A(\lambda)$	Blue Light $B(\lambda)$	Retinal Thermal $R(\lambda)$
632		0.001	0.001	1.000
633		0.001	0.001	1.000
634		0.001	0.001	1.000
635		0.001	0.001	1.000
636		0.001	0.001	1.000
637		0.001	0.001	1.000
638		0.001	0.001	1.000
639		0.001	0.001	1.000
640		0.001	0.001	1.000
641		0.001	0.001	1.000
642		0.001	0.001	1.000
643		0.001	0.001	1.000
644		0.001	0.001	1.000
645		0.001	0.001	1.000
646		0.001	0.001	1.000
647		0.001	0.001	1.000
648		0.001	0.001	1.000
649		0.001	0.001	1.000
650		0.001	0.001	1.000
651		0.001	0.001	1.000
652		0.001	0.001	1.000
653		0.001	0.001	1.000
654		0.001	0.001	1.000
655		0.001	0.001	1.000
656		0.001	0.001	1.000
657		0.001	0.001	1.000
658		0.001	0.001	1.000
659		0.001	0.001	1.000
660		0.001	0.001	1.000
661		0.001	0.001	1.000
662		0.001	0.001	1.000
663		0.001	0.001	1.000
664		0.001	0.001	1.000
665		0.001	0.001	1.000
666		0.001	0.001	1.000
667		0.001	0.001	1.000
668		0.001	0.001	1.000
669		0.001	0.001	1.000
670		0.001	0.001	1.000
671		0.001	0.001	1.000
672		0.001	0.001	1.000
673		0.001	0.001	1.000
674		0.001	0.001	1.000
675		0.001	0.001	1.000
676		0.001	0.001	1.000
677		0.001	0.001	1.000
678		0.001	0.001	1.000
679		0.001	0.001	1.000
680		0.001	0.001	1.000
681		0.001	0.001	1.000
682		0.001	0.001	1.000
683		0.001	0.001	1.000
684		0.001	0.001	1.000
685		0.001	0.001	1.000
686		0.001	0.001	1.000
687		0.001	0.001	1.000
688		0.001	0.001	1.000
689		0.001	0.001	1.000
690		0.001	0.001	1.000
691		0.001	0.001	1.000
692		0.001	0.001	1.000
693		0.001	0.001	1.000
694		0.001	0.001	1.000
695		0.001	0.001	1.000
696		0.001	0.001	1.000
697		0.001	0.001	1.000

* *Table note:*

Interpolated to 1 nm between 200 and 700 nm, to 5 nm between 705 and 1,200nm, and to 20 nm above 1,200 nm.

(continued on next page)

Table F-1. Table of Weighting Functions

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(*λ*)	Aphakic *A*(*λ*)	Blue Light *B*(*λ*)	Retinal Thermal *R*(*λ*)
698		0.001	0.001	1.000
699		0.001	0.001	1.000
700		0.001	0.001	1.000
705				0.977
710				0.955
715				0.933
720				0.912
725				0.891
730				0.871
735				0.851
740				0.832
745				0.813
750				0.794
755				0.776
760				0.759
765				0.741
770				0.724
775				0.708
780				0.692
785				0.676
790				0.661
795				0.646
800				0.631
805				0.617
810				0.603
815				0.589
820				0.575
825				0.562
830				0.550
835				0.537
840				0.525
845				0.513
850				0.501
855				0.490
860				0.479
865				0.468
870				0.457
875				0.447
880				0.437
885				0.427
890				0.417
895				0.407
900				0.398
905				0.389
910				0.380
915				0.372
920				0.363
925				0.355
930				0.347
935				0.339
940				0.331
945				0.324
950				0.316
955				0.309
960				0.302
965				0.295
970				0.288

Table F-1. Table of Weighting Functions *(continued)*

Wavelength [λ], nm	Weighting Functions*			
	Actinic UV *S*(*λ*)	Aphakic *A*(*λ*)	Blue Light *B*(*λ*)	Retinal Thermal *R*(*λ*)
975				0.282
980				0.275
985				0.269
990				0.263
995				0.257
1,000				0.251
1,005				0.245
1,010				0.240
1,015				0.234
1,020				0.229
1,025				0.224
1,030				0.219
1,035				0.214
1,040				0.209
1,045				0.204
1,050				0.200
1,055				0.200
1,060				0.200
1,065				0.200
1,070				0.200
1,075				0.200
1,080				0.200
1,085				0.200
1,090				0.200
1,095				0.200
1,100				0.200
1,105				0.200
1,110				0.200
1,115				0.200
1,120				0.200
1,125				0.200
1,130				0.200
1,135				0.200
1,140				0.200
1,145				0.200
1,150				0.200
1,155				0.159
1,160				0.126
1,165				0.100
1,170				0.080
1,175				0.063
1,180				0.050
1,185				0.040
1,190				0.032
1,195				0.025
1,200				0.020
1,220				0.020
1,240				0.020
1,260				0.020
1,280				0.020
1,300				0.020
1,320				0.020
1,340				0.020
1,360				0.020
1,380				0.020
1,400				0.020

* *Table note:*
Interpolated to 1 nm between 200 and 700 nm, to 5 nm between 705 and 1,200nm, and to 20 nm above 1,200 nm.

Annex G – Units and Conversion Factors

G.1 Luminance Conversion Factors

1 nit = 1 candela/square meter

1 stilb* = 1 candela/square centimeter

1 apostilb* (international) = 0.1 millilambert = 1 blondel

1 apostilb* (German Hefner) = 0.09 millilambert

1 lambert* = 1000 millilamberts

Table G-1. Luminance Conversion Factors

Multiply Number of →to Obtain Number of ↓ By ↘	Footlambert*	Candela/ square meter	Millilambert*	Candela/ square inch*	Candela/ square foot*	Stilb*
Footlambert*	1	0.2919	0.929	452	3.142	2,919
Candela/square meter	3.426	1	3.183	1,550	10.76	10,000
Millilambert*	1.076	0.3142	1	487	3.382	3,142
Candela/square inch*	0.00221	0.000645	0.000205	1	0.000694	6.45
Candela/square foot*	0.3183	0.0929	0.2957	144	1	929
Stilb*	0.00034	0.0001	0.00032	0.155	0.00108	1

G.2 Illuminance Conversion Factors

1 lumen = 1/683 light-watt*

1 lumen-hour = 60 lumen-minutes

1 footcandle* = 1 lumen/square foot*

1 watt-second=107 ergs*

1 phot*= 1 lumen/square centimeter

1 lux= 1 lumen/square meter= 1 metercandle*

Table G-2. Illuminance Conversion Factors

Multiply Number of → to Obtain Number of ↓ By ↘	Footcandles	Lux	Phot*	Milliphot*
Footcandles*	1	0.0929	929	0.929
Lux	10.76	1	10,000	10
Phot*	0.00108	0.0001	1	0.001
Milliphot*	1.076	0.1	1,000	1

* Deprecated unit

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________
Affiliation: ______________________________
Address: ______________________________
City: ______________ State: ________ Zip: ________ Country: ______________
Telephone: ______________________________
Fax: ______________________________
E-mail: ______________________________

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: ______________________________ Date: ______________

2. Title of publications and year published ______________________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):
[ ] Change to read as follows
[ ] Delete and substitute as follows
[ ] Add new text as follows
[ ] Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
____
____

6. Reason and substantiation:

____
____
____

Select as applicable:
[ ] Additional pages are attached. Number of additional pages:
[ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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